TRIM AND STABILITY BOOKLET
OF
KB-24

SEE LETTER : E-151850-203473

APPROVED
FOR COMPLIANCE WITH (SEE LETTER) REQUIREMENTS
ON BEHALF OF GOVT. OF INDIA
INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 02 May 2022

PLAN(S) NOW SENT SUPERSEDE THE APPROPRIATE PLAN(S) PREVIOUSLY APPROVED/~~NOTED~~ ON 04/08/2020 .

AS AMENDED ON PAGE 02 , 24, 51-54 & 56 ONLY.

Rev.	Description	Date	Prepared	Review

Issued for : Approval

Client:	Project:
K.B. SHIPPING & CO.(JAMNAGAR)	**KB-24 (PONTOON)**
Builder: **CHISTI MARINE ENGINEERING WORKS**	

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CONTENTS

General

Page No

Ship's Data

Loading Conditions

~~NON SAILING CONDITIONS~~

FOR 250T CRAWLER CRANE (NON SAILING CONDITIONS)

Appendix

GENERAL PARTICULARS

Name of the Vessel	KB-24
Client	K.B. SHIPPING & CO. (JAMNAGAR)
Builder	CHISTI MARINE ENGINEERING WORKS
Type of Vessel	PONTOON
Class Notation	卐SU,PONTOON,"FOR CRANE OPERATION" "DECK STRENGTHENED FOR 10 T/M^2"

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	17.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement(Ext.)	1768.98 T
Lightship	356.33 T
Deadweight	1412.65 T
Gross Tonnage	648
Net Tonnage	194

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.

2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.

3) The minimum range of stability should be:
   For L <= 100m: 20^0
   For L >= 150m: 15^0
   For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta C \leq 15°$
- $GZC \leq 0,6\ GZMAX$
- $A1 \geq 0.4\ ATOT$

where:

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC, GZMAX: Defined in Fig 1

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle equal to the lesser of:

- heeling angle θR of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θF, corresponding to flooding of unprotected openings

ATOT : Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b=(Pd-Zz)/\Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z: Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM.The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

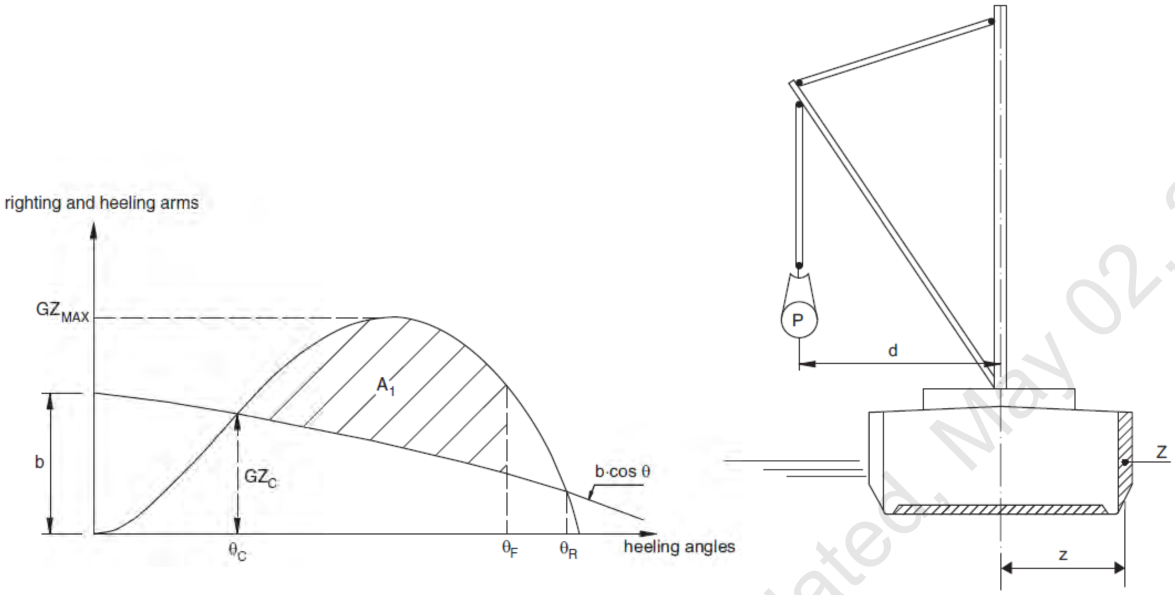


FIGURE 1: CARGO LIFTING

Intact stability criteria in the event of sudden loss of cargo during lifting

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- A2 / A1 ≥ 1
- θ2 - θ3 ≥ 20°

where:

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ1 to the heeling angle θC (see Fig 2)

A2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle θ2 (see Fig 2)

A3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle θ3 (see Fig 2)

θ1 : Heeling angle of equilibrium during lifting (see Fig 2)

θ2 : Heeling angle corresponding to the lesser of θR and θF

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

θ3 : Maximum heeling angle due to roll, at which A3 = A1, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θR : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θF : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$b = Zz \cos\theta/\Delta$

HEELING AND RIGHTING ARMS

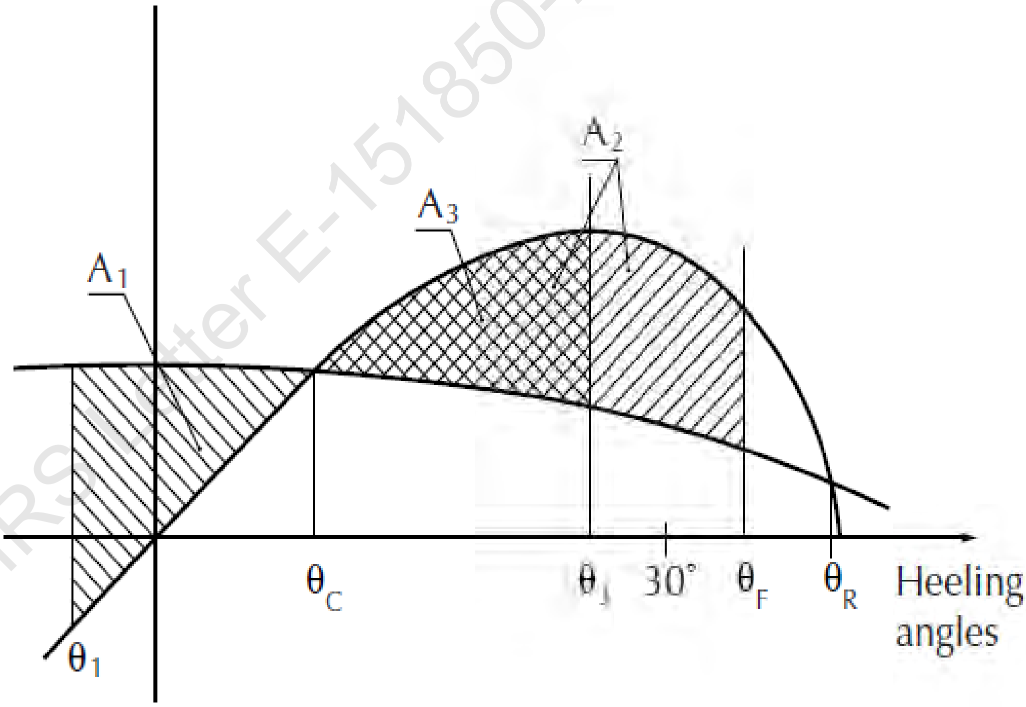


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.

2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship’s stability in various loading conditions.

3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.

4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.

5. In this booklet hydrostatic particulars and cross curves are given at following trims.

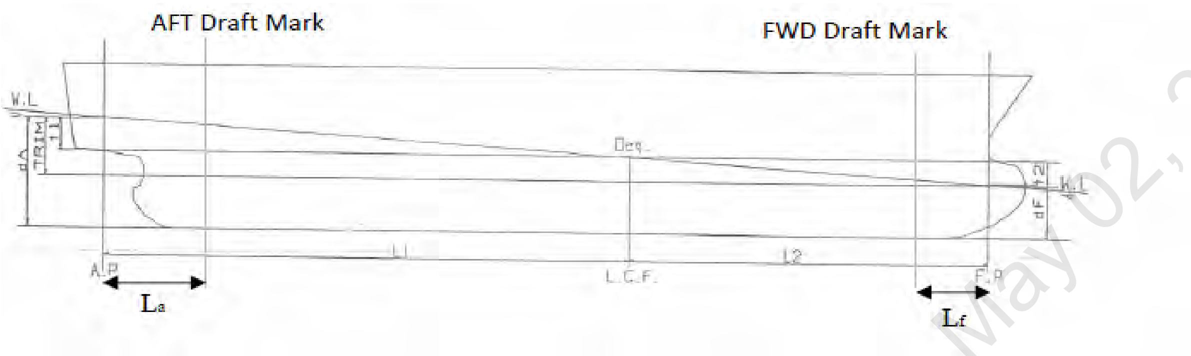
   -1.00m,-0.50m, 0.00m,0.50m, 1.00m

   These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

a) Longitudinal Centre of gravity of the ship calculated as follows:

$$LCG = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

b) Longitudinal centre of buoyancy, LCB

c) Displacement, △

d) Moment to change trim, MCT1cm

e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$Trim = \frac{\Delta * (LCB - LCG)}{MCT1cm * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{LCF * Trim}{LBP.} + T\,(extreme)$$

$$T_{FP} = T_{AP} - Trim$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{Trim * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{Trim * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

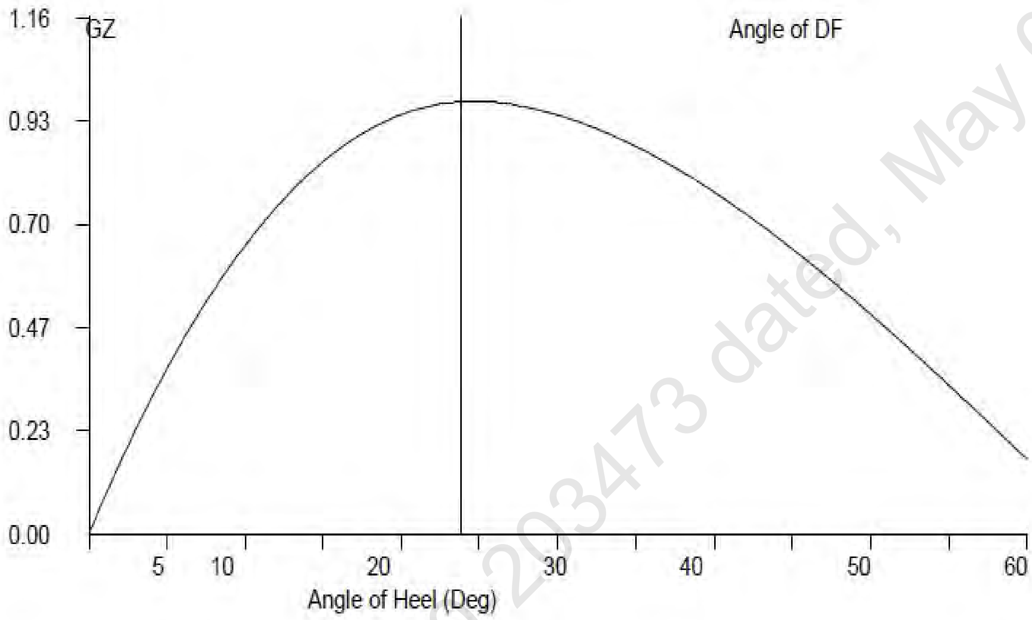
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

1) GZ = KN - KGo Sin (theta)
2) Area under 0-10 = 10*0.01745*[GZ0 + GZ10] / 2
3) Area under 10-20 = 10*0.01745*[GZ10 + GZ20] / 2
4) Area under 20-30 = 10*0.01745*[GZ20 + GZ30] / 2
5) Area under 30-40 = 10*0.01745*[GZ30 + GZ40] / 2
6) Area under 40-50 = 10*0.01745*[GZ40 + GZ50] / 2
7) Area under 50-60 = 10*0.01745*[GZ50 + GZ60] / 2
8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples

1 Loading the Vessel

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 1: Lightship + Crane + Permanent Ballast

Displacement	870.93	Tonnes
LCG	25.000	M Fwd of AP
VCG (KG)	3.176	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

Free Surface moment for a Tank = Moment of Inertia (M^4) for a Tank x Density of Liquid in Tank

KGo (Fluid) = KG (Solid) + Free Surface Correction.

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

Trim = (LCB-LCG) X displacement / (MCT x 100) + Trim*

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 1 : Lightship + Crane + Permanent Ballast

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
03_PFWBTK.C	132.30	1.000	132.30	16.000	2116.80	0.000	0.00	1.500	198.45	91.87
05_PFWBTK.C	132.30	1.000	132.30	34.000	4498.20	0.000	0.00	1.500	198.45	91.87
TOTAL TANK			264.60	25.000	6615.00	0.000	0.00	1.500	396.90	183.74

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			356.33	25.000	8,908.25	0.000	0.00	3.000	1,068.99
CRANE			250.00	25.000	6,250.00	0.000	0.00	5.200	1,300.00
TOTAL FIXED WEIGHT			606.33	25.000	15,158.25	0.000	0.00	3.907	2,368.99

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			264.600	25.000	6615.000	0.000	0.000	1.500	396.900
TOTAL FIXED WEIGHT			606.330	25.000	15158.250	0.000	0.000	3.907	2368.990
TOTAL WEIGHT			870.930	25.000	21773.250	0.000	0.000	3.176	2765.890

Condition 1 : Lightship + Crane + Permanent Ballast

Free Surface Correction for VCG = FSM/Displacement

	0.211	m
VCGc	3.387	m

From the MaxVCG table,

Displacement	870.93	T
Allowable Max VCG	14.657	m

VCGc < Maxvcg PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	870.930	T
Draft ,Tm @ 25.0f =	1.151	m
LCB =	25.021	m
LCF =	25.000	m
MCT =	29.453	T-m/cm
KMT =	23.008	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)
0.006 m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid	1.151	m
Draft @ 50.0f (FP),Tfwd	=Tm - Trim/2	
	1.148	m
Draft@0(AP),Taft	=Tm + Trim/2	
	1.154	m

Condition 1 : Lightship + Crane + Permanent Ballast

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	29.453	T-m	Draft (Mean)	1.151 m
LCB	25.021	m	Draft AP (USK)	1.154 m
			Draft FP (USK)	1.148 m
Metacentric Height				
KMt	23.008	m		
KG	3.176	m		
KG0	3.387	m		
GMt	19.621	m	**Trim**	0.006 m

The condition shown above is first worked
using Even Keel hydrostatic perticulars
Program then interpolates the hydrostatic perticulars
from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	28.900	T-m	Draft (Mean)	2.212 m
LCB	25.000	m	Draft AP (USK)	2.308 m
			Draft FP (USK)	2.116 m
Metacentric Height				
KMt	22.754	m		
KG	3.176	m		
GMt	19.578	m	**Trim**	0.192 m

Trim is calculaed as follows:
Trim = (LCB - LCG) X displacement / (MCT X 100) + Trim*
Where Trim* is trim at which hydrostatics were calculated.

Condition 1 : Lightship + Crane + Permanent Ballast

Intact Stability (Primary Criterion)

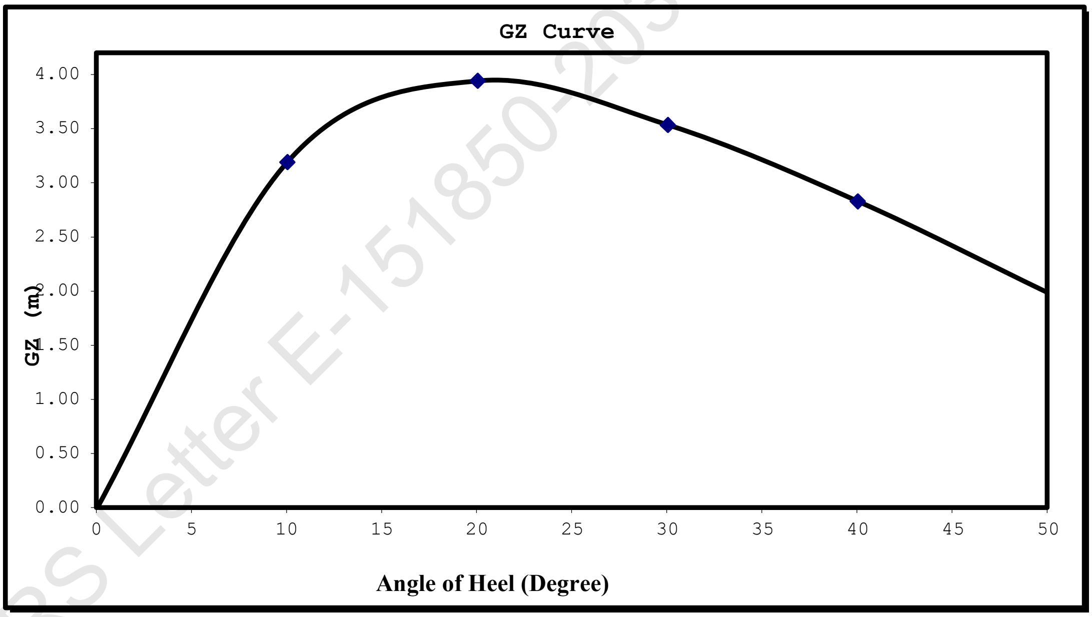
			Minimum Required
Displacement	870.93	T	
KG0	0.000	M	
GMt	19.578	M	$\boxed{0.150\ M}$

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	-0.017	3.192	3.942	3.535	2.830	1.984
GZ (M)	-0.017	3.192	3.942	3.535	2.830	1.984

	Computed values		Minimum Required	
Residual Area from abs 0 deg to MAxRA	0.8614	m-rad	0.080	m-rad
Angle from Equilibrium to RAZero or flood	LARGE	m-rad	20.00	Deg.
Angle from Equilibrium to 50% DK Imm Angle	6.74	m-rad	0.00	Deg.

Plot GZ values against the angles of heel as above.
At 57.3 degree erect an ordinate equal to GM fluid on the same
scale as GZ ordinates.Join the point to the origin.
This is the Tangent to the GZ Curve at the Origin.Find the Maxima of GZ
curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding
If maximum GZ occurs at an angle less than 30 degrees
then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	1.151	M
LCF from AP	24.994	M
LCB from AP	25.000	M
MCT	28.900	T-M/Cm
Trim	0.006	M
Draft (AP)	1.154	M
Draft (FP)	1.148	M
KG	3.176	M
GoMt	19.578	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
TOTAL TANK										

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			356.330	25.000	8,908.250	0.000	0.000	3.000	1,068.990
CRANE									
DECK CARGO									
LOAD LIFTED									
TOTAL FIXED WEIGHT									

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK									
TOTAL FIXED WEIGHT									
TOTAL WEIGHT									

Free Surface Correction for VCG = FSM/Displacement

m

VCGc m

From the MaxVCG table,

Displacement T

Allowable Max VCG m

$\boxed{\text{VCGc} < \text{Maxvcg}}$

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact

Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement = T

Draft ,Tm @ 25.0f = m

LCB = m

LCF = m

MCT = T-m/cm

KMT = m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)

m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2

m

Draft@0(AP),Taft =Tm + Trim/2

m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

MULTIPLY BY	TO CONVERT FROM	TO OBTAIN	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	TO OBTAIN	TO CONVERT FROM	MULTIPLY

Relation between Weight and volume

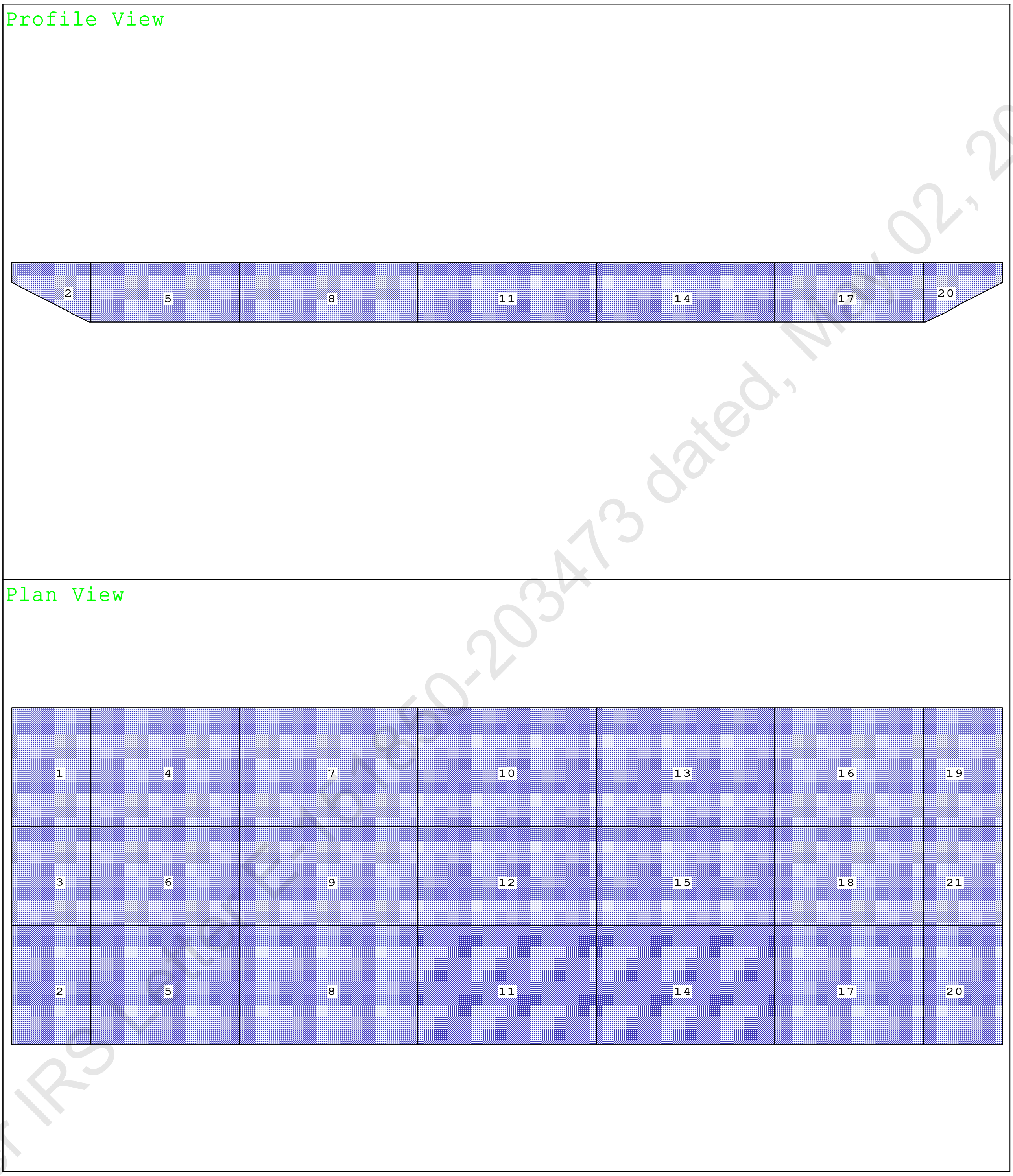
1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill off this opening is a `potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
   If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

   As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

CAPACITY PLAN AND TABLE



Tanks

1 01_VOID.P...100% FRESH WATER	8 03_VOID.S...100% FRESH WATER	16 06_VOID.P...100% FRESH WATER
2 01_VOID.S...100% FRESH WATER	9 03_PFWBTK.C.100% FRESH WATER	17 06_VOID.S...100% FRESH WATER
3 01_VOID.C...100% FRESH WATER	10 04_VOID.P...100% FRESH WATER	18 06_VOID.C...100% FRESH WATER
4 02_VOID.P...100% FRESH WATER	11 04_VOID.S...100% FRESH WATER	19 07_VOID.P...100% FRESH WATER
5 02_VOID.S...100% FRESH WATER	12 04_VOID.C...100% FRESH WATER	20 07_VOID.S...100% FRESH WATER
6 02_VOID.C...100% FRESH WATER	13 05_VOID.P...100% FRESH WATER	21 07_VOID.C...100% FRESH WATER
7 03_VOID.P...100% FRESH WATER	14 05_VOID.S...100% FRESH WATER	
	15 05_PFWBTK.C.100% FRESH WATER	

25-04-22 09:28:19 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

TANK STATUS
Trim: zero, Heel: zero

Part	Cu.M.	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
01_VOID.P	47.2	~~1.000~~	47.16	2.341f	5.470p	1.909	70.56*
01_VOID.S	47.2	~~1.000~~	47.16	2.341f	5.470s	1.909	70.56*
01_VOID.C	39.7	~~1.000~~	39.71	2.337f	0.000	1.901	40.83*
02_VOID.P	131.4	~~1.000~~	131.38	7.750f	5.480p	1.509	132.30*
02_VOID.S	131.4	~~1.000~~	131.38	7.750f	5.480s	1.509	132.30*
02_VOID.C	110.2	~~1.000~~	110.25	7.750f	0.000	1.500	76.56*
03_VOID.P	157.7	~~1.000~~	157.66	16.000f	5.480p	1.509	158.76*
03_VOID.S	157.7	~~1.000~~	157.66	16.000f	5.480s	1.509	158.76*
03_PFWBTK.C	132.3	1.000	132.30	16.000f	0.000	1.500	91.87*
04_VOID.P	157.7	~~1.000~~	157.66	25.000f	5.480p	1.509	158.76*
04_VOID.S	157.7	~~1.000~~	157.66	25.000f	5.480s	1.509	158.76*
04_VOID.C	132.3	~~1.000~~	132.30	25.000f	0.000	1.500	91.87*
05_VOID.P	157.7	~~1.000~~	157.66	34.000f	5.480p	1.509	158.76*
05_VOID.S	157.7	~~1.000~~	157.66	34.000f	5.480s	1.509	158.76*
05_PFWBTK.C	132.3	1.000	132.30	34.000f	0.000	1.500	91.87*
06_VOID.P	131.4	~~1.000~~	131.38	42.250f	5.480p	1.509	132.30*
06_VOID.S	131.4	~~1.000~~	131.38	42.250f	5.480s	1.509	132.30*
06_VOID.C	110.2	~~1.000~~	110.25	42.250f	0.000	1.500	76.56*
07_VOID.P	47.4	~~1.000~~	47.44	47.655f	5.470p	1.900	70.56*
07_VOID.S	47.4	~~1.000~~	47.44	47.655f	5.470s	1.900	70.56*
07_VOID.C	39.9	~~1.000~~	39.94	47.659f	0.000	1.892	40.83*
Total Tanks--------->			2,357.70	25.007f	0.000	1.552	2274.40

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

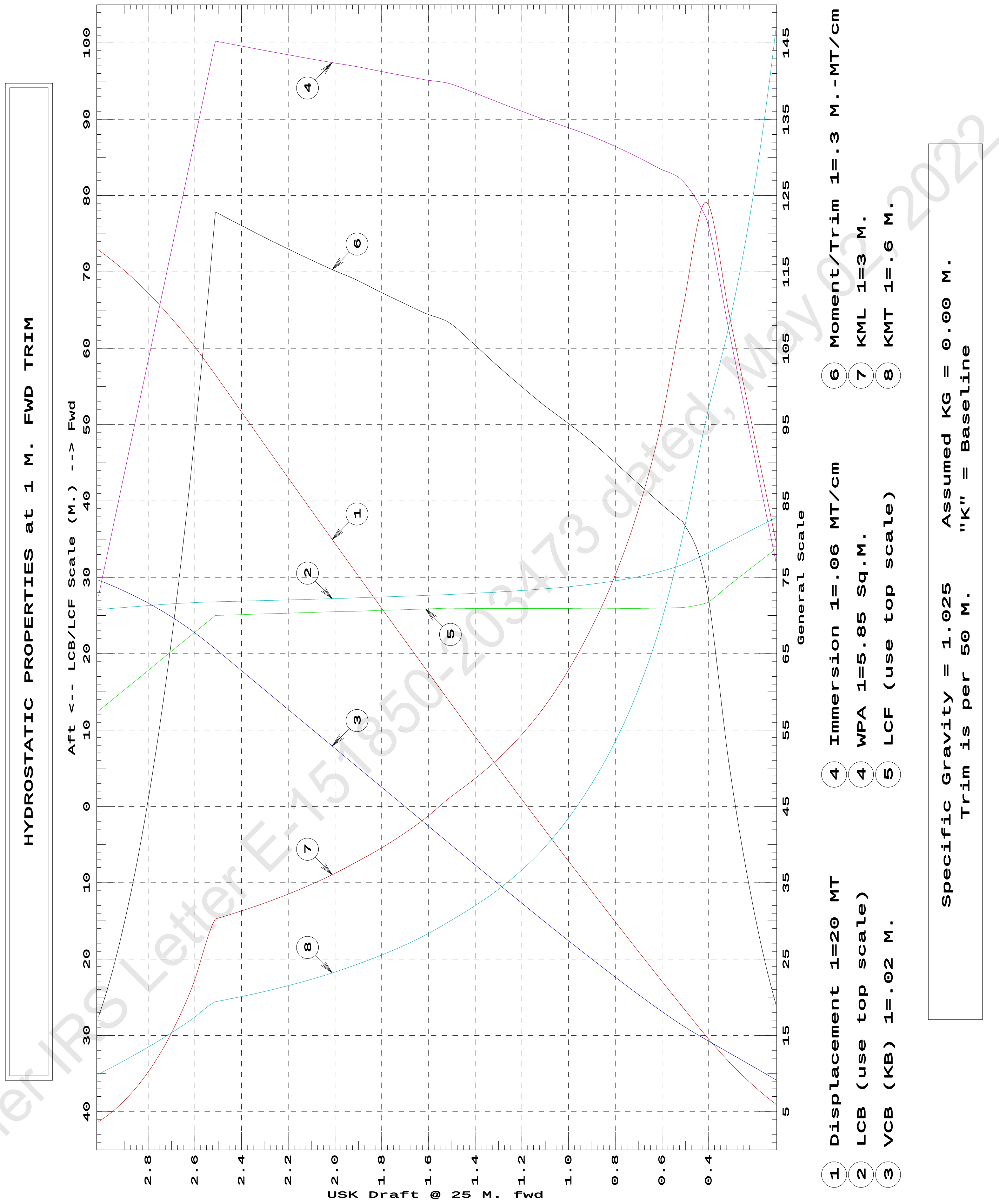
HYDROSTATIC PARTICULARS

HYDROSTATIC PROPERTIES
Trim: Fwd 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	119.34	37.840f	0.183	4.61	33.752f	5.70	238.67	88.197
0.212	169.99	36.258f	0.219	5.52	31.380f	9.64	283.35	74.801
0.312	229.67	34.673f	0.254	6.42	29.000f	15.06	327.84	64.950
0.412	298.40	33.086f	0.289	7.33	26.616f	22.23	372.38	57.417
0.512	373.75	31.683f	0.326	7.61	26.047f	24.68	330.14	48.248
0.612	450.36	30.717f	0.367	7.71	25.975f	25.44	282.41	40.981
0.712	527.96	30.016f	0.412	7.81	25.926f	26.27	248.71	35.728
0.812	606.47	29.485f	0.458	7.90	25.894f	27.10	223.35	31.690
0.912	685.79	29.069f	0.505	7.97	25.884f	27.90	203.41	28.472
1.012	765.82	28.738f	0.553	8.04	25.913f	28.62	186.84	25.799
1.112	846.50	28.467f	0.603	8.10	25.906f	29.31	173.10	23.589
1.212	927.86	28.243f	0.652	8.17	25.912f	30.08	162.05	21.802
1.312	1,009.90	28.054f	0.702	8.24	25.918f	30.88	152.84	20.311
1.412	1,092.65	27.893f	0.753	8.31	25.923f	31.72	145.12	19.055
1.512	1,176.09	27.754f	0.803	8.38	25.952f	32.51	138.19	17.941
1.612	1,260.00	27.630f	0.855	8.41	25.849f	32.87	130.41	16.890
1.712	1,344.26	27.515f	0.905	8.44	25.757f	33.30	123.83	16.002
1.812	1,428.86	27.409f	0.956	8.48	25.662f	33.74	118.05	15.226
1.912	1,513.80	27.308f	1.007	8.52	25.563f	34.20	112.95	14.545
2.012	1,599.09	27.213f	1.058	8.55	25.484f	34.59	108.14	13.910
2.112	1,684.69	27.123f	1.109	8.58	25.392f	35.02	103.91	13.353
2.212	1,770.61	27.036f	1.160	8.61	25.297f	35.46	100.10	12.855
2.312	1,856.86	26.953f	1.211	8.64	25.200f	35.91	96.67	12.409
2.412	1,943.46	26.873f	1.261	8.68	25.100f	36.38	93.58	12.009
2.512	2,030.39	26.795f	1.312	8.71	25.000f	36.86	90.74	11.647
2.612	2,113.11	26.677f	1.360	7.84	22.511f	27.01	63.91	10.283
2.712	2,187.13	26.495f	1.401	6.97	20.009f	19.19	43.85	9.066
2.812	2,252.45	26.271f	1.437	6.10	17.506f	13.09	29.06	7.951
2.912	2,309.06	26.027f	1.467	5.23	15.003f	8.52	18.44	6.916
3.012	2,356.98	25.778f	1.492	4.36	12.500f	5.24	11.12	5.944

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.


HYDROSTATIC PROPERTIES at 1 M. FWD TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ 25 M. fwd
1 Displacement 1=20 MT
2 LCB (use top scale)
3 VCB (KB) 1=.02 M.
4 Immersion 1=.06 MT/cm
4 WPA 1=5.85 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.3 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.6 M.
Specific Gravity = 1.025
Trim is per 50 M.
Assumed KG = 0.00 M.
"K" = Baseline

HYDROSTATIC PROPERTIES

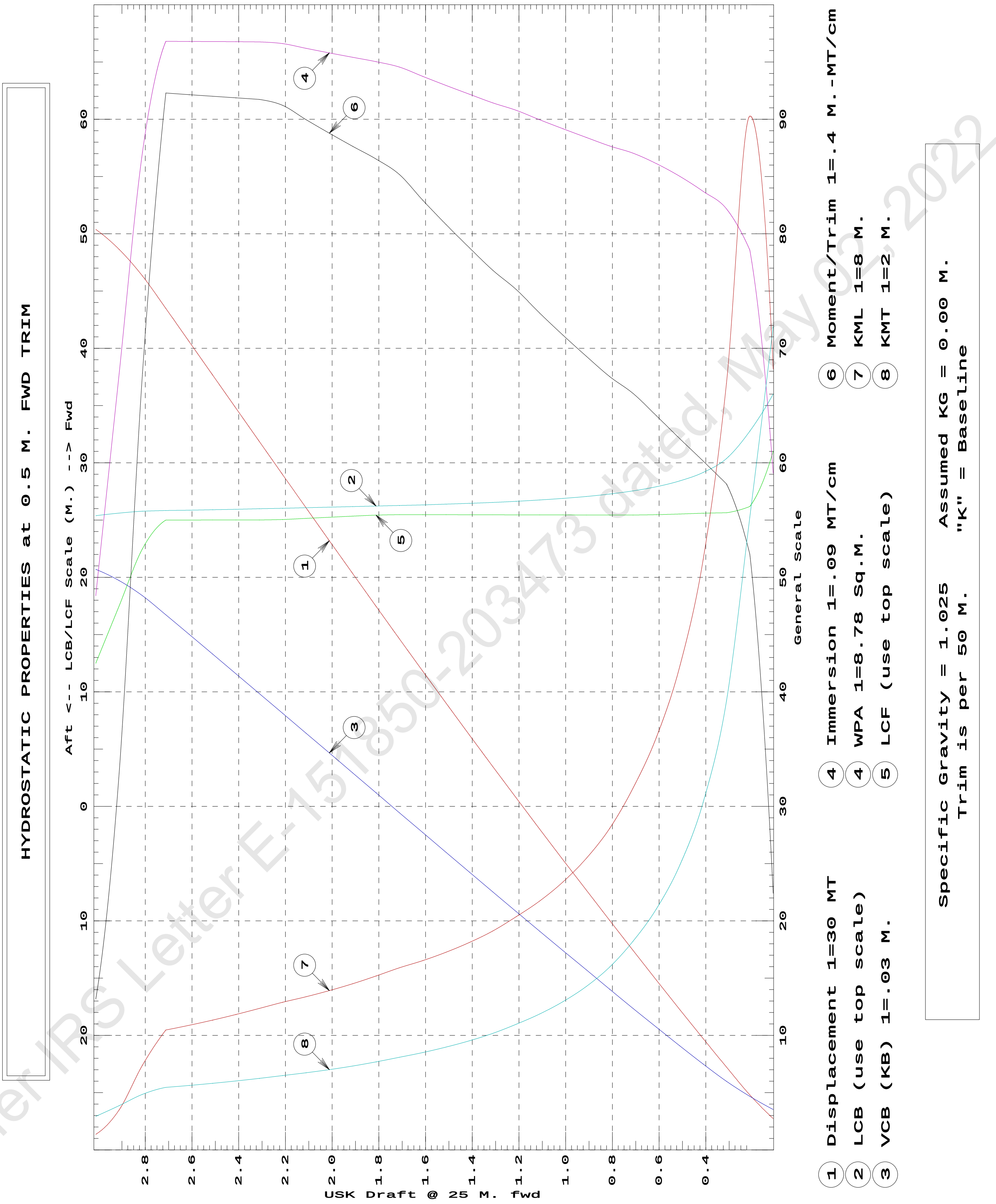
Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	82.14	35.998f	0.106	5.32	31.000f	8.96	545.41	143.815
0.212	144.05	32.764f	0.140	7.07	26.187f	20.81	722.22	110.512
0.312	217.16	30.377f	0.180	7.40	25.639f	23.26	535.62	78.390
0.412	291.82	29.161f	0.225	7.53	25.598f	24.06	412.26	60.691
0.512	367.75	28.419f	0.273	7.65	25.534f	24.84	337.74	49.705
0.612	444.76	27.914f	0.321	7.75	25.473f	25.64	288.26	42.105
0.712	522.67	27.548f	0.371	7.83	25.442f	26.42	252.76	36.454
0.812	601.24	27.273f	0.421	7.89	25.442f	27.01	224.62	31.939
0.912	680.47	27.060f	0.472	7.96	25.445f	27.72	203.68	28.553
1.012	760.38	26.890f	0.522	8.03	25.447f	28.46	187.14	25.876
1.112	840.98	26.752f	0.573	8.10	25.450f	29.24	173.82	23.716
1.212	922.29	26.637f	0.624	8.17	25.452f	30.05	162.93	21.948
1.312	1,004.25	26.541f	0.676	8.23	25.460f	30.73	152.98	20.359
1.412	1,086.87	26.459f	0.728	8.30	25.463f	31.51	144.94	19.060
1.512	1,170.18	26.389f	0.779	8.37	25.466f	32.32	138.10	17.953
1.612	1,254.18	26.327f	0.831	8.44	25.469f	33.17	132.25	17.002
1.712	1,338.88	26.273f	0.883	8.51	25.471f	34.07	127.23	16.184
1.812	1,424.21	26.225f	0.936	8.55	25.433f	34.61	121.50	15.364
1.912	1,509.90	26.177f	0.988	8.59	25.341f	35.06	116.08	14.654
2.012	1,595.94	26.129f	1.040	8.62	25.247f	35.51	111.26	14.027
2.112	1,682.33	26.082f	1.092	8.66	25.150f	35.99	106.96	13.470
2.212	1,769.07	26.034f	1.144	8.69	25.049f	36.49	103.12	12.976
2.312	1,856.11	25.986f	1.196	8.71	25.013f	36.70	98.84	12.483
2.412	1,943.20	25.943f	1.247	8.71	25.010f	36.75	94.55	12.033
2.512	2,030.29	25.903f	1.299	8.71	25.007f	36.80	90.63	11.625
2.612	2,117.40	25.866f	1.351	8.71	25.004f	36.86	87.04	11.255
2.712	2,204.51	25.832f	1.402	8.71	25.001f	36.92	83.73	10.918
2.812	2,289.46	25.779f	1.452	7.84	22.500f	27.13	59.25	9.700
2.912	2,359.15	25.611f	1.491	6.10	17.500f	13.16	27.88	7.717
3.012	2,411.43	25.384f	1.521	4.36	12.500f	5.27	10.93	5.871

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

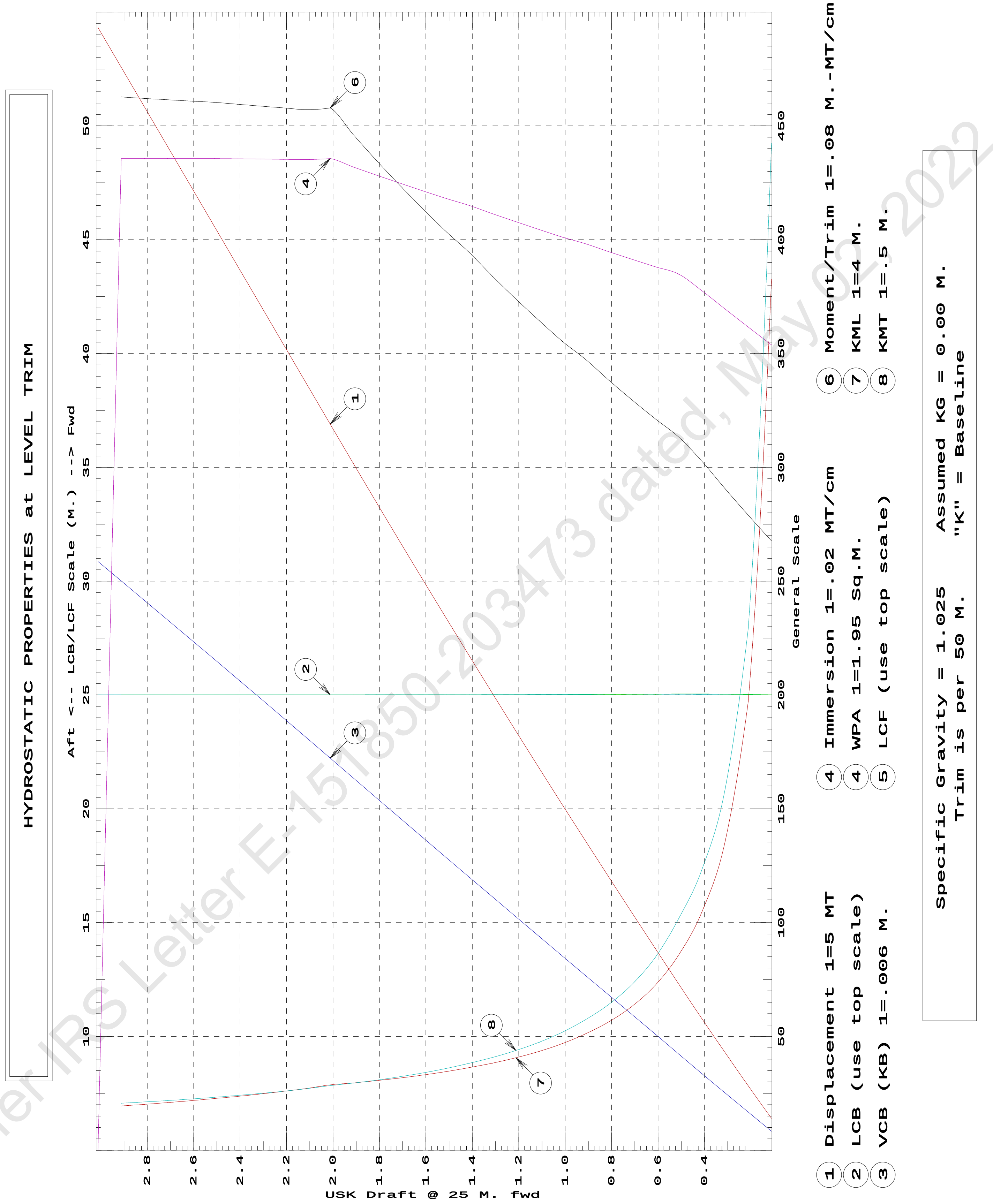

HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ 25 M. fwd
1 Displacement 1=30 MT
2 LCB (use top scale)
3 VCB (KB) 1=.03 M.
4 Immersion 1=.09 MT/cm
4 WPA 1=8.78 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.4 M.-MT/cm
7 KML 1=8 M.
8 KMT 1=2 M.
Specific Gravity = 1.025
Trim is per 50 M.
Assumed KG = 0.00 M.
"K" = Baseline

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	69.98	25.006f	0.050	7.08	25.011f	21.42	1530.11	221.181
0.212	141.52	25.011f	0.101	7.23	25.023f	22.32	788.57	114.651
0.312	214.62	25.017f	0.151	7.39	25.034f	23.25	541.70	79.233
0.412	289.31	25.023f	0.202	7.55	25.045f	24.22	418.51	61.593
0.512	365.56	25.027f	0.254	7.70	25.043f	25.07	342.85	50.844
0.612	442.86	25.029f	0.306	7.76	25.035f	25.70	290.19	42.401
0.712	520.81	25.029f	0.357	7.83	25.027f	26.37	253.12	36.457
0.812	599.41	25.028f	0.409	7.89	25.019f	27.06	225.70	32.060
0.912	678.68	25.027f	0.460	7.96	25.011f	27.79	204.71	28.691
1.012	758.61	25.024f	0.512	8.02	25.000f	28.43	187.36	25.920
1.112	839.18	25.022f	0.564	8.09	25.000f	29.16	173.72	23.717
1.212	920.41	25.020f	0.615	8.16	25.000f	29.91	162.51	21.903
1.312	1,002.33	25.018f	0.667	8.23	24.999f	30.71	153.18	20.391
1.412	1,084.95	25.017f	0.719	8.30	24.999f	31.54	145.35	19.120
1.512	1,168.27	25.016f	0.771	8.36	24.999f	32.28	138.15	17.968
1.612	1,252.22	25.015f	0.824	8.43	24.999f	33.08	132.08	16.991
1.712	1,336.85	25.014f	0.876	8.50	24.999f	33.91	126.83	16.142
1.812	1,422.16	25.013f	0.928	8.57	25.000f	34.78	122.27	15.403
1.912	1,508.17	25.012f	0.981	8.64	25.000f	35.69	118.31	14.757
2.012	1,594.89	25.011f	1.033	8.71	25.000f	36.63	114.84	14.189
2.112	1,681.93	25.011f	1.086	8.70	25.000f	36.57	108.71	13.527
2.212	1,768.98	25.010f	1.139	8.71	25.000f	36.63	103.53	12.975
2.312	1,856.05	25.010f	1.191	8.71	25.000f	36.69	98.85	12.480
2.412	1,943.15	25.009f	1.243	8.71	25.000f	36.76	94.59	12.033
2.512	2,030.26	25.009f	1.295	8.71	25.000f	36.83	90.70	11.629
2.612	2,117.38	25.009f	1.346	8.71	25.000f	36.87	87.07	11.256
2.712	2,204.51	25.008f	1.398	8.71	25.000f	36.92	83.73	10.916
2.812	2,291.64	25.008f	1.449	8.71	25.000f	36.97	80.65	10.605
2.912	2,378.76	25.008f	1.500	8.71	25.000f	37.02	77.81	10.321
3.012	2,465.88	25.007f	1.552	0.00				

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.


HYDROSTATIC PROPERTIES at LEVEL TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ 25 M. fwd
1 Displacement 1=5 MT
2 LCB (use top scale)
3 VCB (KB) 1=.006 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.08 M.-MT/cm
7 KML 1=4 M.
8 KMT 1=.5 M.
Specific Gravity = 1.025
Trim is per 50 M.
Assumed KG = 0.00 M.
"K" = Baseline

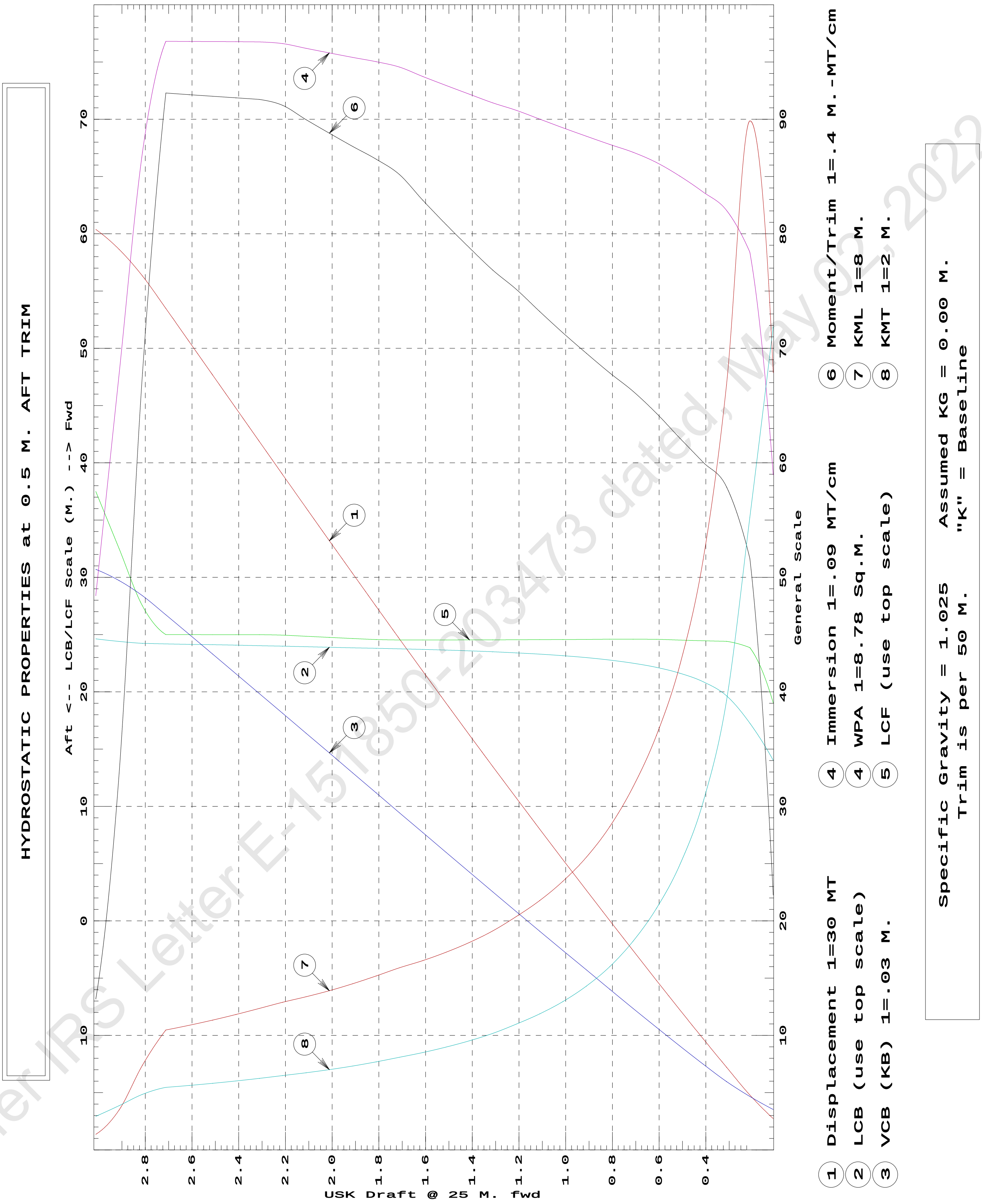
HYDROSTATIC PROPERTIES
Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	81.95	14.026f	0.106	5.31	19.038f	8.90	542.86	143.791
0.212	143.72	17.268f	0.140	7.06	23.862f	20.66	718.85	110.489
0.312	216.69	19.662f	0.179	7.38	24.413f	23.16	534.39	78.437
0.412	291.26	20.883f	0.225	7.53	24.456f	24.02	412.30	60.758
0.512	367.18	21.628f	0.272	7.65	24.523f	24.86	338.53	49.780
0.612	444.25	22.135f	0.321	7.76	24.587f	25.73	289.56	42.184
0.712	522.26	22.503f	0.371	7.84	24.598f	26.50	253.69	36.455
0.812	600.97	22.778f	0.421	7.90	24.594f	27.14	225.80	32.005
0.912	680.31	22.989f	0.472	7.97	24.583f	27.83	204.50	28.599
1.012	760.30	23.156f	0.522	8.03	24.573f	28.54	187.68	25.909
1.112	840.96	23.291f	0.573	8.10	24.563f	29.29	174.14	23.740
1.212	922.29	23.403f	0.624	8.17	24.548f	30.06	162.97	21.951
1.312	1,004.26	23.495f	0.676	8.23	24.539f	30.73	152.99	20.359
1.412	1,086.89	23.574f	0.728	8.30	24.536f	31.51	144.95	19.061
1.512	1,170.19	23.643f	0.779	8.37	24.533f	32.32	138.10	17.953
1.612	1,254.19	23.702f	0.831	8.44	24.530f	33.17	132.24	17.002
1.712	1,338.89	23.754f	0.883	8.51	24.527f	34.07	127.22	16.183
1.812	1,424.22	23.801f	0.936	8.55	24.566f	34.61	121.49	15.364
1.912	1,509.91	23.847f	0.988	8.59	24.658f	35.05	116.07	14.654
2.012	1,595.94	23.893f	1.040	8.62	24.753f	35.51	111.25	14.026
2.112	1,682.33	23.940f	1.092	8.66	24.850f	35.99	106.96	13.470
2.212	1,769.07	23.987f	1.144	8.69	24.951f	36.49	103.12	12.975
2.312	1,856.11	24.033f	1.196	8.71	24.987f	36.70	98.84	12.483
2.412	1,943.20	24.076f	1.247	8.71	24.990f	36.75	94.55	12.032
2.512	2,030.29	24.115f	1.299	8.71	24.993f	36.80	90.63	11.625
2.612	2,117.40	24.151f	1.351	8.71	24.996f	36.86	87.04	11.255
2.712	2,204.51	24.185f	1.402	8.71	24.999f	36.92	83.73	10.918
2.812	2,289.46	24.237f	1.452	7.84	27.500f	27.13	59.25	9.700
2.912	2,359.16	24.404f	1.491	6.10	32.500f	13.16	27.88	7.717
3.012	2,411.43	24.631f	1.521	4.36	37.500f	5.27	10.93	5.871

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

23-06-20 16:53:22 Cybermarine Technologies PTE. Ltd.
GHS 17.02B PONTOON (YARD-CHISTI/11)



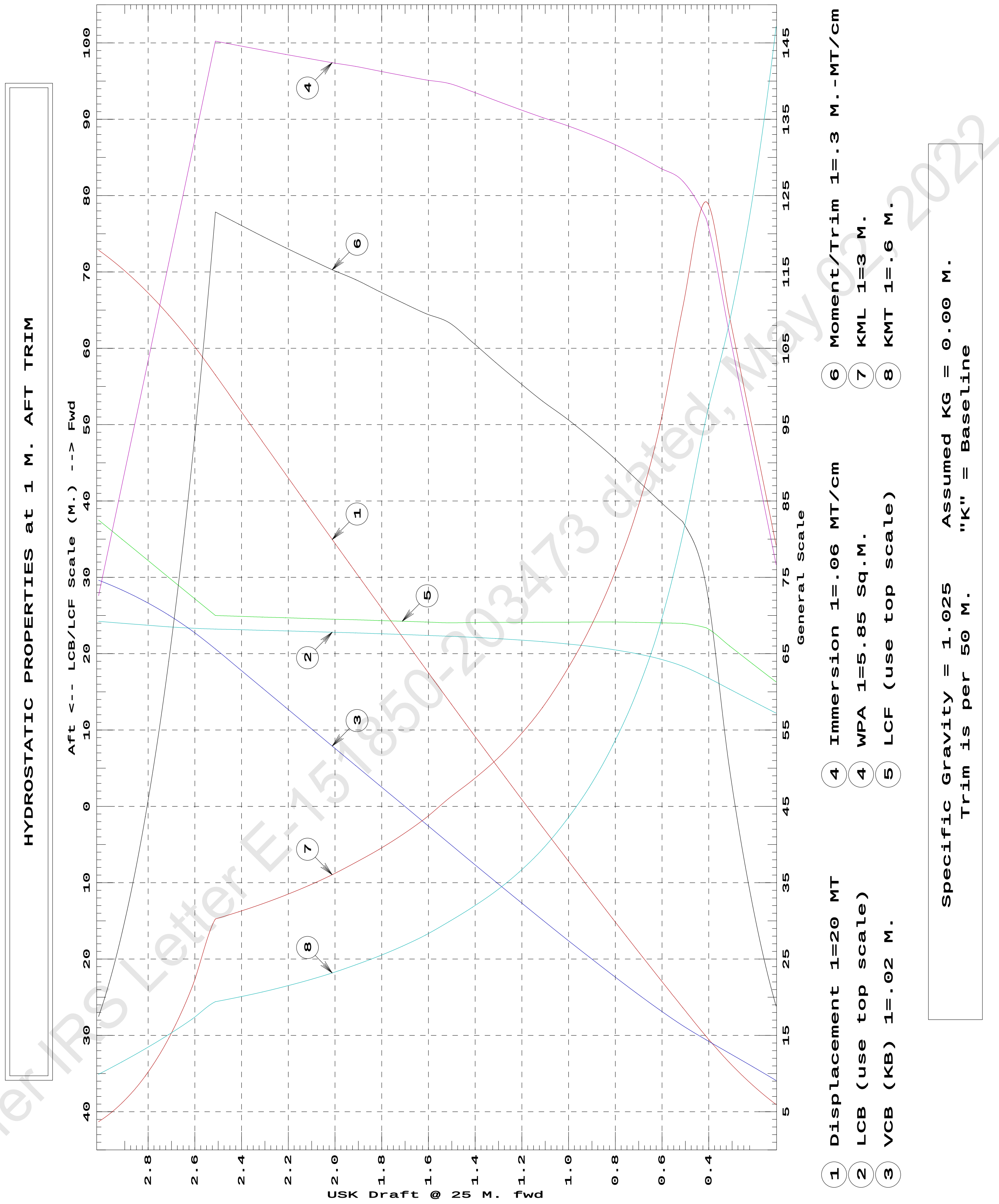
23-06-20 16:53:22 Cybermarine Technologies PTE. Ltd.
GHS 17.02B **PONTOON (YARD-CHISTI/11)**

HYDROSTATIC PROPERTIES
Trim: Aft 1.000/50.000, No Heel, Fixed VCG = 0.000

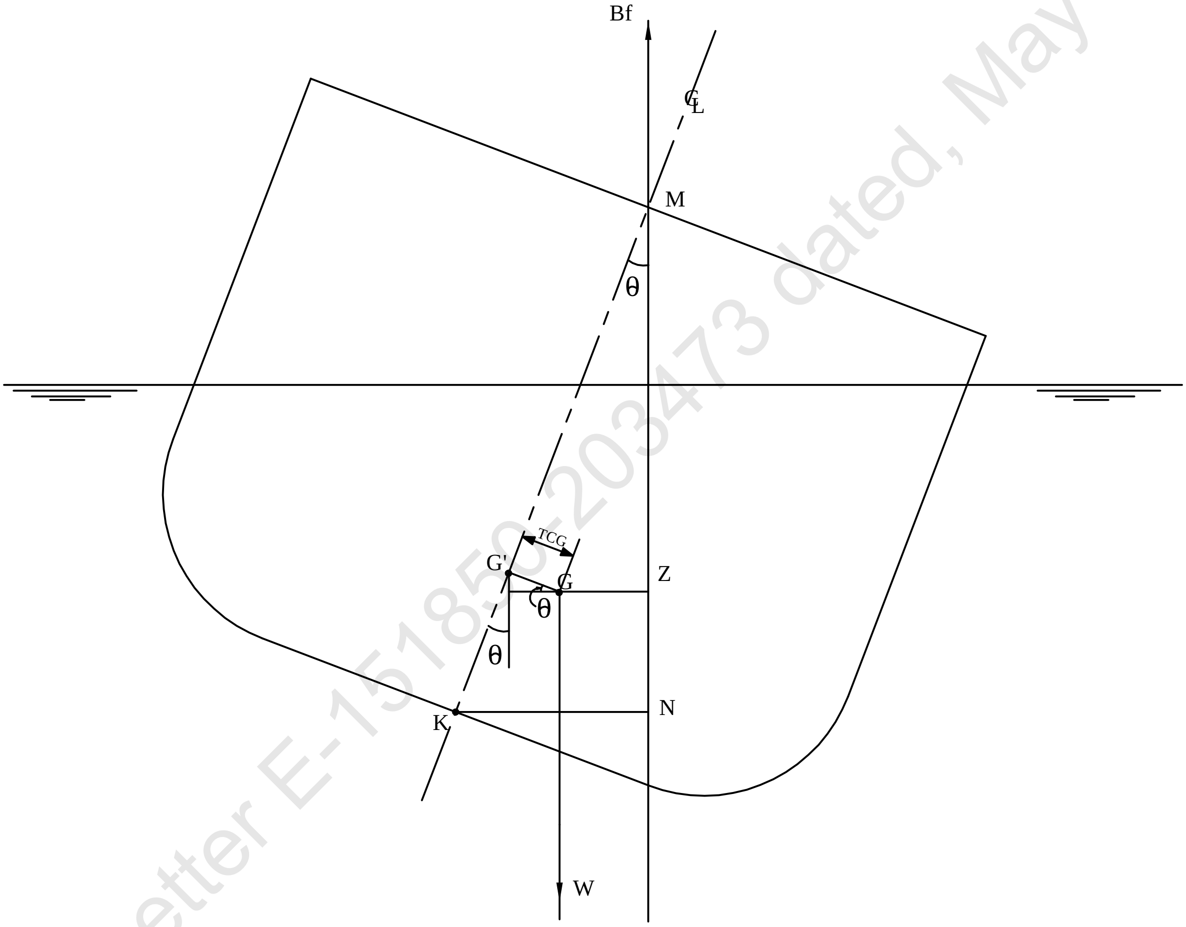
Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	118.82	12.199f	0.182	4.60	16.289f	5.65	237.55	88.297
0.212	169.35	13.782f	0.218	5.51	18.653f	9.58	282.73	74.914
0.312	228.93	15.366f	0.253	6.41	21.024f	15.01	327.65	65.052
0.412	297.59	16.952f	0.289	7.32	23.400f	22.18	372.59	57.503
0.512	372.93	18.353f	0.325	7.61	23.968f	24.69	330.96	48.332
0.612	449.58	19.317f	0.367	7.72	24.043f	25.51	283.60	41.083
0.712	527.26	20.018f	0.412	7.82	24.102f	26.37	250.01	35.819
0.812	605.88	20.550f	0.458	7.91	24.145f	27.24	224.74	31.774
0.912	685.35	20.968f	0.505	7.99	24.148f	28.03	204.43	28.499
1.012	765.52	21.300f	0.554	8.05	24.123f	28.76	187.83	25.850
1.112	846.32	21.570f	0.603	8.11	24.123f	29.43	173.82	23.627
1.212	927.77	21.794f	0.652	8.18	24.110f	30.17	162.54	21.830
1.312	1,009.88	21.981f	0.702	8.25	24.097f	30.94	153.15	20.331
1.412	1,092.66	22.141f	0.753	8.32	24.082f	31.74	145.22	19.065
1.512	1,176.10	22.277f	0.803	8.38	24.047f	32.51	138.19	17.941
1.612	1,260.02	22.399f	0.855	8.41	24.151f	32.87	130.41	16.890
1.712	1,344.27	22.512f	0.905	8.44	24.243f	33.30	123.82	16.002
1.812	1,428.87	22.617f	0.956	8.48	24.338f	33.74	118.04	15.226
1.912	1,513.81	22.716f	1.007	8.52	24.436f	34.20	112.94	14.545
2.012	1,599.10	22.810f	1.058	8.55	24.515f	34.59	108.13	13.909
2.112	1,684.69	22.899f	1.109	8.58	24.607f	35.01	103.90	13.352
2.212	1,770.61	22.984f	1.160	8.61	24.702f	35.45	100.09	12.854
2.312	1,856.87	23.066f	1.211	8.64	24.799f	35.91	96.67	12.408
2.412	1,943.46	23.145f	1.261	8.68	24.900f	36.38	93.58	12.009
2.512	2,030.39	23.223f	1.312	8.71	25.000f	36.86	90.74	11.647
2.612	2,113.11	23.340f	1.360	7.84	27.489f	27.01	63.91	10.283
2.712	2,187.13	23.522f	1.401	6.97	29.991f	19.19	43.85	9.066
2.812	2,252.45	23.745f	1.437	6.10	32.494f	13.09	29.06	7.951
2.912	2,309.06	23.989f	1.467	5.23	34.997f	8.52	18.44	6.916
3.012	2,356.98	24.238f	1.492	4.36	37.500f	5.24	11.12	5.944

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.


HYDROSTATIC PROPERTIES at 1 M. AFT TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ 25 M. fwd
1 Displacement 1=20 MT
2 LCB (use top scale)
3 VCB (KB) 1=.02 M.
4 Immersion 1=.06 MT/cm
4 WPA 1=5.85 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.3 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.6 M.
Specific Gravity = 1.025
Trim is per 50 M.
Assumed KG = 0.00 M.
"K" = Baseline

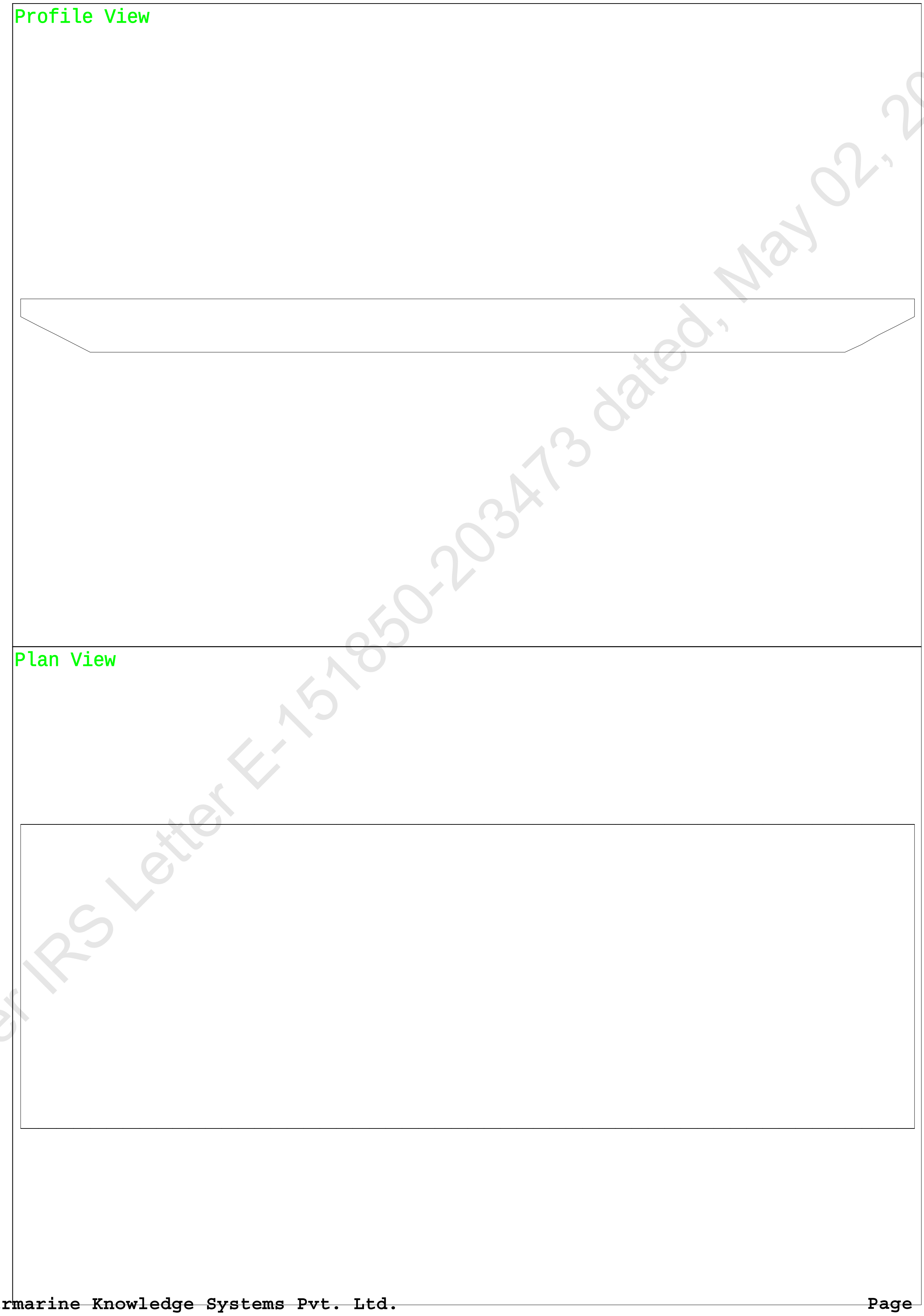
KN CONCEPT



$$GZ = KN - KG' \sin\theta - TCG \cos\theta$$

VOLUME USED FOR KN COMPUTATION

Condition Graphic

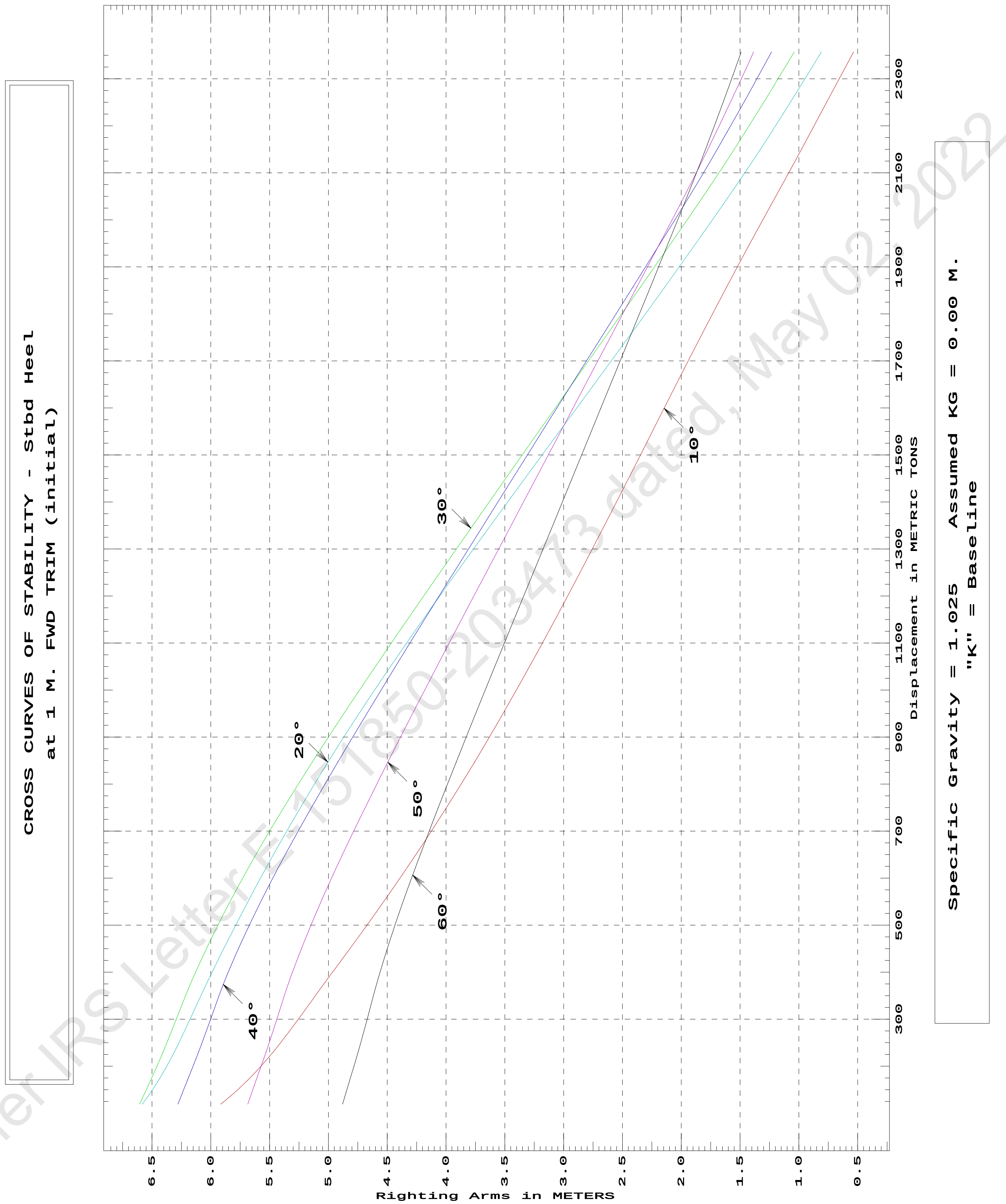


CROSS CURVE TABLES

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
119.34	5.916s	6.582s	6.606s	6.279s	5.685s	4.879s
169.99	5.688s	6.444s	6.511s	6.199s	5.615s	4.819s
229.67	5.471s	6.311s	6.406s	6.105s	5.531s	4.748s
298.40	5.260s	6.179s	6.295s	6.004s	5.440s	4.672s
373.75	5.042s	6.039s	6.175s	5.893s	5.343s	4.591s
450.36	4.816s	5.889s	6.036s	5.764s	5.228s	4.499s
527.96	4.592s	5.730s	5.882s	5.618s	5.101s	4.396s
606.47	4.373s	5.563s	5.714s	5.459s	4.961s	4.283s
685.79	4.162s	5.386s	5.533s	5.288s	4.812s	4.165s
765.82	3.956s	5.200s	5.340s	5.107s	4.656s	4.041s
846.50	3.758s	5.004s	5.138s	4.920s	4.495s	3.914s
927.86	3.566s	4.797s	4.926s	4.726s	4.329s	3.783s
1,009.90	3.379s	4.581s	4.708s	4.527s	4.160s	3.651s
1,092.65	3.196s	4.355s	4.485s	4.324s	3.988s	3.515s
1,176.09	3.017s	4.121s	4.256s	4.118s	3.813s	3.378s
1,260.00	2.840s	3.882s	4.023s	3.909s	3.636s	3.240s
1,344.26	2.665s	3.638s	3.788s	3.698s	3.459s	3.102s
1,428.86	2.491s	3.391s	3.552s	3.486s	3.280s	2.963s
1,513.80	2.318s	3.142s	3.313s	3.273s	3.101s	2.824s
1,599.09	2.146s	2.891s	3.073s	3.059s	2.921s	2.684s
1,684.69	1.972s	2.639s	2.831s	2.844s	2.741s	2.544s
1,770.61	1.795s	2.389s	2.589s	2.628s	2.560s	2.404s
1,856.86	1.614s	2.141s	2.347s	2.411s	2.378s	2.263s
1,943.46	1.430s	1.894s	2.106s	2.195s	2.196s	2.122s
2,030.39	1.239s	1.650s	1.866s	1.979s	2.015s	1.981s
2,113.11	1.057s	1.425s	1.646s	1.780s	1.846s	1.850s
2,187.13	0.898s	1.231s	1.456s	1.609s	1.702s	1.737s
2,252.45	0.759s	1.066s	1.293s	1.462s	1.578s	1.641s
2,309.06	0.638s	0.926s	1.155s	1.337s	1.472s	1.558s
2,356.98	0.534s	0.809s	1.040s	1.232s	1.384s	1.490s

Distances in METERS.---Specific Gravity = 1.025.-------------

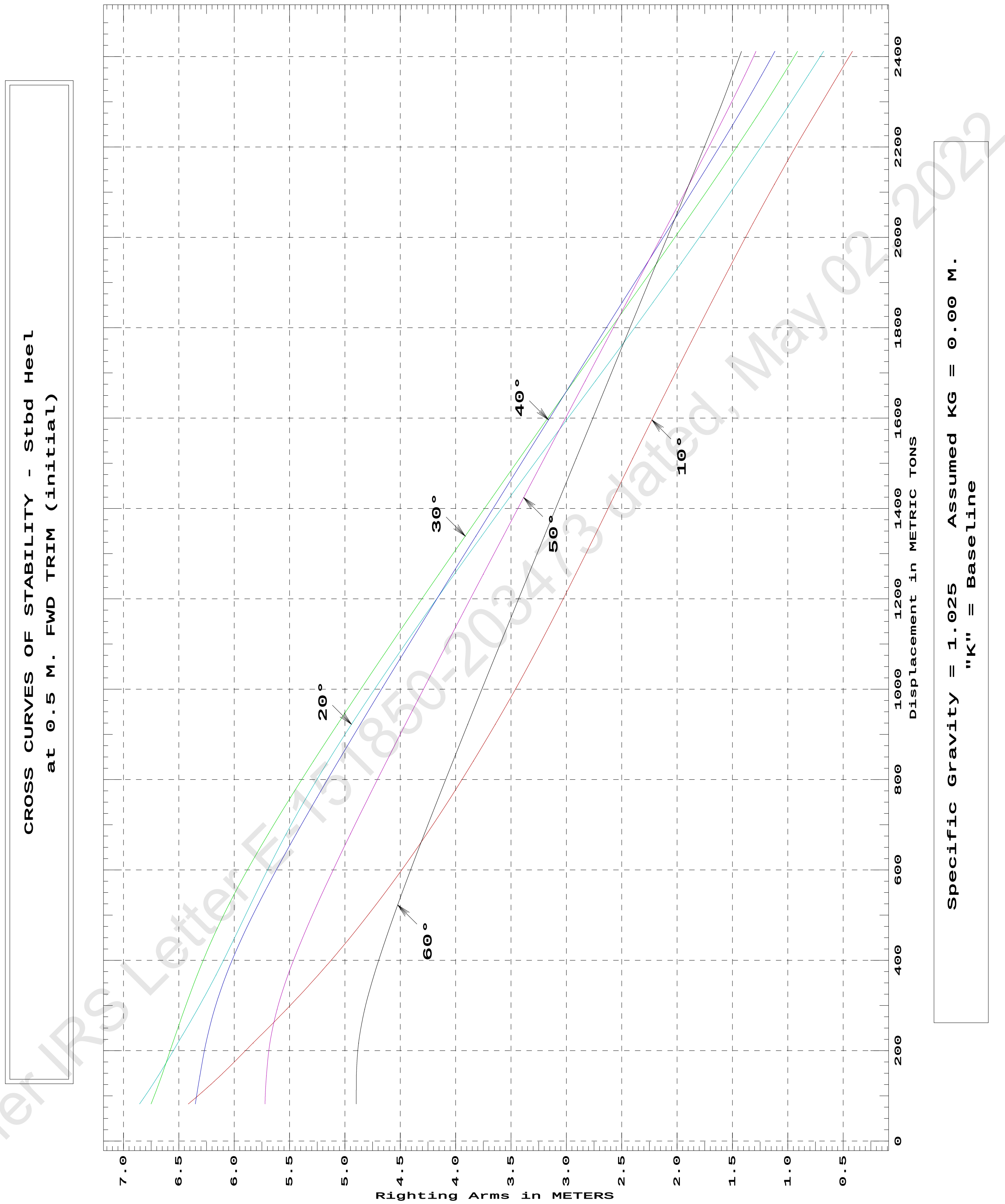

CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. FWD TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = Baseline
10°
20°
30°
40°
50°
60°
Displacement in METRIC TONS
300
500
700
900
1100
1300
1500
1700
1900
2100
2300
Righting Arms in METERS
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
82.14	6.415s	6.854s	6.751s	6.350s	5.721s	4.898s
144.05	6.129s	6.687s	6.655s	6.309s	5.709s	4.896s
217.16	5.829s	6.509s	6.554s	6.254s	5.677s	4.876s
291.82	5.526s	6.331s	6.448s	6.177s	5.612s	4.821s
367.75	5.240s	6.164s	6.333s	6.072s	5.514s	4.736s
444.76	4.972s	6.007s	6.202s	5.940s	5.391s	4.635s
522.67	4.721s	5.854s	6.050s	5.786s	5.252s	4.523s
601.24	4.484s	5.696s	5.878s	5.617s	5.104s	4.404s
680.47	4.259s	5.529s	5.691s	5.438s	4.949s	4.281s
760.38	4.045s	5.347s	5.491s	5.251s	4.788s	4.154s
840.98	3.840s	5.150s	5.283s	5.059s	4.623s	4.024s
922.29	3.642s	4.940s	5.067s	4.861s	4.454s	3.891s
1,004.25	3.450s	4.720s	4.845s	4.659s	4.282s	3.756s
1,086.87	3.267s	4.492s	4.619s	4.453s	4.107s	3.619s
1,170.18	3.089s	4.256s	4.387s	4.244s	3.930s	3.481s
1,254.18	2.916s	4.013s	4.152s	4.032s	3.751s	3.341s
1,338.88	2.745s	3.765s	3.913s	3.817s	3.569s	3.199s
1,424.21	2.575s	3.513s	3.670s	3.600s	3.386s	3.056s
1,509.90	2.402s	3.256s	3.426s	3.381s	3.201s	2.912s
1,595.94	2.228s	2.997s	3.180s	3.161s	3.016s	2.767s
1,682.33	2.051s	2.736s	2.932s	2.940s	2.830s	2.622s
1,769.07	1.871s	2.475s	2.682s	2.718s	2.643s	2.477s
1,856.11	1.689s	2.217s	2.432s	2.495s	2.455s	2.331s
1,943.20	1.504s	1.962s	2.182s	2.270s	2.267s	2.185s
2,030.29	1.315s	1.713s	1.934s	2.047s	2.078s	2.038s
2,117.40	1.121s	1.469s	1.688s	1.825s	1.891s	1.891s
2,204.51	0.920s	1.229s	1.449s	1.606s	1.704s	1.746s
2,289.46	0.715s	0.998s	1.222s	1.399s	1.527s	1.607s
2,359.15	0.545s	0.815s	1.043s	1.235s	1.388s	1.498s
2,411.43	0.416s	0.678s	0.913s	1.117s	1.288s	1.419s

Heel Angles in Degrees

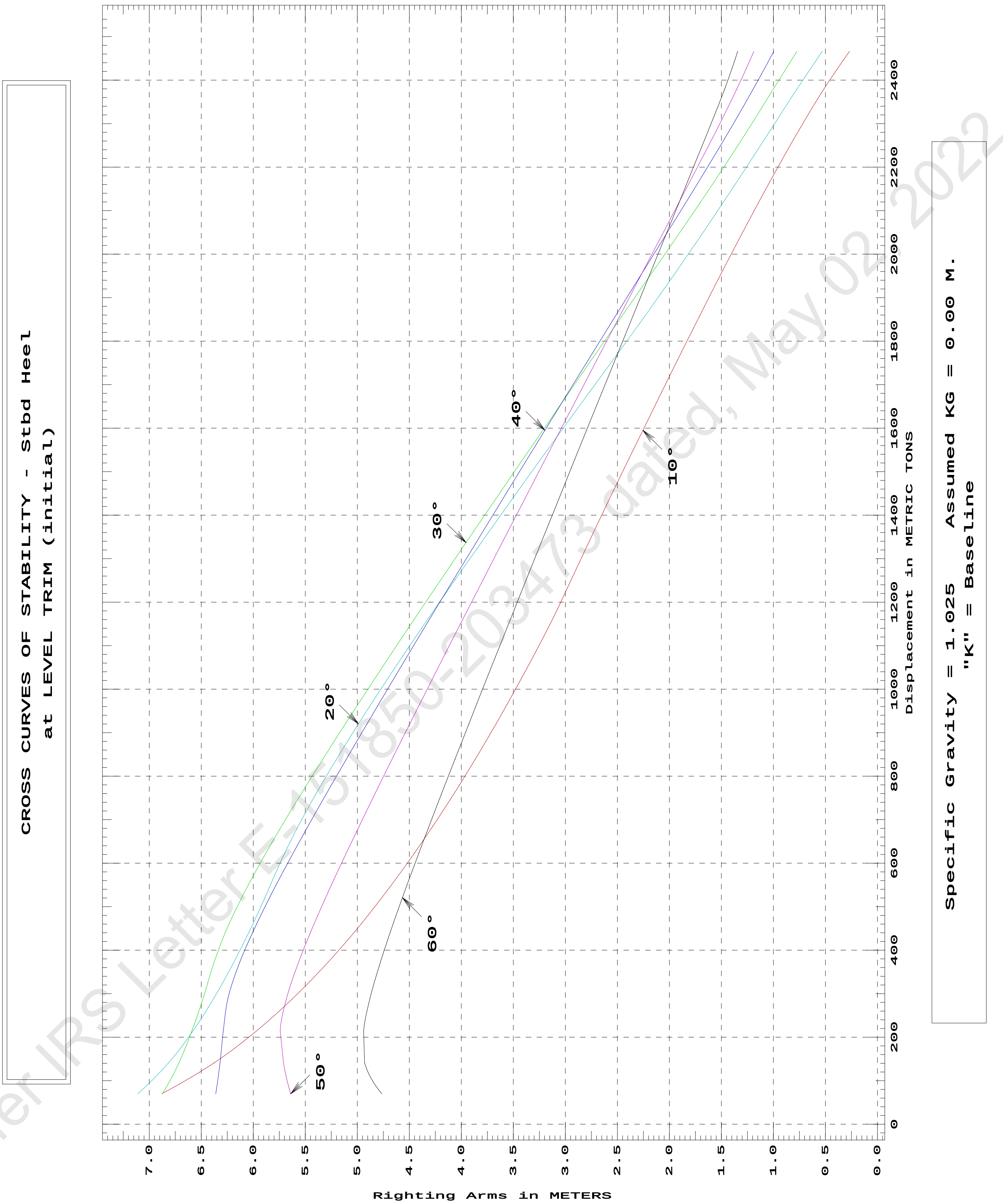
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. FWD TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = Baseline
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
7.0
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
69.98	6.880s	7.110s	6.873s	6.361s	5.640s	4.765s
141.52	6.375s	6.814s	6.714s	6.320s	5.710s	4.931s
214.62	5.968s	6.580s	6.590s	6.288s	5.741s	4.940s
289.31	5.620s	6.381s	6.483s	6.244s	5.681s	4.878s
365.56	5.308s	6.202s	6.386s	6.141s	5.576s	4.788s
442.86	5.024s	6.039s	6.266s	6.003s	5.447s	4.682s
520.81	4.764s	5.888s	6.111s	5.844s	5.304s	4.567s
599.41	4.522s	5.744s	5.935s	5.671s	5.153s	4.446s
678.68	4.292s	5.582s	5.744s	5.489s	4.995s	4.321s
758.61	4.075s	5.399s	5.543s	5.300s	4.833s	4.192s
839.18	3.867s	5.200s	5.332s	5.106s	4.666s	4.061s
920.41	3.667s	4.989s	5.115s	4.907s	4.496s	3.928s
1,002.33	3.474s	4.768s	4.892s	4.703s	4.323s	3.792s
1,084.95	3.287s	4.538s	4.664s	4.497s	4.148s	3.655s
1,168.27	3.108s	4.301s	4.432s	4.287s	3.970s	3.516s
1,252.22	2.943s	4.058s	4.195s	4.074s	3.790s	3.375s
1,336.85	2.776s	3.809s	3.955s	3.858s	3.608s	3.233s
1,422.16	2.606s	3.556s	3.712s	3.640s	3.424s	3.089s
1,508.17	2.433s	3.298s	3.466s	3.420s	3.237s	2.944s
1,594.89	2.255s	3.036s	3.217s	3.197s	3.049s	2.797s
1,681.93	2.076s	2.771s	2.966s	2.973s	2.860s	2.650s
1,768.98	1.896s	2.505s	2.714s	2.748s	2.671s	2.502s
1,856.05	1.714s	2.241s	2.461s	2.523s	2.481s	2.354s
1,943.15	1.529s	1.984s	2.208s	2.297s	2.292s	2.206s
2,030.26	1.340s	1.732s	1.955s	2.071s	2.101s	2.058s
2,117.38	1.146s	1.487s	1.707s	1.845s	1.911s	1.910s
2,204.51	0.946s	1.248s	1.465s	1.622s	1.721s	1.761s
2,291.64	0.738s	1.012s	1.230s	1.404s	1.533s	1.613s
2,378.76	0.516s	0.775s	1.002s	1.196s	1.354s	1.470s
2,465.88	0.270s	0.531s	0.776s	0.997s	1.189s	1.344s

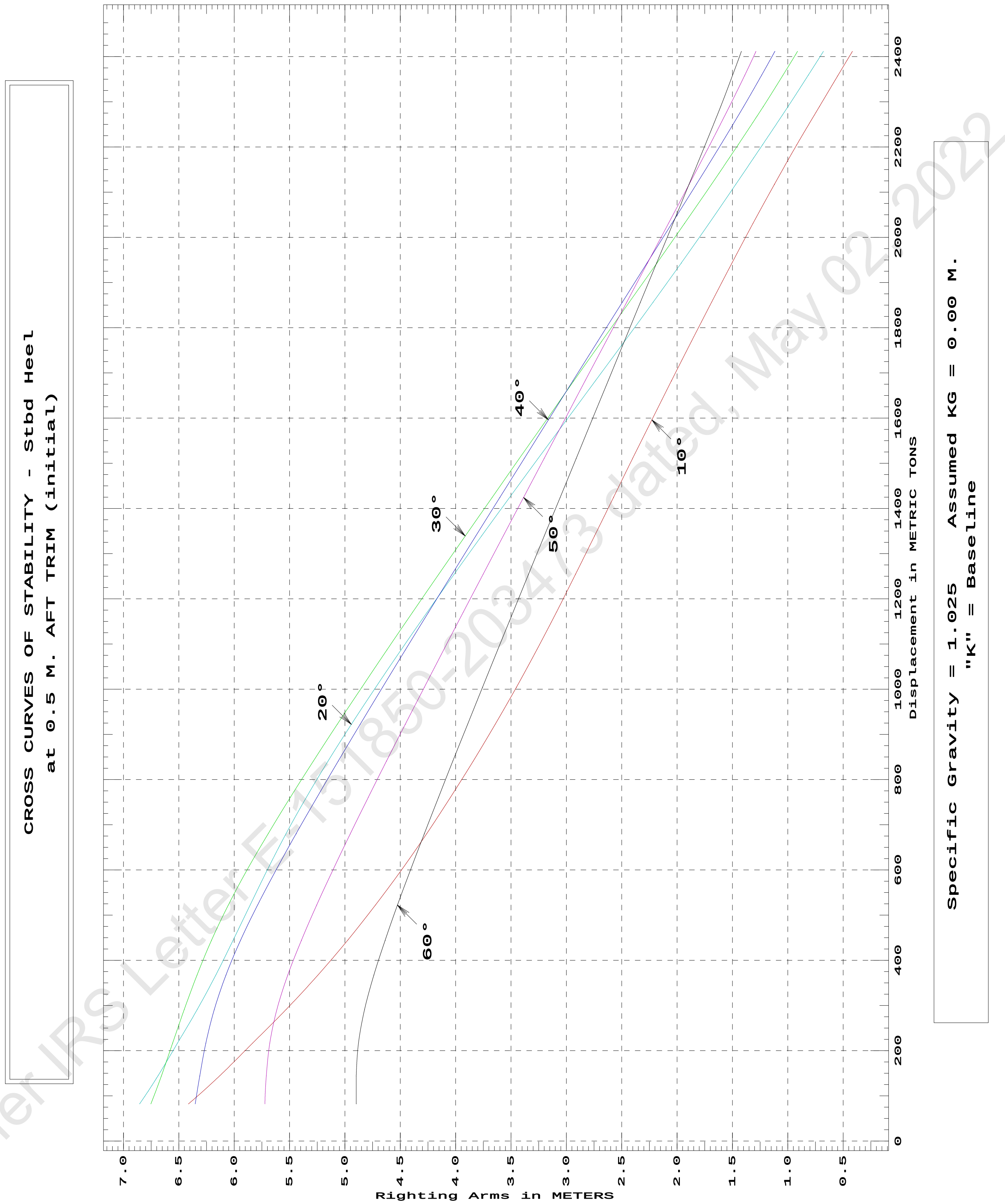
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = Baseline
Displacement in METRIC TONS
0 200 400 600 800 1000 1200 1400 1600 1800 2000 2200 2400
Righting Arms in METERS
7.0 6.5 6.0 5.5 5.0 4.5 4.0 3.5 3.0 2.5 2.0 1.5 1.0 0.5 0.0
10°
20°
30°
40°
50°
60°

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
81.95	6.414s	6.854s	6.752s	6.351s	5.722s	4.898s
143.72	6.133s	6.689s	6.657s	6.311s	5.710s	4.897s
216.69	5.834s	6.511s	6.556s	6.256s	5.678s	4.877s
291.26	5.530s	6.333s	6.450s	6.179s	5.613s	4.822s
367.18	5.244s	6.166s	6.336s	6.074s	5.516s	4.738s
444.25	4.976s	6.009s	6.204s	5.942s	5.393s	4.637s
522.26	4.724s	5.856s	6.053s	5.788s	5.255s	4.525s
600.97	4.486s	5.699s	5.881s	5.619s	5.106s	4.406s
680.31	4.261s	5.531s	5.693s	5.440s	4.950s	4.283s
760.30	4.047s	5.349s	5.493s	5.253s	4.789s	4.155s
840.96	3.841s	5.152s	5.285s	5.060s	4.624s	4.025s
922.29	3.643s	4.942s	5.069s	4.862s	4.455s	3.892s
1,004.26	3.451s	4.721s	4.847s	4.660s	4.283s	3.757s
1,086.89	3.268s	4.493s	4.620s	4.454s	4.108s	3.620s
1,170.19	3.090s	4.257s	4.388s	4.245s	3.931s	3.482s
1,254.19	2.917s	4.014s	4.153s	4.033s	3.751s	3.341s
1,338.89	2.746s	3.766s	3.913s	3.818s	3.570s	3.199s
1,424.22	2.575s	3.513s	3.671s	3.600s	3.386s	3.056s
1,509.91	2.403s	3.257s	3.426s	3.382s	3.202s	2.912s
1,595.94	2.228s	2.998s	3.180s	3.162s	3.016s	2.768s
1,682.33	2.051s	2.738s	2.932s	2.940s	2.830s	2.623s
1,769.07	1.871s	2.476s	2.683s	2.718s	2.643s	2.477s
1,856.11	1.689s	2.217s	2.432s	2.495s	2.455s	2.331s
1,943.20	1.504s	1.962s	2.182s	2.271s	2.267s	2.185s
2,030.29	1.315s	1.713s	1.934s	2.048s	2.079s	2.038s
2,117.40	1.121s	1.469s	1.688s	1.825s	1.891s	1.892s
2,204.51	0.919s	1.228s	1.449s	1.606s	1.704s	1.746s
2,289.46	0.715s	0.998s	1.222s	1.399s	1.527s	1.606s
2,359.16	0.545s	0.815s	1.043s	1.235s	1.388s	1.498s
2,411.43	0.416s	0.679s	0.913s	1.117s	1.288s	1.419s

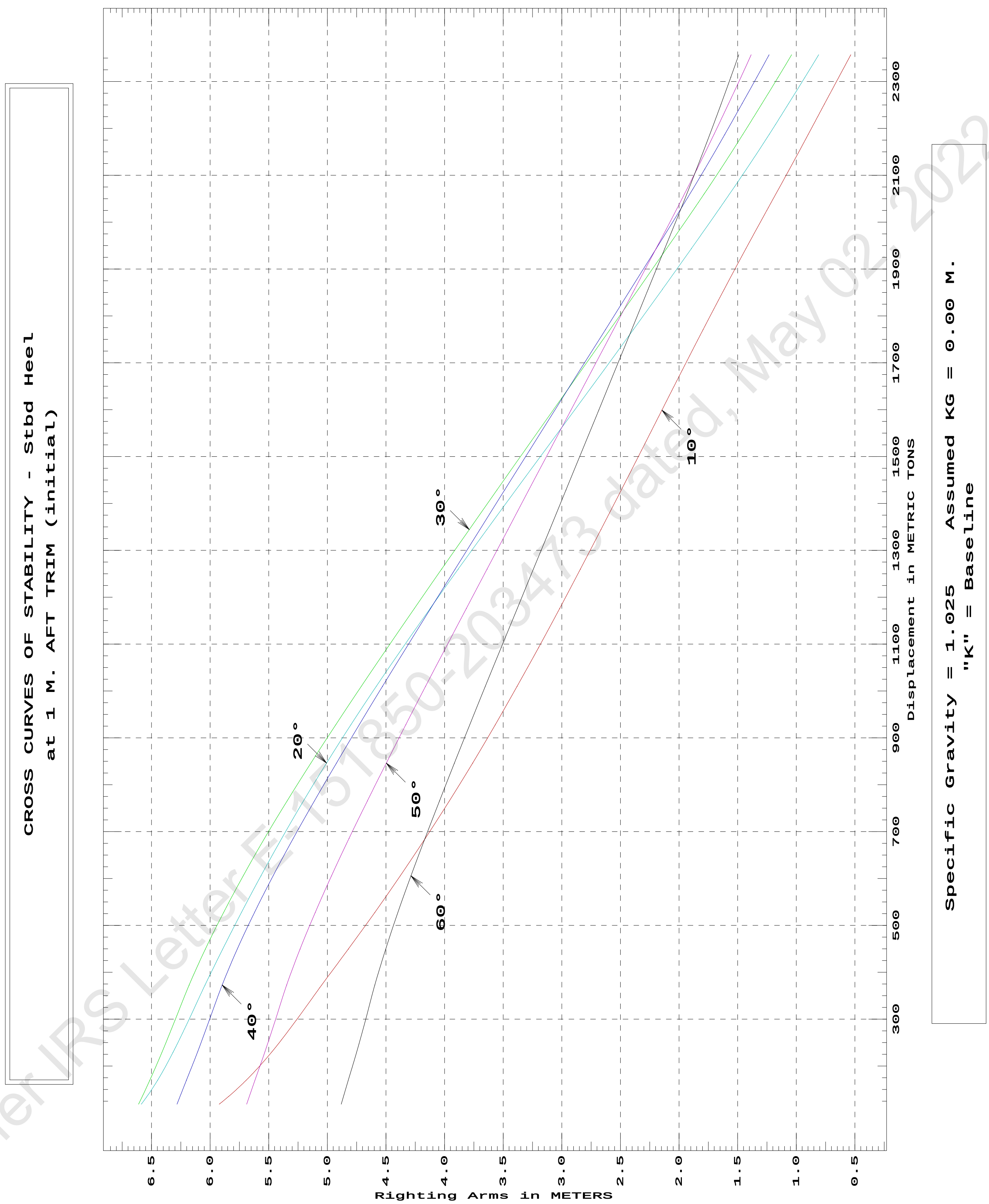
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)
10°
20°
30°
40°
50°
60°
Righting Arms in METERS
7.0
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
Displacement in METRIC TONS
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = Baseline

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
118.82	5.923s	6.587s	6.610s	6.282s	5.688s	4.881s
169.35	5.694s	6.448s	6.515s	6.202s	5.618s	4.822s
228.93	5.477s	6.317s	6.411s	6.108s	5.534s	4.751s
297.59	5.266s	6.184s	6.299s	6.008s	5.444s	4.675s
372.93	5.047s	6.043s	6.178s	5.897s	5.346s	4.595s
449.58	4.821s	5.892s	6.040s	5.767s	5.232s	4.502s
527.26	4.596s	5.733s	5.885s	5.622s	5.104s	4.399s
605.88	4.377s	5.566s	5.717s	5.462s	4.964s	4.286s
685.35	4.165s	5.390s	5.536s	5.291s	4.815s	4.167s
765.52	3.959s	5.203s	5.343s	5.110s	4.659s	4.043s
846.32	3.761s	5.006s	5.140s	4.922s	4.497s	3.916s
927.77	3.568s	4.799s	4.929s	4.728s	4.331s	3.785s
1,009.88	3.381s	4.583s	4.710s	4.529s	4.162s	3.652s
1,092.66	3.198s	4.357s	4.486s	4.326s	3.990s	3.517s
1,176.10	3.018s	4.123s	4.257s	4.119s	3.814s	3.380s
1,260.02	2.841s	3.883s	4.024s	3.910s	3.637s	3.242s
1,344.27	2.666s	3.639s	3.789s	3.699s	3.460s	3.103s
1,428.87	2.492s	3.392s	3.552s	3.487s	3.281s	2.964s
1,513.81	2.319s	3.143s	3.314s	3.274s	3.102s	2.824s
1,599.10	2.146s	2.891s	3.074s	3.060s	2.922s	2.685s
1,684.69	1.972s	2.640s	2.831s	2.845s	2.741s	2.545s
1,770.61	1.795s	2.390s	2.590s	2.628s	2.560s	2.404s
1,856.87	1.615s	2.142s	2.347s	2.412s	2.378s	2.263s
1,943.46	1.430s	1.895s	2.106s	2.196s	2.197s	2.122s
2,030.39	1.239s	1.651s	1.867s	1.980s	2.015s	1.981s
2,113.11	1.058s	1.425s	1.646s	1.781s	1.847s	1.850s
2,187.13	0.898s	1.231s	1.456s	1.610s	1.702s	1.738s
2,252.45	0.759s	1.066s	1.293s	1.462s	1.578s	1.641s
2,309.06	0.638s	0.926s	1.155s	1.337s	1.472s	1.558s
2,356.98	0.534s	0.809s	1.040s	1.232s	1.384s	1.491s

Distances in METERS.---Specific Gravity = 1.025.-------------

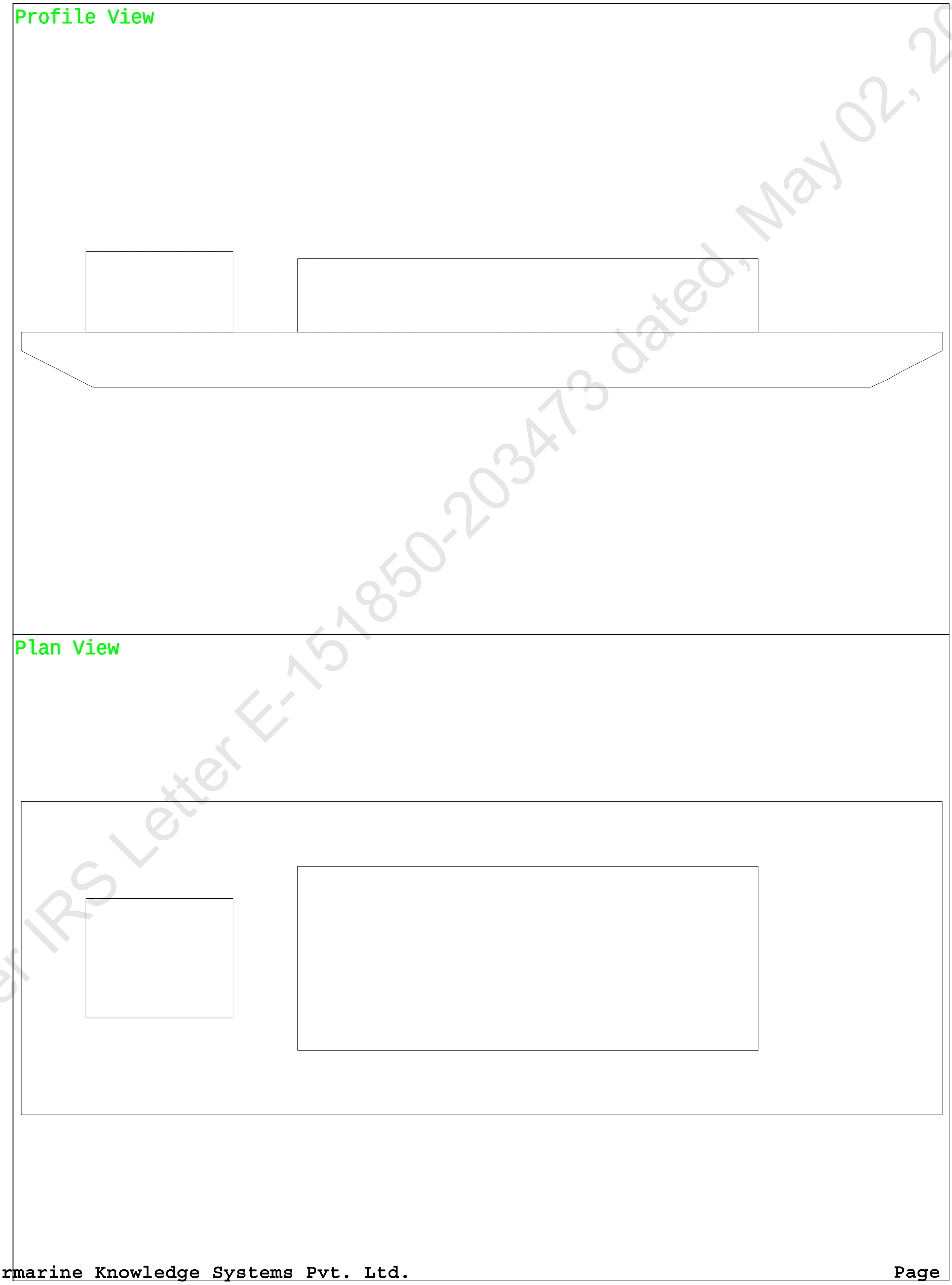

CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. AFT TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = Baseline
Displacement in METRIC TONS
300
500
700
900
1100
1300
1500
1700
1900
2100
2300
Righting Arms in METERS
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
10°
20°
30°
40°
50°
60°

DEADWEIGHT TABLE

Draft M	Disp T	Dwt T	
0.712	520.81	164.48	
0.812	599.41	243.08	
0.912	678.68	322.35	
1.012	758.61	402.28	
1.112	839.18	482.85	
1.212	920.41	564.08	
1.312	1002.33	646.00	
1.412	1084.95	728.62	
1.512	1168.27	811.94	
1.612	1252.22	895.89	
1.712	1336.85	980.52	
1.812	1422.16	1065.83	
1.912	1508.17	1151.84	
2.012	1594.89	1238.56	
2.112	1681.93	1325.60	
2.212	**1768.98**	**1412.65**	**Full Load Displacement**

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition Graphic



~~MAXIMUM VCG vs. DISPLACEMENT~~

~~Heeling moment is present from: wind~~

~~Trim = Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)~~

~~Displacement METRIC TONS~~	~~Max VCG~~	~~--- Margins --- LIM1~~	~~LIM2~~	~~LIM3~~
~~300.00~~	~~17.478~~	~~121%~~	~~0d~~	~~8d~~
~~400.00~~	~~17.029~~	~~102%~~	~~0d~~	~~7d~~
~~500.00~~	~~16.486~~	~~97%~~	~~0d~~	~~6d~~
~~600.00~~	~~15.892~~	~~87%~~	~~0d~~	~~6d~~
~~700.00~~	~~15.252~~	~~77%~~	~~0d~~	~~5d~~
~~800.00~~	~~14.639~~	~~69%~~	~~0d~~	~~5d~~
~~900.00~~	~~13.904~~	~~57%~~	~~0d~~	~~4d~~
~~1,000.00~~	~~13.125~~	~~47%~~	~~0d~~	~~4d~~
~~1,100.00~~	~~12.395~~	~~35%~~	~~0d~~	~~3d~~
~~1,200.00~~	~~11.582~~	~~25%~~	~~0d~~	~~3d~~
~~1,300.00~~	~~10.709~~	~~17%~~	~~0d~~	~~3d~~
~~1,400.00~~	~~9.890~~	~~9%~~	~~0d~~	~~2d~~
~~1,500.00~~	~~9.063~~	~~0%~~	~~0d~~	~~2d~~
~~1,600.00~~	~~7.919~~	~~1%~~	~~1d~~	~~1d~~
~~1,700.00~~	~~6.781~~	~~0%~~	~~3d~~	~~1d~~
~~1,800.00~~	~~5.647~~	~~0%~~	~~5d~~	~~1d~~

~~Heeling moment is present from: wind~~

~~Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)~~

~~Displacement METRIC TONS~~	~~Max VCG~~	~~--- Margins --- LIM1~~	~~LIM2~~	~~LIM3~~
~~300.00~~	~~17.910~~	~~116%~~	~~0d~~	~~11d~~
~~400.00~~	~~17.384~~	~~120%~~	~~0d~~	~~9d~~
~~500.00~~	~~16.858~~	~~106%~~	~~0d~~	~~8d~~
~~600.00~~	~~16.276~~	~~81%~~	~~0d~~	~~7d~~
~~700.00~~	~~15.651~~	~~71%~~	~~0d~~	~~6d~~
~~800.00~~	~~15.060~~	~~56%~~	~~0d~~	~~6d~~
~~900.00~~	~~14.339~~	~~51%~~	~~0d~~	~~5d~~
~~1,000.00~~	~~13.532~~	~~50%~~	~~0d~~	~~5d~~
~~1,100.00~~	~~12.775~~	~~41%~~	~~0d~~	~~4d~~
~~1,200.00~~	~~11.964~~	~~31%~~	~~0d~~	~~4d~~
~~1,300.00~~	~~11.100~~	~~21%~~	~~0d~~	~~3d~~
~~1,400.00~~	~~10.260~~	~~9%~~	~~0d~~	~~3d~~
~~1,500.00~~	~~9.256~~	~~0%~~	~~1d~~	~~3d~~
~~1,600.00~~	~~8.047~~	~~0%~~	~~2d~~	~~2d~~
~~1,700.00~~	~~7.028~~	~~0%~~	~~3d~~	~~2d~~
~~1,800.00~~	~~5.870~~	~~0%~~	~~5d~~	~~2d~~

~~NOTE:~~

~~The permissible VCG calculation does not cover Crane Operation or lifting of weights and it is valid only for unmanned tow with~~ crane in stowed position.
maximum deck cargo height = 4m above deck level

~~Heeling moment is present from: wind~~

~~Trim = zero at zero heel (trim righting arm held at zero)~~

~~Displacement~~		~~--- Margins ---~~		
~~METRIC TONS~~	~~Max VCG~~	~~LIM1~~	~~LIM2~~	~~LIM3~~
~~300.00~~	~~18.062~~	~~113%~~	~~0d~~	~~17d~~
~~400.00~~	~~17.462~~	~~121%~~	~~0d~~	~~14d~~
~~500.00~~	~~16.908~~	~~107%~~	~~0d~~	~~11d~~
~~600.00~~	~~16.399~~	~~86%~~	~~0d~~	~~9d~~
~~700.00~~	~~15.813~~	~~64%~~	~~0d~~	~~8d~~
~~800.00~~	~~15.194~~	~~56%~~	~~0d~~	~~7d~~
~~900.00~~	~~14.437~~	~~50%~~	~~0d~~	~~6d~~
~~1,000.00~~	~~13.642~~	~~51%~~	~~0d~~	~~6d~~
~~1,100.00~~	~~12.918~~	~~43%~~	~~0d~~	~~5d~~
~~1,200.00~~	~~12.053~~	~~34%~~	~~0d~~	~~5d~~
~~1,300.00~~	~~11.264~~	~~21%~~	~~0d~~	~~4d~~
~~1,400.00~~	~~10.410~~	~~5%~~	~~0d~~	~~4d~~
~~1,500.00~~	~~9.358~~	~~0%~~	~~1d~~	~~4d~~
~~1,600.00~~	~~8.128~~	~~0%~~	~~2d~~	~~3d~~
~~1,700.00~~	~~6.987~~	~~0%~~	~~3d~~	~~3d~~
~~1,800.00~~	~~5.934~~	~~0%~~	~~5d~~	~~2d~~

~~Heeling moment is present from: wind~~

~~Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)~~

~~Displacement~~		~~--- Margins ---~~		
~~METRIC TONS~~	~~Max VCG~~	~~LIM1~~	~~LIM2~~	~~LIM3~~
~~300.00~~	~~17.915~~	~~116%~~	~~0d~~	~~11d~~
~~400.00~~	~~17.389~~	~~120%~~	~~0d~~	~~10d~~
~~500.00~~	~~16.857~~	~~107%~~	~~0d~~	~~8d~~
~~600.00~~	~~16.281~~	~~81%~~	~~0d~~	~~7d~~
~~700.00~~	~~15.657~~	~~71%~~	~~0d~~	~~6d~~
~~800.00~~	~~15.067~~	~~56%~~	~~0d~~	~~6d~~
~~900.00~~	~~14.345~~	~~52%~~	~~0d~~	~~5d~~
~~1,000.00~~	~~13.536~~	~~50%~~	~~0d~~	~~5d~~
~~1,100.00~~	~~12.779~~	~~40%~~	~~0d~~	~~4d~~
~~1,200.00~~	~~11.967~~	~~30%~~	~~0d~~	~~4d~~
~~1,300.00~~	~~11.103~~	~~21%~~	~~0d~~	~~3d~~
~~1,400.00~~	~~10.259~~	~~9%~~	~~0d~~	~~3d~~
~~1,500.00~~	~~9.267~~	~~0%~~	~~1d~~	~~3d~~
~~1,600.00~~	~~8.045~~	~~0%~~	~~2d~~	~~2d~~
~~1,700.00~~	~~7.028~~	~~0%~~	~~3d~~	~~2d~~
~~1,800.00~~	~~5.866~~	~~0%~~	~~5d~~	~~2d~~

~~NOTE:~~

~~The permissible VCG calculation does not cover Crane Operation or lifting of weights and it is valid only for unmanned tow with~~ crane in stowed position.

maximum deck cargo height = 4m above deck level

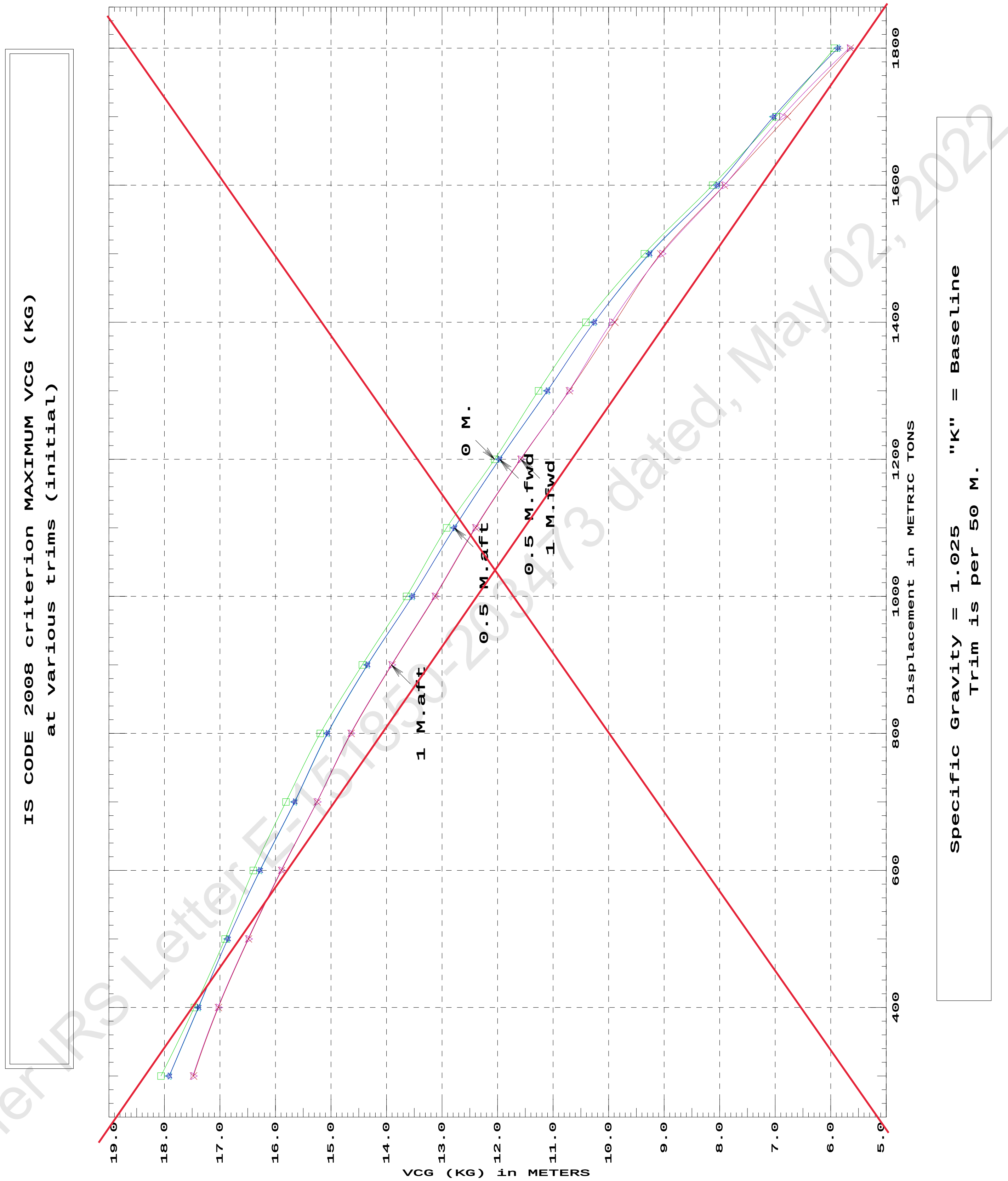
~~Heeling moment is present from: wind~~
~~Trim = Aft 1.000/50.000 at zero heel (trim righting arm held at zero)~~

~~Displacement METRIC TONS~~	~~Max VCG~~	~~--- Margins --- LIM1~~	~~LIM2~~	~~LIM3~~
~~300.00~~	~~17.488~~	~~121%~~	~~0d~~	~~8d~~
~~400.00~~	~~17.036~~	~~102%~~	~~0d~~	~~7d~~
~~500.00~~	~~16.494~~	~~97%~~	~~0d~~	~~6d~~
~~600.00~~	~~15.901~~	~~88%~~	~~0d~~	~~6d~~
~~700.00~~	~~15.254~~	~~77%~~	~~0d~~	~~5d~~
~~800.00~~	~~14.647~~	~~69%~~	~~0d~~	~~5d~~
~~900.00~~	~~13.913~~	~~57%~~	~~0d~~	~~4d~~
~~1,000.00~~	~~13.135~~	~~47%~~	~~0d~~	~~4d~~
~~1,100.00~~	~~12.405~~	~~35%~~	~~0d~~	~~3d~~
~~1,200.00~~	~~11.589~~	~~26%~~	~~0d~~	~~3d~~
~~1,300.00~~	~~10.715~~	~~18%~~	~~0d~~	~~3d~~
~~1,400.00~~	~~9.948~~	~~7%~~	~~0d~~	~~2d~~
~~1,500.00~~	~~9.047~~	~~0%~~	~~0d~~	~~2d~~
~~1,600.00~~	~~7.916~~	~~1%~~	~~1d~~	~~1d~~
~~1,700.00~~	~~6.883~~	~~0%~~	~~2d~~	~~1d~~
~~1,800.00~~	~~5.667~~	~~0%~~	~~5d~~	~~1d~~

~~Distances in METERS.---Specific Gravity = 1.025.---d = degrees.~~

~~LIM~~	~~IS CODE 2008 CRITERION~~		~~Min/Max~~
~~(1)~~	~~Residual Area from abs 0 deg to MaxRA~~	~~>~~	~~0.0800 m.-Rad~~
~~(2)~~	~~Angle from Equilibrium to RAzero or Flood~~	~~>~~	~~20.00 deg~~
~~(3)~~	~~Angle from Equilibrium to 50% Dk Imm Angle~~	~~>~~	~~0.00 deg~~

~~NOTE:~~
~~The permissible VCG calculation does not cover Crane Operation or lifting of weights and it is valid only for unmanned tow with~~ crane in stowed position.
maximum deck cargo height = 4m above deck level


IS CODE 2008 criterion MAXIMUM VCG (KG)
at various trims (initial)
0 M.
0.5 M.aft
1 M.aft
0.5 M.fwd
1 M.fwd
19.0
18.0
17.0
16.0
15.0
14.0
13.0
12.0
11.0
10.0
9.0
8.0
7.0
6.0
5.0
VCG (KG) in METERS
400
600
800
1000
1200
1400
1600
1800
Displacement in METRIC TONS
Specific Gravity = 1.025 "K" = Baseline
Trim is per 50 M.

LOADING CONDITIONS

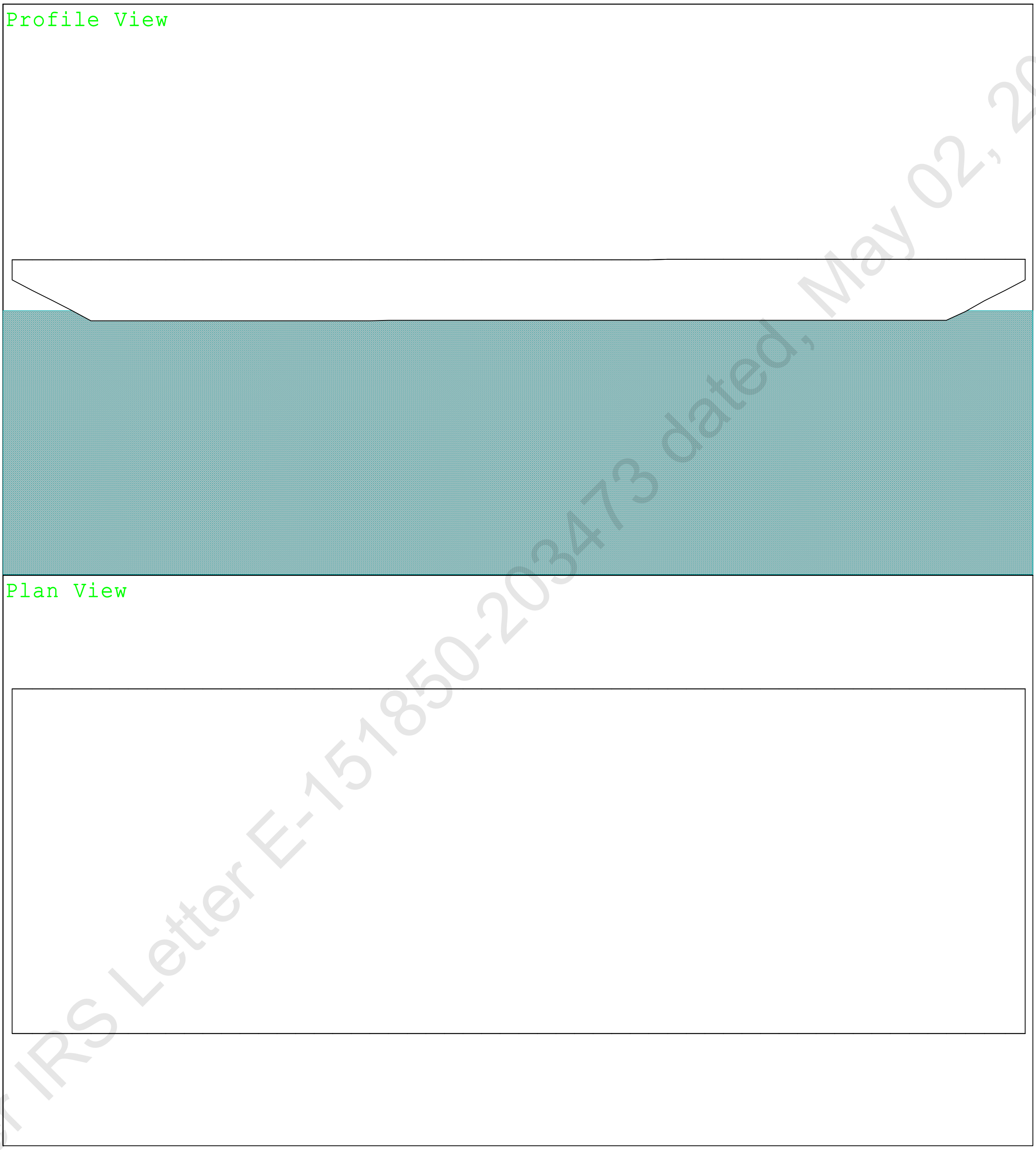
SUMMARY OF CONDITIONS						
~~CONDITION NO.~~	~~1~~	~~2~~	~~3~~	~~4~~	~~5~~	~~6~~
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Permanent Ballast	Condition 2 : Lightship + Permanent Ballast + Max.Deck Cargo Towing Condition with Max. draft 2.212m	Condition 3 : Lightship + Load at Aft + Ballast Crane Lifting 36.60T @ 18m Radius Aft	Condition 4 : Lightship + Load at Stbd + Ballast Crane Lifting 36.60T @ 18m Radius Stbd	Condition 5 : Lightship + Load at Fwd + Ballast Crane Lifting 36.60T @ 18m Radius Fwd
TOTAL TANK LOAD (T)	0.000	264.600	264.600	264.600	264.600	264.600
DWT CONST.(T)	0.000	250.000	1148.050	286.600	286.600	286.600
DEADWEIGHT (T)	0.000	514.600	1412.650	551.200	551.200	551.200
LIGHT WEIGHT (T)	356.330	356.330	356.330	356.330	356.330	356.330
DISPLACEMENT (T)	356.330	870.930	1768.980	907.530	907.530	907.530
FWD DRAFT (M)	0.498	1.148	2.209	1.078	1.193	1.308
AFT DRAFT (M)	0.502	1.154	2.215	1.314	1.199	1.084
MEAN DRAFT (M)	0.500	1.151	2.212	1.196	1.196	1.196
TRIM (M)	0.000	0.006a	0.005a	0.236a	0.006a	0.224f
L.C.G. (FROM AP) (M)	25.000f	25.000f	25.000f	24.274f	25.000f	25.726f
L.C.B. (FROM AP) (M)	25.000f	25.000f	25.000f	24.248f	25.000f	25.752f
M.C.T. (T-CM)	24.750	28.900	35.190	28.690	28.680	28.690
KG (M)	3.000	3.176	4.074	6.152	6.152	6.152
GM_T (M)	48.885	19.578	8.798	15.825	15.813	15.824
MAX. GZ (M)	5.173	3.950	1.186	2.986	2.995	2.986

25-04-22 10:23:35 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 00 : Lightship

Lightship condition

Condition Graphic - Draft: 0.498 @ 50.000f, 0.502 @ 0.000 Heel: zero



25-04-22 10:23:35 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 00 : Lightship

Lightship condition

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 0.498 @ 50.00f, 0.500 @ 25.00f, 0.502 @ 0.00
                       Trim: 0.000/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
WEIGHT                               356.33  25.000f  0.000   3.000
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->                0.00                                0.00
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        356.33  25.000f  0.000   0.247
-------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         749.3  25.039f  0.000    349.98  51.637
                                    MT/cm--------m.-MT/cm-------GML-----GMT
                                       7.68            24.75    347.22  48.885
Distances in METERS.-----------------------------------------Moments in m.-MT.
```

SUMMARY OF LOADING

0.0 Cu.M. (0%) FRESH WATER

DRAFT AT FP (M) = 0.498
DRAFT AT MS (M) = 0.500
DRAFT AT AP (M) = 0.502
DRAFT AT FWD MARK (M) = 0.498
DRAFT AT AFT MARK (M) = 0.502

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GHS 18.00

KB-24

Condition 00 : Lightship
Lightship condition

HYDROSTATIC PROPERTIES
Trim: 0.000/50.000, No Heel, VCG = 3.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
0.500	356.33	25.000f	0.247	7.68	25.039f	24.75	347.22	48.885

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK.

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 0.498 @ 50.00f, 0.500 @ 25.00f, 0.502 @ 0.00
Trim: 0.000/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	121.1	1.284	1.525	0.05500	10.16

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.000f TCG = 0.000 VCG = 3.000

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
0.490	0.00a	0.00	356.33	0.000	-0.029	0.0000	15.141
0.490	0.00a	0.03s	356.33	0.000	0.000	-0.0000	15.114
0.443	0.01a	5.03s	356.31	0.000	3.691	0.1610	10.114
0.173	0.01a	10.03s	356.27	0.000	4.798	0.5415	5.114
-0.217	0.01a	15.03s	356.33	0.000	5.121	0.9800	0.114
-0.226	0.01a	15.14s	356.33	0.000	5.124	0.9902	0.000
-0.553	0.01a	18.80s	356.20	0.000	5.173	1.3194	-3.654
-0.668	0.01a	20.03s	356.33	0.000	5.168	1.4306	-4.886
-1.157	0.01a	25.03s	356.54	0.000	5.066	1.8785	-9.886
-1.669	0.01a	30.03s	356.54	0.000	4.867	2.3126	-14.886
-2.185	0.01a	35.03s	356.16	0.000	4.577	2.7253	-19.886
-2.683	0.01a	40.03s	356.18	0.000	4.197	3.1088	-24.886
-3.160	0.01a	45.03s	356.19	0.000	3.753	3.4561	-29.886
-3.613	0.01a	50.03s	356.20	0.000	3.261	3.7625	-34.886
-4.039	0.01a	55.03s	356.21	0.000	2.730	4.0242	-39.886
-4.435	0.01a	60.03s	356.22	0.000	2.170	4.2382	-44.886

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 10.16 (constant)

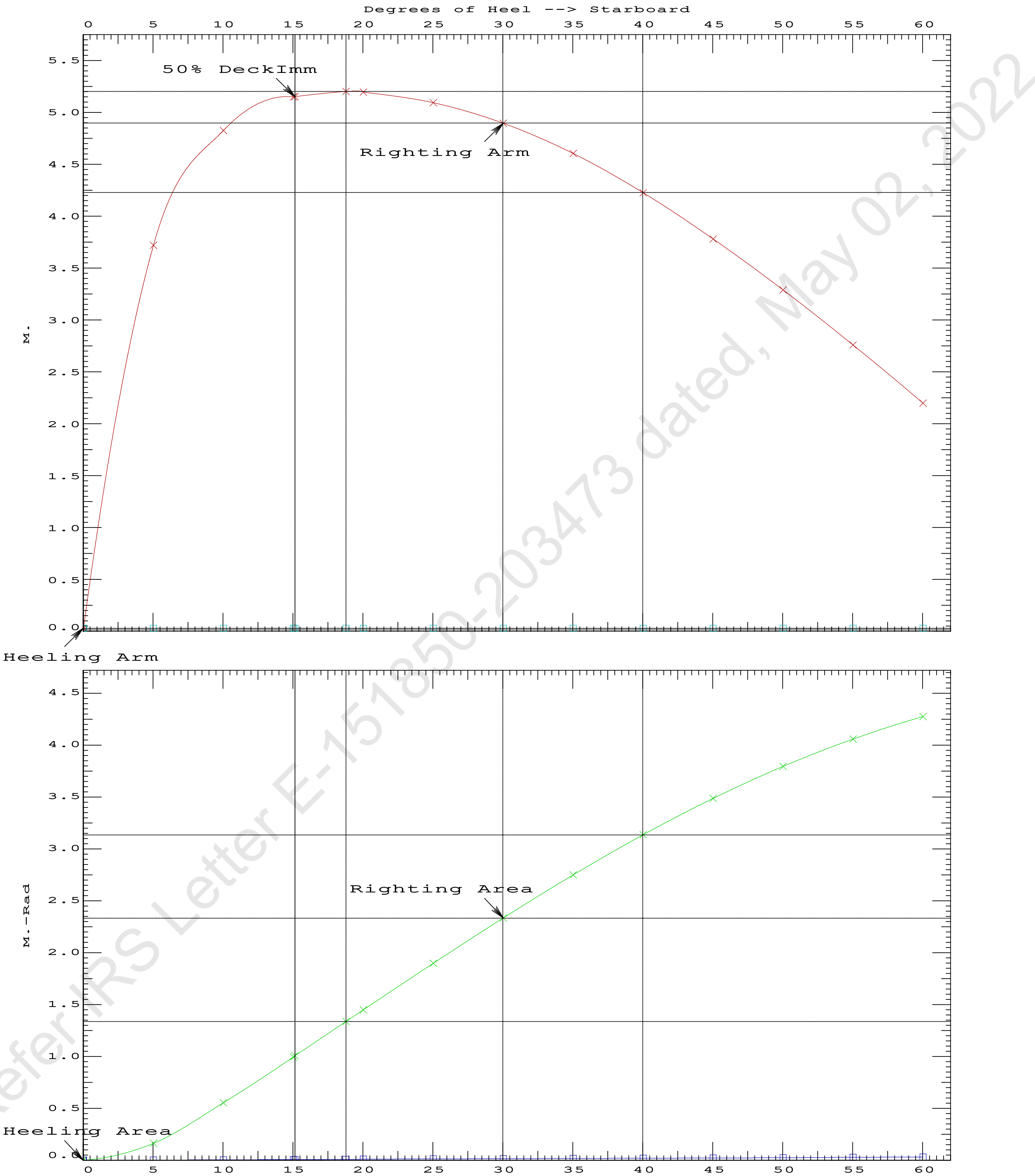
LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained
(1)	Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	1.3194 P
(2)	Angle from Equilibrium to RAzero or Flood	>	20.00	deg	LARGE
(3)	Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	15.11 P

25-04-22 10:23:35 Cybermarine Technologies PTE. Ltd.
GHS 18.00

KB-24

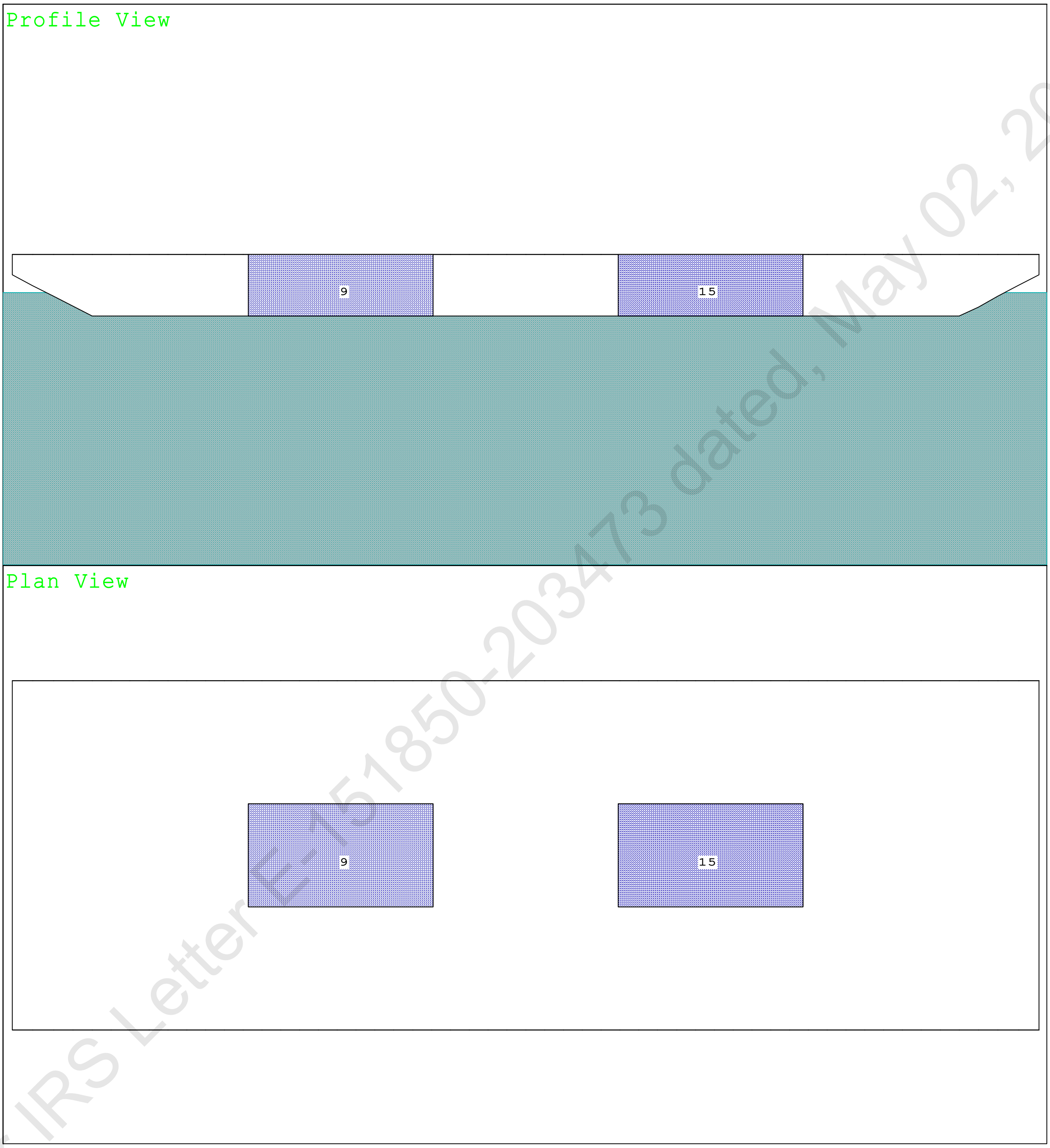
Condition 00 : Lightship

Lightship condition



Condition 1 : Lightship + Crane + Permanent Ballast

Condition Graphic - Draft: 1.148 @ 50.000f, 1.154 @ 0.000 Heel: zero



Tanks
9 03_PFWBTK.C.100% FRESH WATER 15 05_PFWBTK.C.100% FRESH WATER

```
25-04-22 10:23:37    Cybermarine Technologies PTE. Ltd.
GHS 18.00                              KB-24
```

Condition 1 : Lightship + Crane + Permanent Ballast

```
                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
            USK draft: 1.148 @ 50.00f, 1.151 @ 25.00f, 1.154 @ 0.00
                      Trim: Aft 0.006/50.000,  Heel: zero
Part----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          356.33  25.000f  0.000   3.000
Crane                               250.00  25.000f  0.000   5.200
  Total Fixed--------->             606.33  25.000f  0.000   3.907
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000       132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000       132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->             264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->             870.93  25.000f  0.000   3.176
                                 Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       870.93  25.000f  0.000   0.584
-----------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         791.8  24.994f  0.000    168.49  22.170*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.12            28.90    165.90  19.578
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.
```

SUMMARY OF LOADING

264.6 Cu.M. (11%) FRESH WATER

250.00 MT Crane

```
DRAFT AT FP (M) =    1.148
DRAFT AT MS (M) =    1.151
DRAFT AT AP (M) =    1.154
DRAFT AT FWD MARK (M) =    1.149
DRAFT AT AFT MARK (M) =    1.154
```

25-04-22 10:23:37 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 1 : Lightship + Crane + Permanent Ballast

HYDROSTATIC PROPERTIES
Trim: Aft 0.006/50.000, No Heel, VCG = 3.176

Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.151	870.93	25.000f	0.584	8.12	24.994f	28.90	165.90	19.578

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.148 @ 50.00f, 1.151 @ 25.00f, 1.154 @ 0.00
Trim: Aft 0.006/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	91.5	0.942	1.500	0.05500	7.55
CRANE.C	32.0	3.861	4.419	0.05500	7.78
Total wind heeling moment to starboard------------->					15.33

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.000f TCG = 0.000 VCG = 3.176
Free Surface Adjustment: 0.211
Adjusted CG: LCG = 25.000f TCG = 0.000 VCG = 3.387

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% Area	DeckImm Angle
1.142	0.01a	0.00	870.93	0.000	-0.017	0.0000	6.790
1.142	0.01a	0.05s	870.93	0.000	0.000	-0.0000	6.741
1.128	0.01a	5.05s	870.93	0.000	1.720	0.0750	1.741
1.115	0.01a	6.79s	870.92	0.000	2.295	0.1361	0.000
1.059	0.01a	10.05s	870.93	0.000	3.192	0.2928	-3.259
0.849	0.01a	15.05s	870.93	0.000	3.846	0.6044	-8.259
0.665	0.01a	18.80s	870.93	0.000	3.950	0.8614	-12.009
0.603	0.01a	20.05s	870.93	0.000	3.942	0.9475	-13.259
0.351	0.01a	25.05s	870.70	0.000	3.796	1.2877	-18.259
0.097	0.01a	30.05s	870.74	0.000	3.535	1.6083	-23.259
-0.157	0.01a	35.05s	870.93	0.000	3.205	1.9029	-28.259
-0.410	0.01a	40.05s	870.93	0.000	2.830	2.1666	-33.259
-0.661	0.01a	45.05s	870.93	0.000	2.420	2.3959	-38.259
-0.906	0.01a	50.05s	870.93	0.000	1.984	2.5882	-43.259
-1.144	0.01a	55.05s	870.93	0.000	1.526	2.7415	-48.259
-1.374	0.02a	60.05s	870.93	0.000	1.054	2.8542	-53.259

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 15.33 (constant)

25-04-22 10:23:37 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 1 : Lightship + Crane + Permanent Ballast

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA          >   0.0800 m.-Rad 0.8614 P
(2) Angle from Equilibrium to RAzero or Flood      >    20.00  deg      LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle     >     0.00  deg     6.74 P
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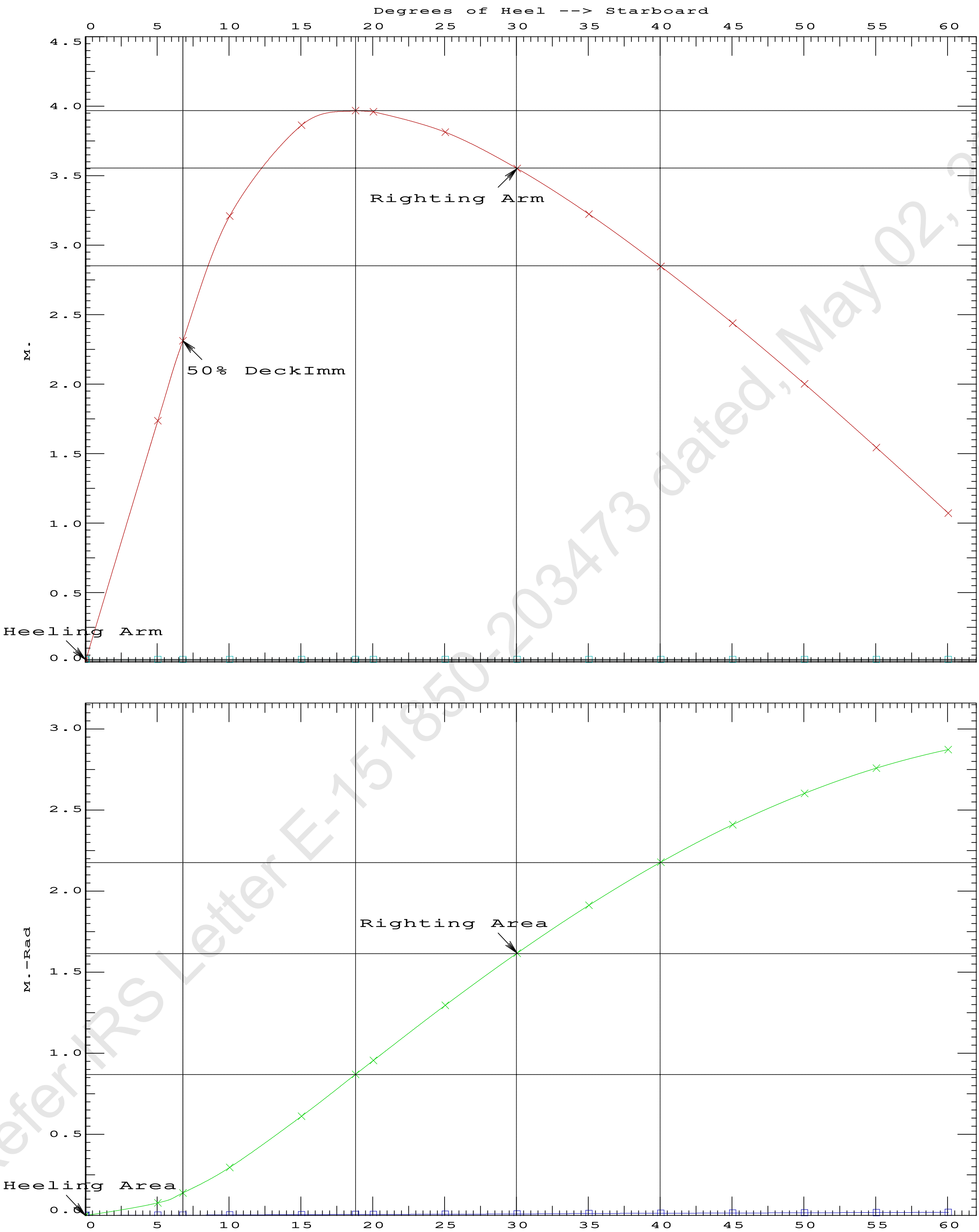
KB-24

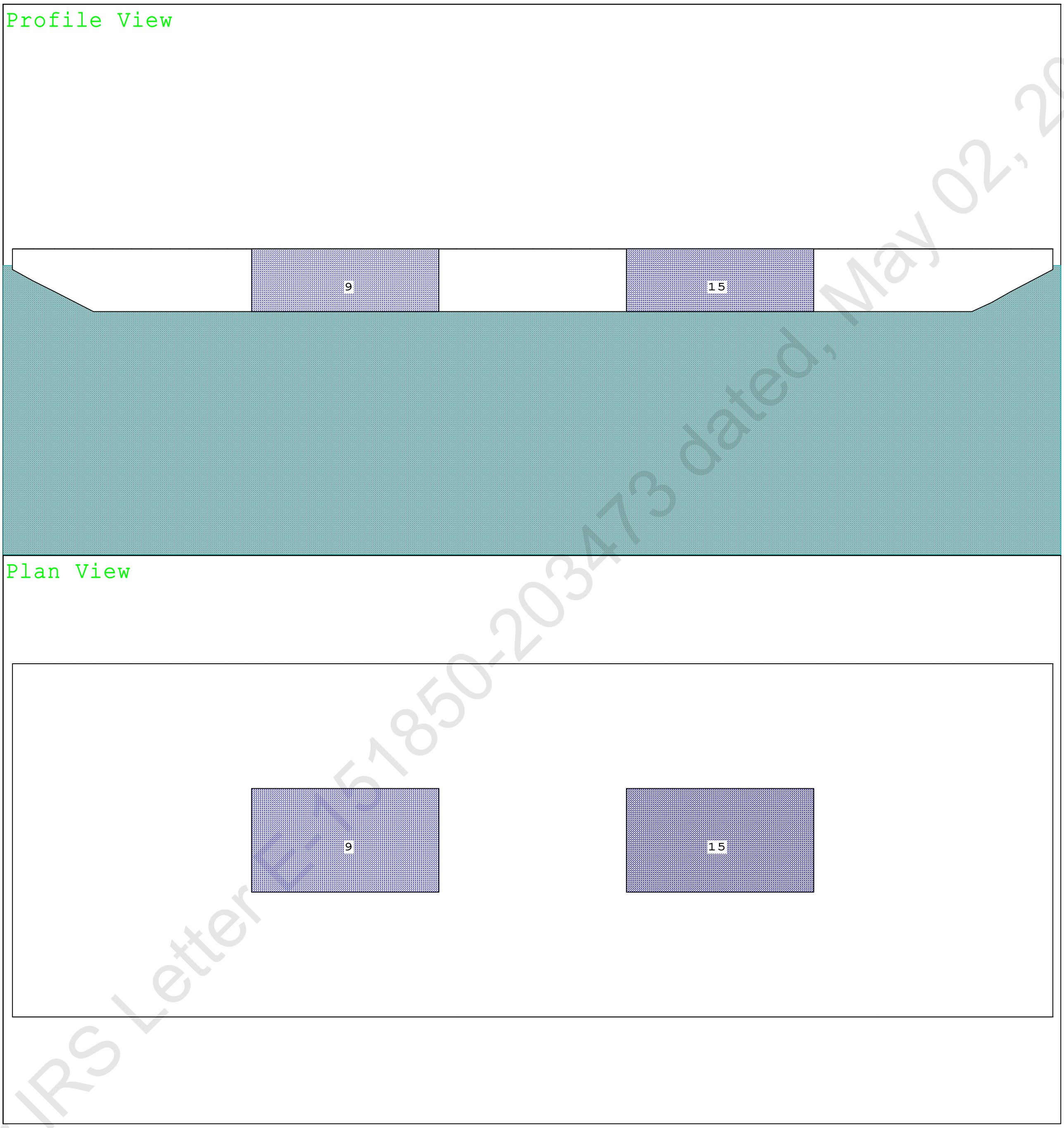
Condition 1 : Lightship + Crane + Permanent Ballast



Condition 2 : Lightship + Permanent Ballast + Max.Deck Cargo

Towing Condition with Max. draft 2.212m

Condition Graphic - Draft: 2.209 @ 50.000f, 2.215 @ 0.000 Heel: zero



Tanks	
	9 03_PFWBTK.C.100% FRESH WATER 15 05_PFWBTK.C.100% FRESH WATER

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25-04-22 10:23:40    Cybermarine Technologies PTE. Ltd.
GHS 18.00                         KB-24
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Condition 2 : Lightship + Permanent Ballast + Max.Deck Cargo

Towing Condition with Max. draft 2.212m

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.209 @ 50.00f, 2.212 @ 25.00f, 2.215 @ 0.00
                     Trim: Aft 0.005/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           356.33  25.000f  0.000   3.000
Deck Cargo                         1,148.05  25.000f  0.000   5.000
  Total Fixed--------->            1,504.38  25.000f  0.000   4.526
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000        132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000        132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->              264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->            1,768.98  25.000f  0.000   4.074
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,768.98  25.000f  0.000   1.139
------------------------------------------------------------------------------
               Righting Arms:                      0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         849.4  25.000f  0.000    102.40  11.733*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.71           35.19     99.46   8.798
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            SUMMARY OF LOADING
       264.6 Cu.M.  (11%) FRESH WATER

    1,148.05 MT  Deck Cargo

DRAFT AT FP (M) =    2.209
DRAFT AT MS (M) =    2.212
DRAFT AT AP (M) =    2.215
DRAFT AT FWD MARK (M) =    2.210
DRAFT AT AFT MARK (M) =    2.214
```

25-04-22 10:23:40 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 2 : Lightship + Permanent Ballast + Max.Deck Cargo
Towing Condition with Max. draft 2.212m

HYDROSTATIC PROPERTIES
Trim: Aft 0.005/50.000, No Heel, VCG = 4.074

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,768.98	25.000f	1.139	8.71	25.000f	35.19	99.46	8.798

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.209 @ 50.00f, 2.212 @ 25.00f, 2.215 @ 0.00
Trim: Aft 0.005/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.0	0.400	1.466	0.05500	3.23
DECKCARGO.C	160.0	2.800	3.866	0.05500	34.02
Total wind heeling moment to starboard------------->					37.25

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.000f TCG = 0.000 VCG = 4.074
Free Surface Adjustment: 0.104
Adjusted CG: LCG = 25.000f TCG = 0.000 VCG = 4.177

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
2.203	0.01a	0.00	1,769.0	0.000	-0.021	0.0000	2.688
2.203	0.01a	0.13s	1,769.0	0.000	0.000	-0.0000	2.555
2.200	0.01a	2.69s	1,769.0	0.000	0.396	0.0088	0.000
2.191	0.01a	5.13s	1,769.0	0.000	0.770	0.0337	-2.445
2.270	0.01a	10.13s	1,769.0	0.000	1.153	0.1223	-7.445
2.382	0.01a	13.09s	1,769.0	0.000	1.186	0.1834	-10.407
2.473	0.01a	15.13s	1,769.0	0.000	1.171	0.2253	-12.445
2.721	0.01a	20.13s	1,769.0	0.000	1.051	0.3238	-17.445
2.966	0.02a	25.13s	1,769.0	0.000	0.848	0.4073	-22.445
3.189	0.02a	30.13s	1,769.0	0.000	0.597	0.4707	-27.445
3.387	0.02a	35.13s	1,769.0	0.000	0.322	0.5110	-32.445
3.560	0.02a	40.13s	1,769.0	0.000	0.034	0.5266	-37.445
3.579	0.02a	40.72s	1,769.0	0.000	0.000	0.5268	-38.027
3.706	0.02a	45.13s	1,769.0	0.000	-0.261	0.5168	-42.445
3.823	0.03a	50.13s	1,769.0	0.000	-0.558	0.4811	-47.445
3.912	0.03a	55.13s	1,769.0	0.000	-0.853	0.4195	-52.445
3.970	0.03a	60.13s	1,769.0	0.000	-1.144	0.3323	-57.445

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

25-04-22 10:23:40 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 2 : Lightship + Permanent Ballast + Max.Deck Cargo
Towing Condition with Max. draft 2.212m

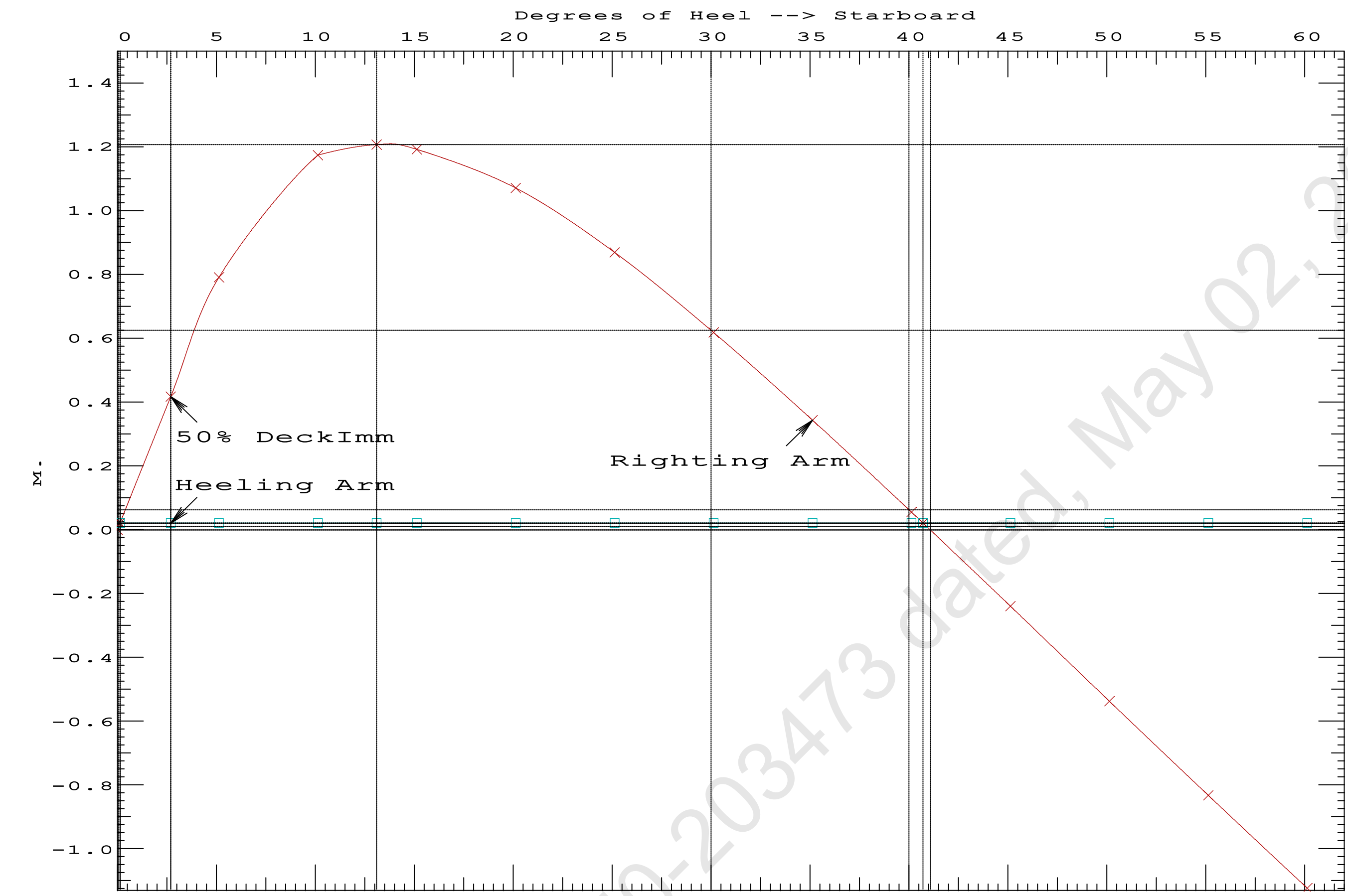
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 37.25 (constant)

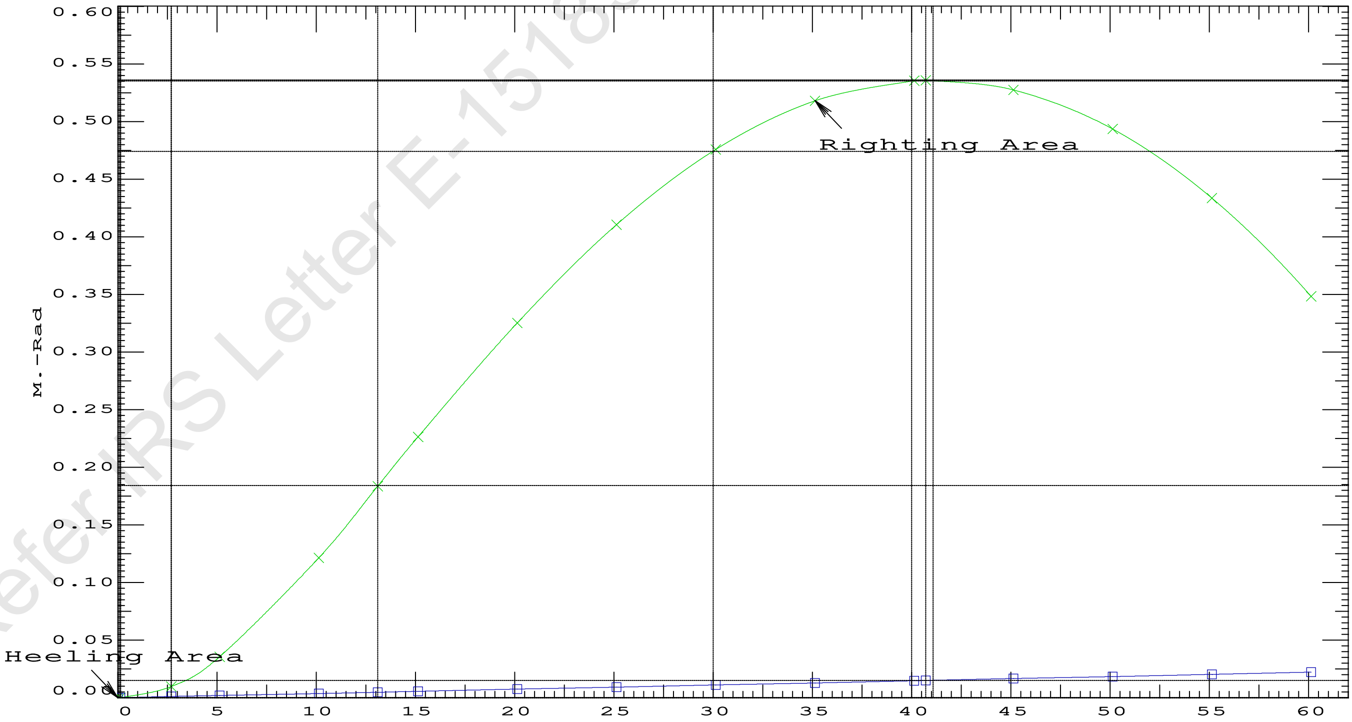
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1834	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	40.58	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.55	P

25-04-22 10:23:40 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 2 : Lightship + Permanent Ballast + Max.Deck Cargo

Towing Condition with Max. draft 2.212m



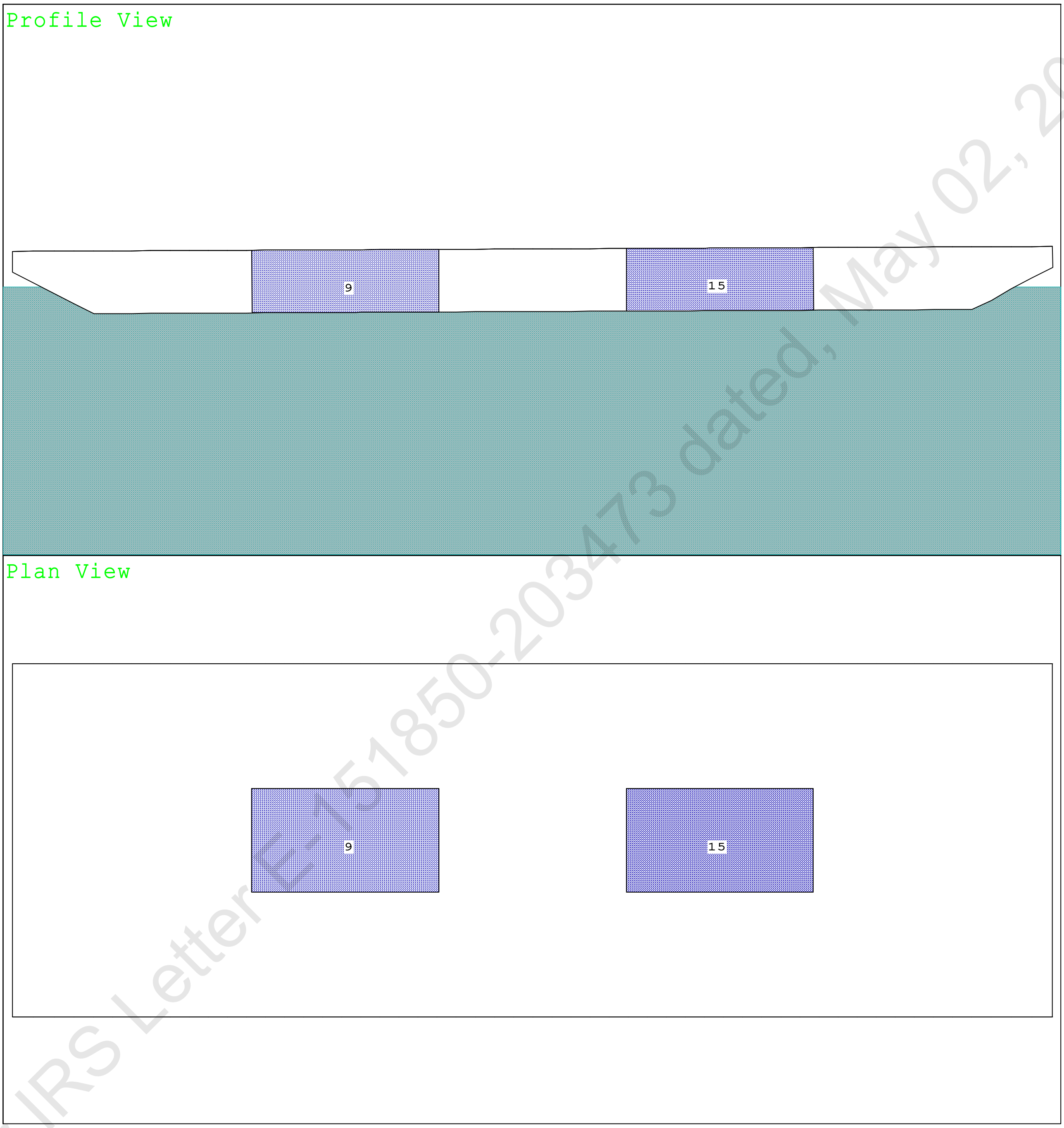


KB-24

Condition 3 : Lightship + Load at Aft + Ballast

Crane Lifting 36.60T @ 18m Radius Aft

Condition Graphic - Draft: 1.078 @ 50.000f, 1.314 @ 0.000 Heel: zero



Tanks	9 03_PFWBTK.C.100% FRESH WATER 15 05_PFWBTK.C.100% FRESH WATER

25-04-22 10:23:46 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 3 : Lightship + Load at Aft + Ballast
Crane Lifting 36.60T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.078 @ 50.00f, 1.196 @ 25.00f, 1.314 @ 0.00
Trim: Aft 0.236/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			356.33	25.000f	0.000	3.000	
Crane			250.00	25.000f	0.000	5.200	
Crane Load			36.60	7.000f	0.000	76.977	
Total Fixed--------->			642.93	23.975f	0.000	8.067	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			907.53	24.274f	0.000	6.152	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	907.48	24.248f	0.000	0.609	
Righting Arms:				0.000	0.000		

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	794.9	24.782f	0.000	163.63	21.368*
		MT/cm	m.-MT/cm		GML	GMT
		8.15		28.69	158.09	15.825

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING
264.6 Cu.M. (11%) FRESH WATER

250.00 MT Crane 36.60 MT Crane Load

DRAFT AT FP (M) = 1.078
DRAFT AT MS (M) = 1.196
DRAFT AT AP (M) = 1.314
DRAFT AT FWD MARK (M) = 1.096
DRAFT AT AFT MARK (M) = 1.295

HYDROSTATIC PROPERTIES
Trim: Aft 0.236/50.000, No Heel, VCG = 6.152

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.197	907.48	24.248f	0.609	8.15	24.782f	28.69	158.09	15.825

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

25-04-22 10:23:46 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 3 : Lightship + Load at Aft + Ballast
Crane Lifting 36.60T @ 18m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.078 @ 50.00f, 1.196 @ 25.00f, 1.314 @ 0.00
Trim: Aft 0.236/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	89.4	0.920	1.502	0.05500	7.38
CRANE.C	32.0	3.816	4.398	0.05500	7.74
Total wind heeling moment to starboard------------->					15.12

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.274f TCG = 0.000 VCG = 6.152
Free Surface Adjustment: 0.202
Adjusted CG: LCG = 24.275f TCG = 0.000 VCG = 6.355

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.302	0.27a	0.00	907.53	0.000	-0.016	0.0000	5.942
1.302	0.27a	0.05s	907.53	0.000	0.000	-0.0000	5.887
1.286	0.27a	5.05s	907.52	0.000	1.392	0.0607	0.887
1.282	0.27a	5.94s	907.51	0.000	1.632	0.0841	0.000
1.236	0.30a	10.05s	907.51	0.000	2.582	0.2364	-4.113
1.078	0.38a	15.05s	907.53	0.000	2.984	0.4836	-9.113
1.065	0.39a	15.44s	907.53	0.000	2.986	0.5036	-9.497
0.910	0.51a	20.05s	907.53	0.000	2.820	0.7406	-14.113
0.737	0.65a	25.05s	907.53	0.000	2.430	0.9713	-19.113
0.561	0.78a	30.05s	907.53	0.000	1.940	1.1627	-24.113
0.383	0.92a	35.05s	907.53	0.000	1.397	1.3087	-29.113
0.204	1.05a	40.05s	907.53	0.000	0.821	1.4057	-34.113
0.025	1.18a	45.05s	907.53	0.000	0.227	1.4516	-39.113
-0.042	1.22a	46.94s	907.53	0.000	0.000	1.4553	-41.002
-0.153	1.30a	50.05s	907.53	0.000	-0.376	1.4451	-44.113
-0.331	1.41a	55.05s	907.53	0.000	-0.982	1.3859	-49.113
-0.508	1.50a	60.05s	907.53	0.000	-1.584	1.2739	-54.113

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

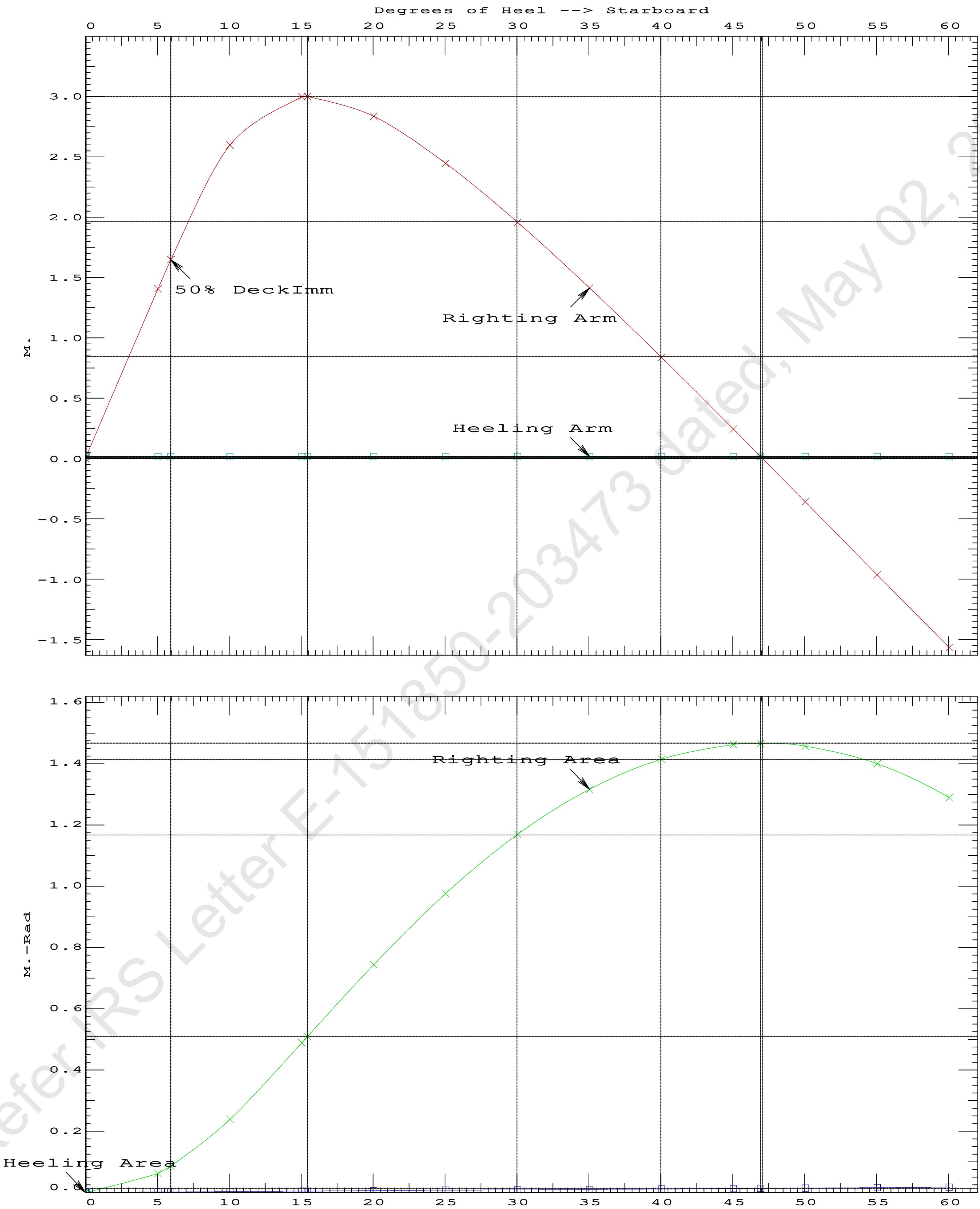
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 15.12 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.5036	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	46.89	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	5.89	P

25-04-22 10:23:46 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 3 : Lightship + Load at Aft + Ballast
Crane Lifting 36.60T @ 18m Radius Aft



25-04-22 10:23:46 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 3 : Lightship + Load at Aft + Ballast
Crane Lifting 36.60T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.274f TCG = 0.000 VCG = 6.152
Free Surface Adjustment: 0.202
Adjusted CG: LCG = 24.275f TCG = 0.000 VCG = 6.355

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.302	0.27a	0.00	907.52	0.000	0.000	0.0000
1.287	0.27a	5.00s	907.52	0.000	1.393	0.0608
1.237	0.30a	10.00s	907.50	0.000	2.589	0.2359
1.080	0.38a	15.00s	907.53	0.000	2.999	0.4851
1.065	0.39a	15.45s	907.53	0.000	3.001	0.5086
0.911	0.51a	20.00s	907.53	0.000	2.839	0.7435
0.739	0.65a	25.00s	907.53	0.000	2.451	0.9768
0.563	0.78a	30.00s	907.53	0.000	1.961	1.1701
0.385	0.92a	35.00s	907.53	0.000	1.418	1.3179
0.206	1.05a	40.00s	907.53	0.000	0.843	1.4168
0.027	1.18a	45.00s	907.53	0.000	0.249	1.4646
-0.047	1.23a	47.07s	907.53	0.000	0.000	1.4691
-0.151	1.30a	50.00s	907.53	0.000	-0.354	1.4601
-0.329	1.40a	55.00s	907.53	0.000	-0.960	1.4028
-0.506	1.50a	60.00s	907.53	0.000	-1.562	1.2927

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

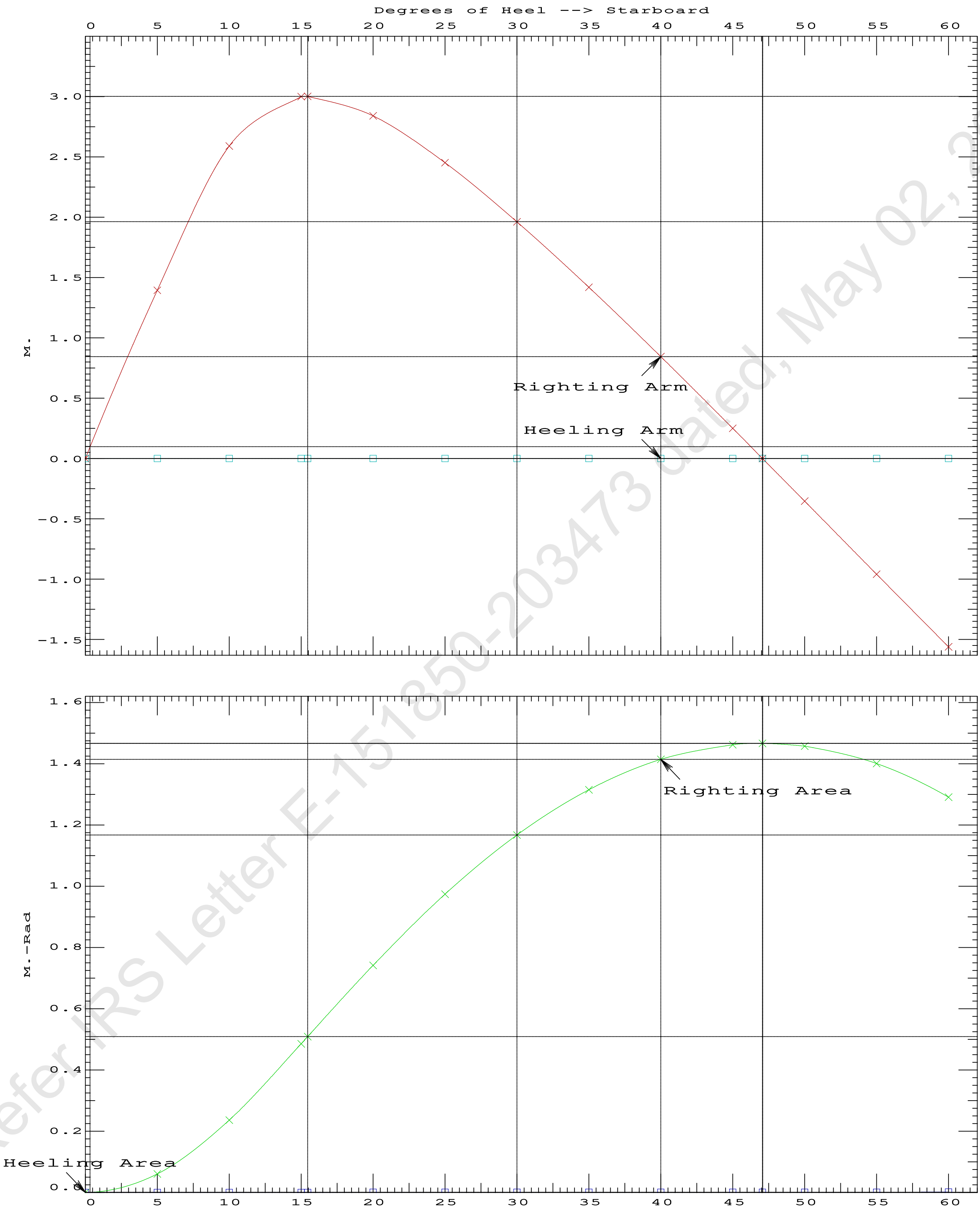
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	272348	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1623.86	P

25-04-22 10:23:46 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 3 : Lightship + Load at Aft + Ballast

Crane Lifting 36.60T @ 18m Radius Aft



25-04-22 10:23:44 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 3 : Lightship + Load at Aft + Ballast
Crane Lifting 36.60T @ 18m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.078 @ 50.00f, 1.196 @ 25.00f, 1.314 @ 0.00
Trim: Aft 0.236/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			356.33	25.000f	0.000	3.000	
Crane			250.00	25.000f	0.000	5.200	
Crane Load			36.60	7.000f	0.000	76.977	
Total Fixed--------->			642.93	23.975f	0.000	8.067	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			907.53	24.274f	0.000	6.152	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	907.53	24.248f	0.001s	0.609

	LCG	TCG
Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001s
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	791.5	24.865f	0.026s	161.53	21.244*
		MT/cm		m.-MT/cm	GML	GMT
		8.11		28.31	155.98	15.701

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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25-04-22 10:23:44    Cybermarine Technologies PTE. Ltd.
GHS 18.00                          KB-24

                         After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.148 @ 50.00f, 1.151 @ 25.00f, 1.154 @ 0.00
                  Trim: Aft 0.006/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           356.33  25.000f  0.000   3.000
Crane                                250.00  25.000f  0.000   5.200
  Total Fixed--------->              606.33  25.000f  0.000   3.907
               Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000        132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000        132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->              264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->              870.93  25.000f  0.000   3.176
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        870.93  25.000f  0.001s  0.584
------------------------------------------------------------------------------
             Righting Arms:                   0.000   0.001s
             External Arms:                   0.000   0.001s
      Residual Righting Arms:                 0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         790.2  24.996f  0.014s   167.45  22.085*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.10             28.72   164.86  19.493
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

            Theta3 is 0.07 degrees, as demonstrated below.

                RESIDUAL RIGHTING ARMS vs HEEL ANGLE
     Fixed CG:  LCG = 25.000f  TCG = 0.000   VCG =  3.907
```

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.142	0.01a	0.00	870.93	0.000	0.000	0.0000
1.142	0.01a	0.07s	870.93	0.000	0.024	0.0000
1.129	0.01a	5.00s	870.93	0.000	1.738	0.0759
1.060	0.01a	10.00s	870.93	0.000	3.235	0.2947
0.851	0.01a	15.00s	870.93	0.000	3.914	0.6126
0.644	0.01a	19.22s	870.93	0.000	4.035	0.9077
0.605	0.01a	20.00s	870.93	0.000	4.031	0.9627

25-04-22 10:23:44 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

0.353	0.01a	25.00s	870.70	0.000	3.903	1.3115
0.099	0.01a	30.00s	870.73	0.000	3.659	1.6423

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 606.33 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

```
LIM--------------------STABILITY CRITERION------------Min/Max--------Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1  >    1.000          LARGE
-----------------------------------------------------------------------------
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 25.000f TCG = 0.000 VCG = 3.907

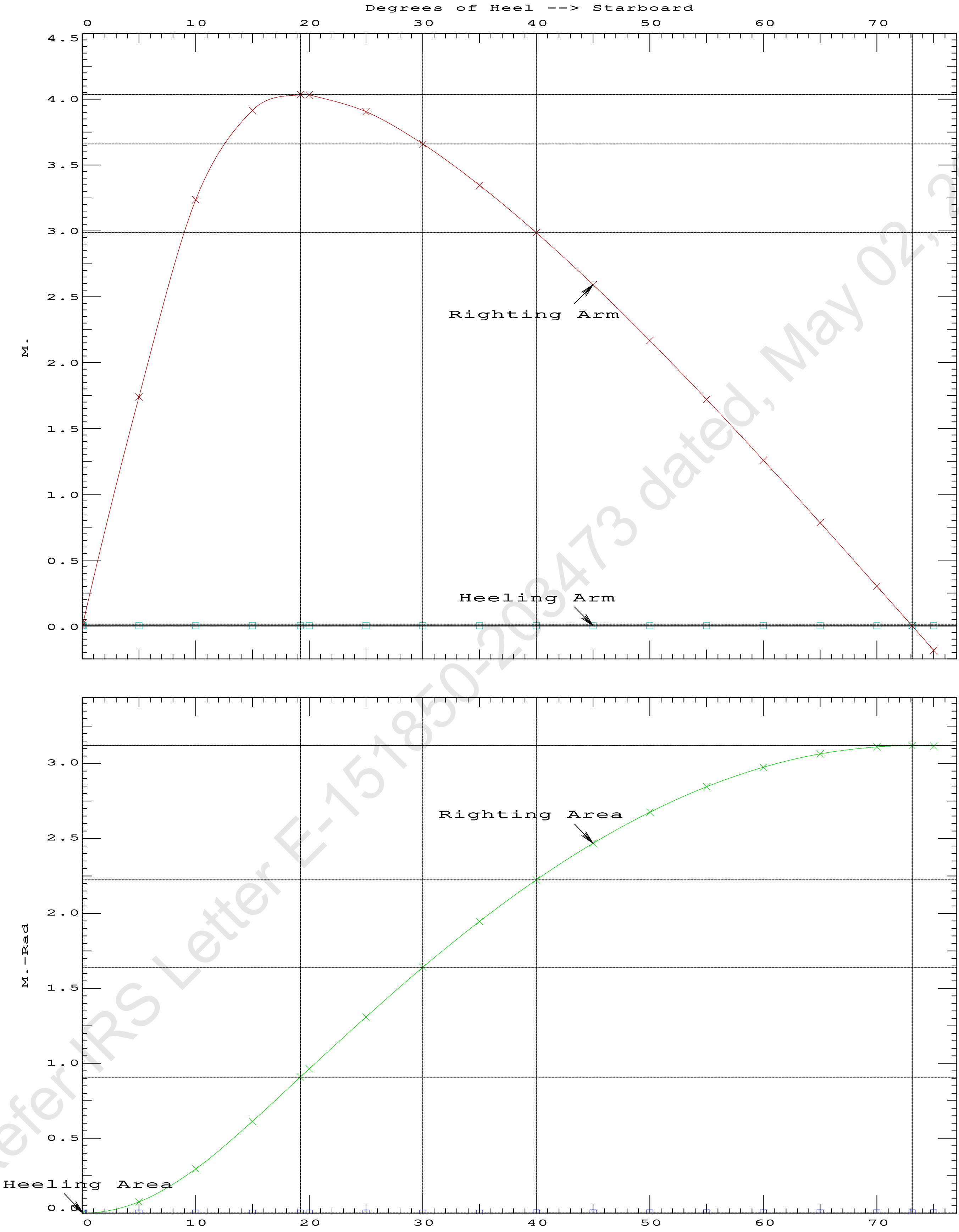
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.142	0.01a	0.00	870.93	0.000	0.000	0.0000
1.142	0.01a	0.07s	870.93	0.000	0.024	0.0000
1.129	0.01a	5.00s	870.93	0.000	1.738	0.0759
1.060	0.01a	10.00s	870.93	0.000	3.235	0.2947
0.851	0.01a	15.00s	870.93	0.000	3.914	0.6126
0.644	0.01a	19.22s	870.93	0.000	4.035	0.9077
0.605	0.01a	20.00s	870.93	0.000	4.031	0.9627
0.353	0.01a	25.00s	870.70	0.000	3.903	1.3115
0.099	0.01a	30.00s	870.73	0.000	3.659	1.6423
-0.155	0.01a	35.00s	870.93	0.000	3.346	1.9485
-0.408	0.01a	40.00s	870.93	0.000	2.986	2.2251
-0.658	0.01a	45.00s	870.93	0.000	2.590	2.4686
-0.904	0.01a	50.00s	870.93	0.000	2.166	2.6763
-1.142	0.01a	55.00s	870.93	0.000	1.720	2.8460
-1.372	0.02a	60.00s	870.93	0.000	1.258	2.9761
-1.591	0.02a	65.00s	870.93	0.000	0.783	3.0652
-1.798	0.02a	70.00s	870.93	0.000	0.301	3.1126
-1.920	0.02a	73.10s	870.93	0.000	0.000	3.1207
-1.992	0.02a	75.00s	870.93	0.000	-0.185	3.1176

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 606.33 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

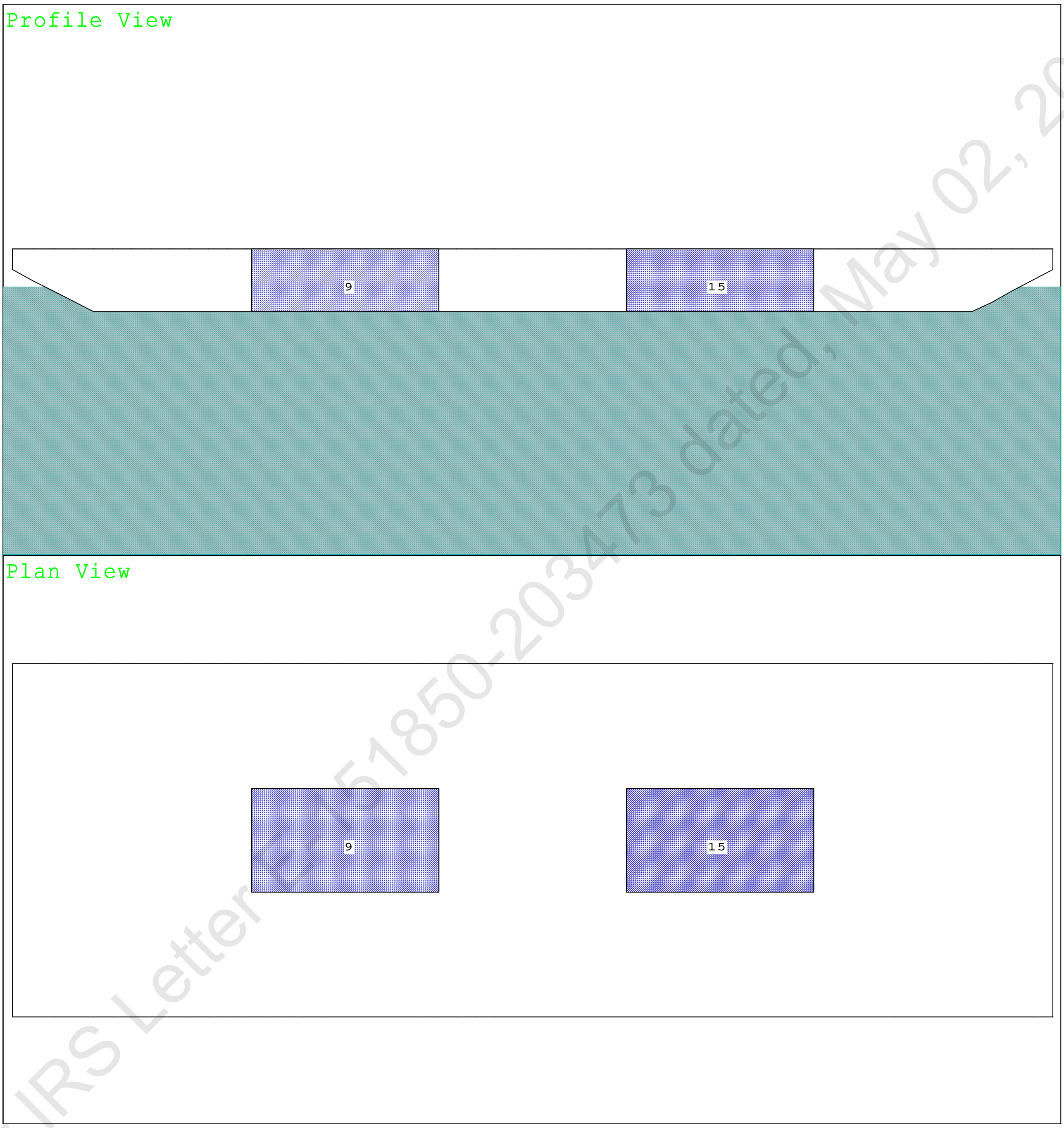
LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	73.03 P


Degrees of Heel --> Starboard
0
10
20
30
40
50
60
70
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
M.
Righting Arm
Heeling Arm
3.0
2.5
2.0
1.5
1.0
0.5
0.0
M.-Rad
Righting Area
Heeling Area

25-04-22 10:23:51 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 4 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.60T @ 18m Radius Stbd

Condition Graphic - Draft: 1.193 @ 50.000f, 1.199 @ 0.000 Heel: zero



Tanks	
	9 03_PFWBTK.C.100% FRESH WATER 15 05_PFWBTK.C.100% FRESH WATER

25-04-22 10:23:51 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 4 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.60T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.193 @ 50.00f, 1.196 @ 25.00f, 1.199 @ 0.00
Trim: Aft 0.006/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			356.33	25.000f	0.000	3.000	
Crane			250.00	25.000f	0.000	5.200	
Crane Load			36.60	25.000f	0.000	76.977	
Total Fixed--------->			642.93	25.000f	0.000	8.067	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			907.53	25.000f	0.000	6.152	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	907.53	24.999f	0.000	0.607	
Righting Arms:				0.000	0.000		

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	794.8	24.994f	0.000	163.54	21.358*
		MT/cm		m.-MT/cm	GML	GMT
		8.15		28.68	157.99	15.813

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

264.6 Cu.M. (11%) FRESH WATER

250.00 MT Crane 36.60 MT Crane Load

DRAFT AT FP (M) = 1.193
DRAFT AT MS (M) = 1.196
DRAFT AT AP (M) = 1.199
DRAFT AT FWD MARK (M) = 1.193
DRAFT AT AFT MARK (M) = 1.199

HYDROSTATIC PROPERTIES
Trim: Aft 0.006/50.000, No Heel, VCG = 6.152

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.196	907.53	24.999f	0.607	8.15	24.994f	28.68	157.99	15.813

Distances in METERS.------Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

25-04-22 10:23:51 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 4 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.60T @ 18m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.193 @ 50.00f, 1.196 @ 25.00f, 1.199 @ 0.00
Trim: Aft 0.006/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	89.4	0.919	1.499	0.05500	7.37
CRANE.C	32.0	3.816	4.396	0.05500	7.74

Total wind heeling moment to starboard-------------> 15.10
Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.000f TCG = 0.000 VCG = 6.152
Free Surface Adjustment: 0.202
Adjusted CG: LCG = 25.000f TCG = 0.000 VCG = 6.355

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.187	0.01a	0.00	907.52	0.000	-0.016	0.0000	6.528
1.187	0.01a	0.06s	907.53	0.000	0.000	-0.0000	6.471
1.173	0.01a	5.06s	907.53	0.000	1.393	0.0608	1.471
1.163	0.01a	6.53s	907.52	0.000	1.788	0.1016	0.000
1.110	0.01a	10.06s	907.53	0.000	2.587	0.2370	-3.529
0.914	0.01a	15.06s	907.53	0.000	2.993	0.4848	-8.529
0.897	0.01a	15.44s	907.53	0.000	2.995	0.5047	-8.911
0.688	0.01a	20.06s	907.53	0.000	2.830	0.7427	-13.529
0.457	0.01a	25.06s	907.30	0.000	2.441	0.9742	-18.529
0.223	0.01a	30.06s	907.53	0.000	1.950	1.1665	-23.529
-0.013	0.01a	35.06s	907.53	0.000	1.406	1.3134	-28.529
-0.249	0.01a	40.06s	907.53	0.000	0.831	1.4112	-33.529
-0.483	0.01a	45.06s	907.53	0.000	0.236	1.4579	-38.529
-0.574	0.01a	47.02s	907.53	0.000	0.000	1.4619	-40.489
-0.713	0.01a	50.06s	907.53	0.000	-0.368	1.4522	-43.529
-0.938	0.02a	55.06s	907.53	0.000	-0.975	1.3936	-48.529
-1.156	0.02a	60.06s	907.53	0.000	-1.577	1.2822	-53.529

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 15.10 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.5047	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	46.96	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	6.47	P

25-04-22 10:23:51 Cybermarine Technologies PTE. Ltd.
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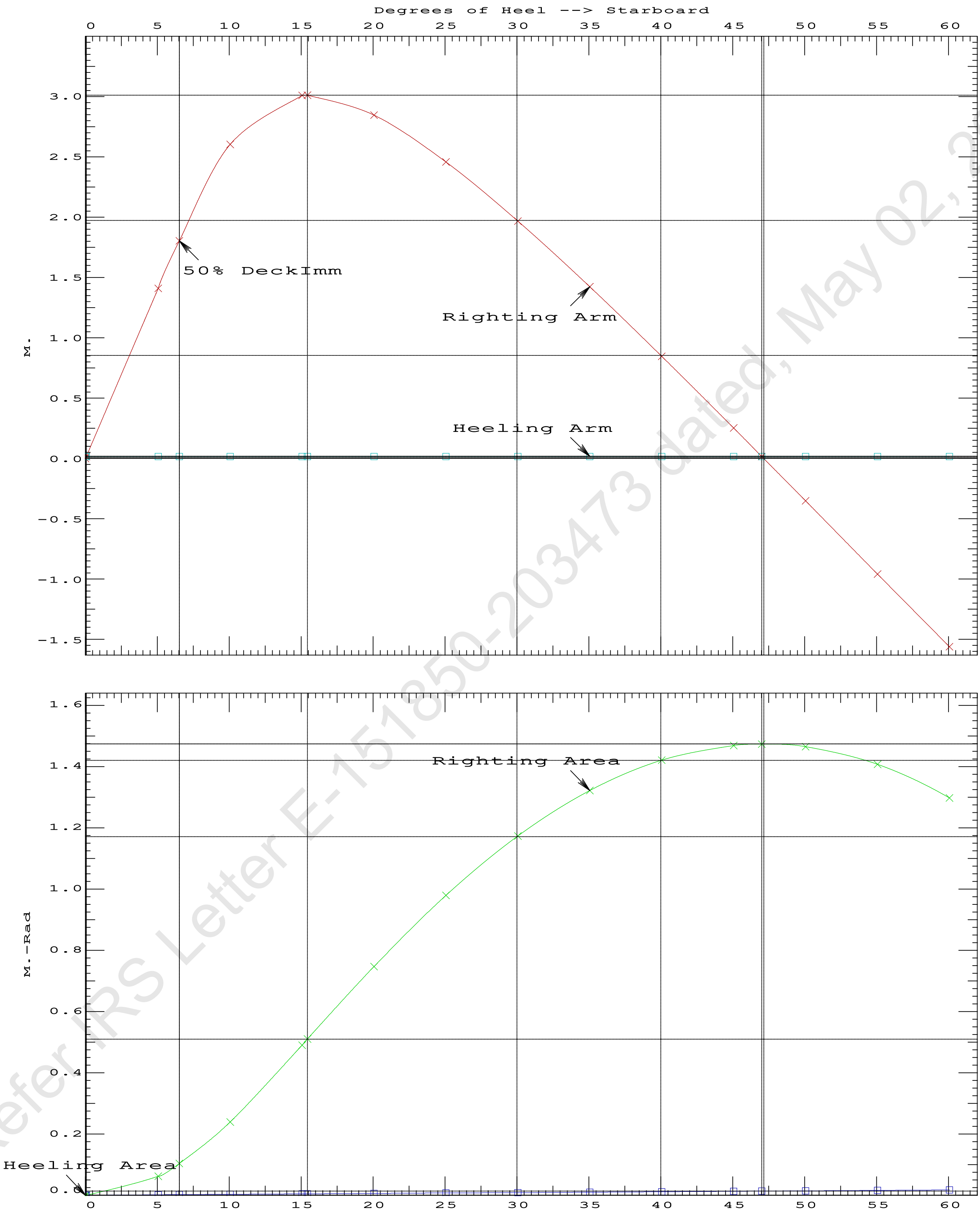
Condition 4 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.60T @ 18m Radius Stbd



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GHS 18.00 **KB-24**

Condition 4 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.60T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.000f TCG = 0.000 VCG = 6.152
Free Surface Adjustment: 0.202
Adjusted CG: LCG = 25.000f TCG = 0.009p VCG = 6.354

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.183	0.01a	2.56s	907.52	0.000	0.000	0.0000
1.153	0.01a	7.56s	907.53	0.000	1.353	0.0590
1.024	0.01a	12.56s	907.54	0.000	2.178	0.2202
0.885	0.01a	15.71s	907.53	0.000	2.294	0.3445
0.802	0.01a	17.56s	907.53	0.000	2.254	0.4180
0.573	0.01a	22.56s	907.53	0.000	1.951	0.6051
0.340	0.01a	27.56s	907.32	0.000	1.503	0.7568
0.105	0.01a	32.56s	907.53	0.000	0.981	0.8658
-0.131	0.01a	37.56s	907.53	0.000	0.419	0.9272
-0.300	0.01a	41.14s	907.53	0.000	0.000	0.9403
-0.366	0.01a	42.56s	907.53	0.000	-0.168	0.9383
-0.599	0.01a	47.56s	907.53	0.000	-0.769	0.8976
-0.826	0.02a	52.56s	907.53	0.000	-1.375	0.8040
-1.048	0.02a	57.56s	907.53	0.000	-1.981	0.6576
-1.262	0.02a	62.56s	907.53	0.000	-2.581	0.4585

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 658.80

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	2.56 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	316.0 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2.923 P

25-04-22 10:23:51 Cybermarine Technologies PTE. Ltd.
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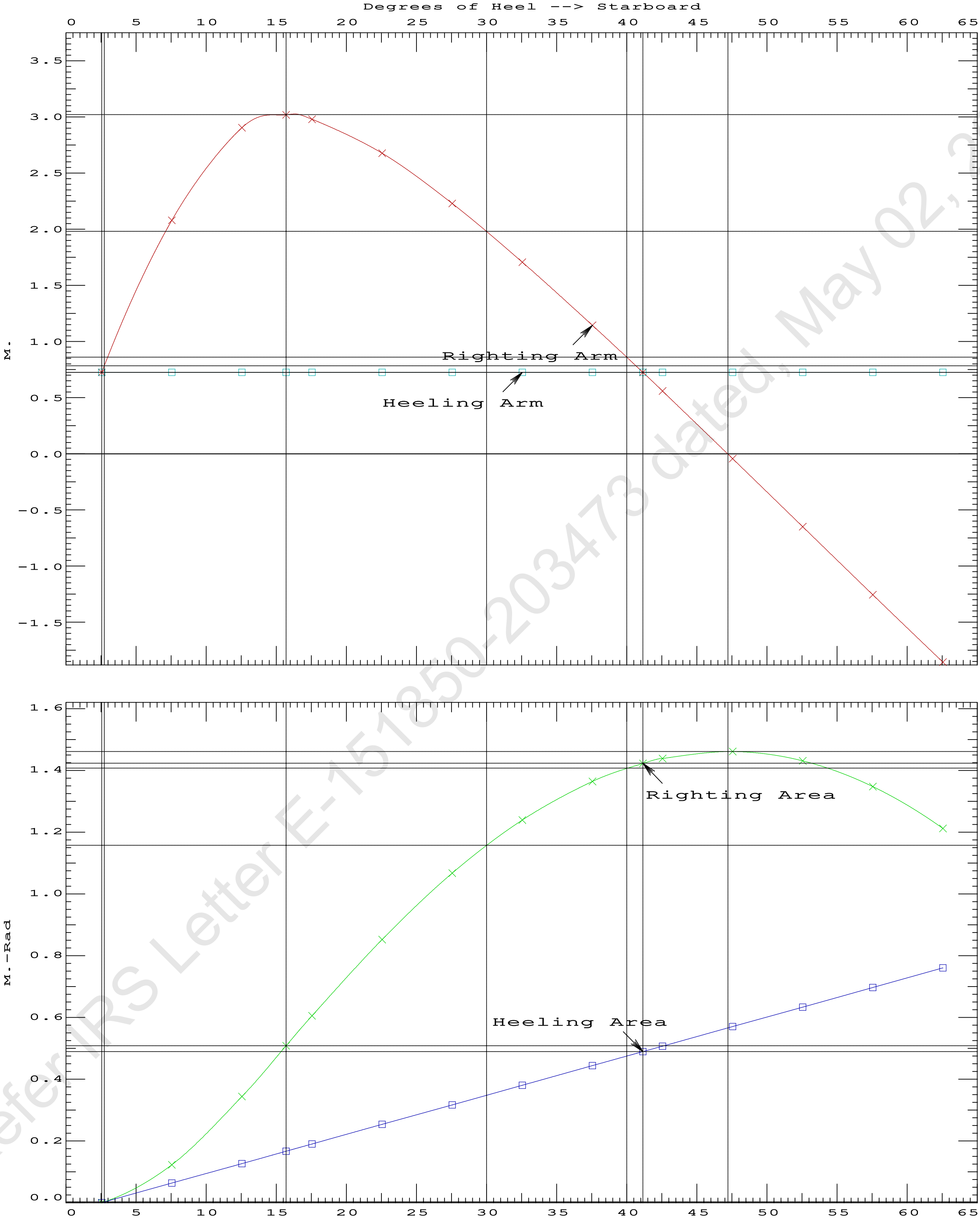
Condition 4 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.60T @ 18m Radius Stbd



25-04-22 10:28:44 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 4 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.60T @ 18m Radius Stbd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.189 @ 50.00f, 1.193 @ 25.00f, 1.196 @ 0.00
Trim: Aft 0.006/50.000, Heel: Stbd 2.56 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			356.33	25.000f	0.000	3.000	
Crane			250.00	25.000f	0.000	5.200	
Crane Load			36.60	25.000f	0.000	76.977	
Total Fixed--------->			642.93	25.000f	0.000	8.067	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			907.53	25.000f	0.000	6.152	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	907.52	24.999f	0.973s	0.629

Righting Arms:	0.000	0.726s
External Arms:	0.000	0.726s
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	793.2	24.994f	0.124s	162.51	21.355*
		MT/cm				
		8.13				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.56

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25-04-22 10:28:44   Cybermarine Technologies PTE. Ltd.
GHS 18.00                        KB-24

                      After hook load drop
           WEIGHT and DISPLACEMENT and WATERPLANE STATUS
       USK draft: 1.148 @ 50.00f, 1.151 @ 25.00f, 1.154 @ 0.00
              Trim: Aft 0.006/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          356.33  25.000f  0.000   3.000
Crane                               250.00  25.000f  0.000   5.200
  Total Fixed--------->             606.33  25.000f  0.000   3.907
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000       132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000       132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->             264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->             870.93  25.000f  0.000   3.176
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       870.93  25.000f  0.001s  0.584
-----------------------------------------------------------------------------
               Righting Arms:                0.000   0.001s
               External Arms:                0.000   0.001s
      Residual Righting Arms:                0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025        790.2  24.996f  0.014s   167.45  22.085*
                                     MT/cm--------m.-MT/cm-------GML-----GMT
                                      8.10            28.72   164.86  19.493
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 8098.83; t3: 30.00; t: 15.00
limmarg: 2958.34; t3: 15.00; t: 7.50
limmarg: 748.01; t3: 7.50; t: 3.75
limmarg: 107.66; t3: 3.70; t: 1.88
limmarg: -50.63; t3: 1.80; t: 0.94
limmarg: 10.85; t3: 2.70; t: 0.47
limmarg: -26.30; t3: 2.20; t: 0.23
limmarg: -12.32; t3: 2.40; t: 0.12
limmarg: -4.87; t3: 2.50; t: 0.06
limmarg: 2.84; t3: 2.60; t: 0.03
limmarg: 2.84; t3: 2.60; t: 0.01

             Theta3 is 2.60 degrees, as demonstrated below.

                   RESIDUAL RIGHTING ARMS vs HEEL ANGLE
          Fixed CG:  LCG = 25.000f  TCG = 0.000   VCG =  3.907

      Origin    Degrees of   Displacement    Residual Arms       Res.
       Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
       1.138   0.01a   2.56p      870.93      0.000   -0.897    0.0000
       1.142   0.01a   0.00       870.92      0.000    0.000   -0.0200
       1.138   0.01a   2.60s      870.93      0.000    0.909    0.0006
       1.129   0.01a   5.00s      870.93      0.000    1.738    0.0561
       1.060   0.01a  10.00s      870.93      0.000    3.235    0.2745
       0.851   0.01a  15.00s      870.93      0.000    3.914    0.5925
       0.644   0.01a  19.22s      870.93      0.000    4.035    0.8876
```

0.605	0.01a	20.00s	870.93	0.000	4.031	0.9426
0.353	0.01a	25.00s	870.70	0.000	3.903	1.2914
0.099	0.01a	30.00s	870.73	0.000	3.659	1.6222

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 606.33 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

```
LIM--------------------STABILITY CRITERION------------Min/Max--------Attained
(1) Res. Area Ratio from abs -2.6 deg to abs 2.6   >    1.000        1.028 P
-----------------------------------------------------------------------------
```

25-04-22 10:28:44 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 25.000f TCG = 0.000 VCG = 3.907

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.138	0.01a	2.56p	870.93	0.000	-0.897	0.0000
1.142	0.01a	0.00	870.92	0.000	0.000	-0.0200
1.138	0.01a	2.60s	870.93	0.000	0.909	0.0006
1.129	0.01a	5.00s	870.93	0.000	1.738	0.0561
1.060	0.01a	10.00s	870.93	0.000	3.235	0.2745
0.851	0.01a	15.00s	870.93	0.000	3.914	0.5925
0.644	0.01a	19.22s	870.93	0.000	4.035	0.8876
0.605	0.01a	20.00s	870.93	0.000	4.031	0.9426
0.353	0.01a	25.00s	870.70	0.000	3.903	1.2914
0.099	0.01a	30.00s	870.73	0.000	3.659	1.6222
-0.155	0.01a	35.00s	870.93	0.000	3.346	1.9284
-0.408	0.01a	40.00s	870.93	0.000	2.986	2.2050
-0.658	0.01a	45.00s	870.93	0.000	2.590	2.4485
-0.904	0.01a	50.00s	870.93	0.000	2.166	2.6562
-1.142	0.01a	55.00s	870.93	0.000	1.720	2.8259
-1.372	0.02a	60.00s	870.93	0.000	1.258	2.9559
-1.591	0.02a	65.00s	870.93	0.000	0.783	3.0451
-1.798	0.02a	70.00s	870.93	0.000	0.301	3.0924
-1.920	0.02a	73.10s	870.93	0.000	0.000	3.1006
-1.992	0.02a	75.00s	870.93	0.000	-0.185	3.0975

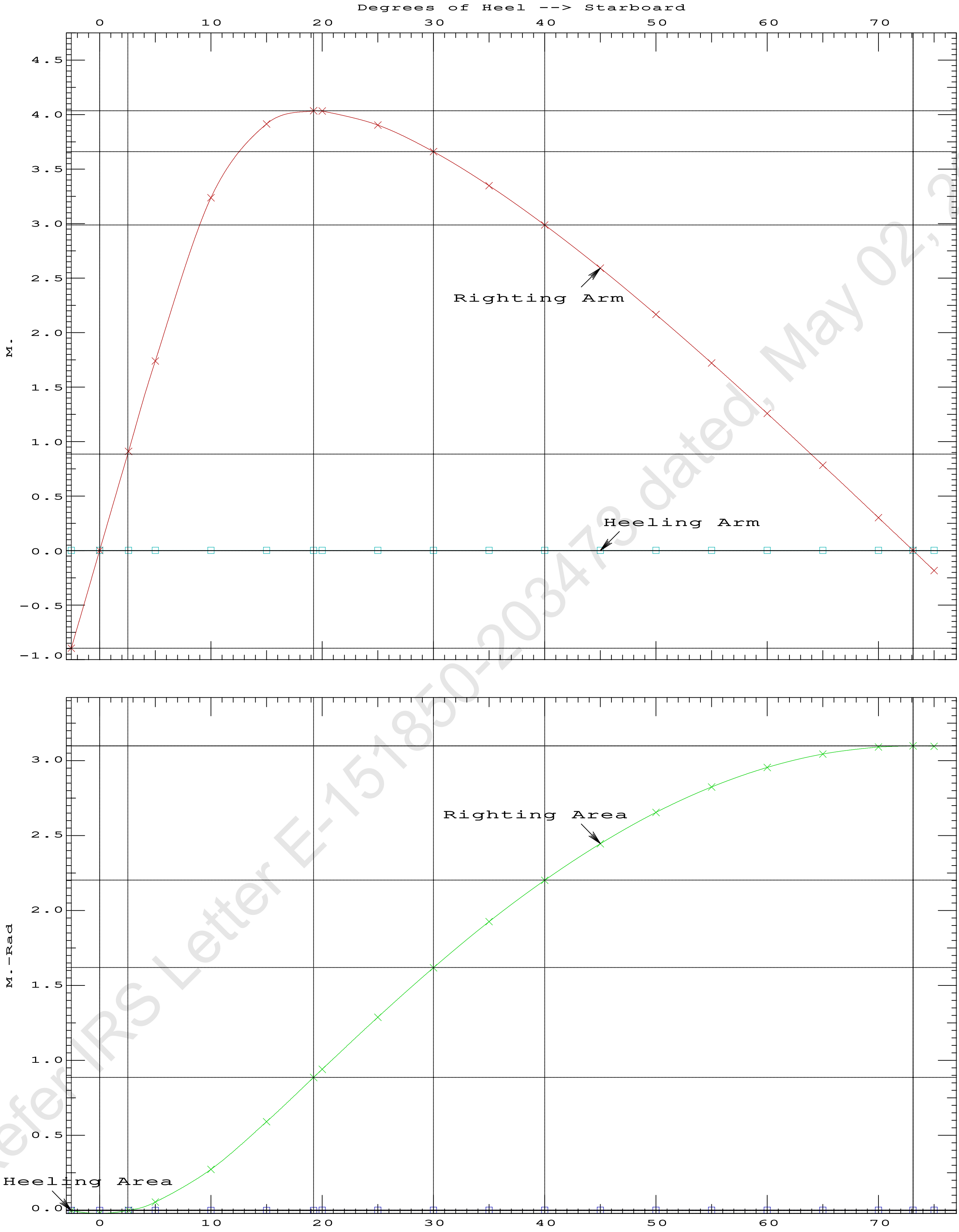
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 606.33 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM---RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -2.6 deg to RAzero	>	1.000		155.776 P
(2) Angle from abs 2.6 deg to RAzero	>	20.00	deg	70.50 P

25-04-22 10:28:44 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

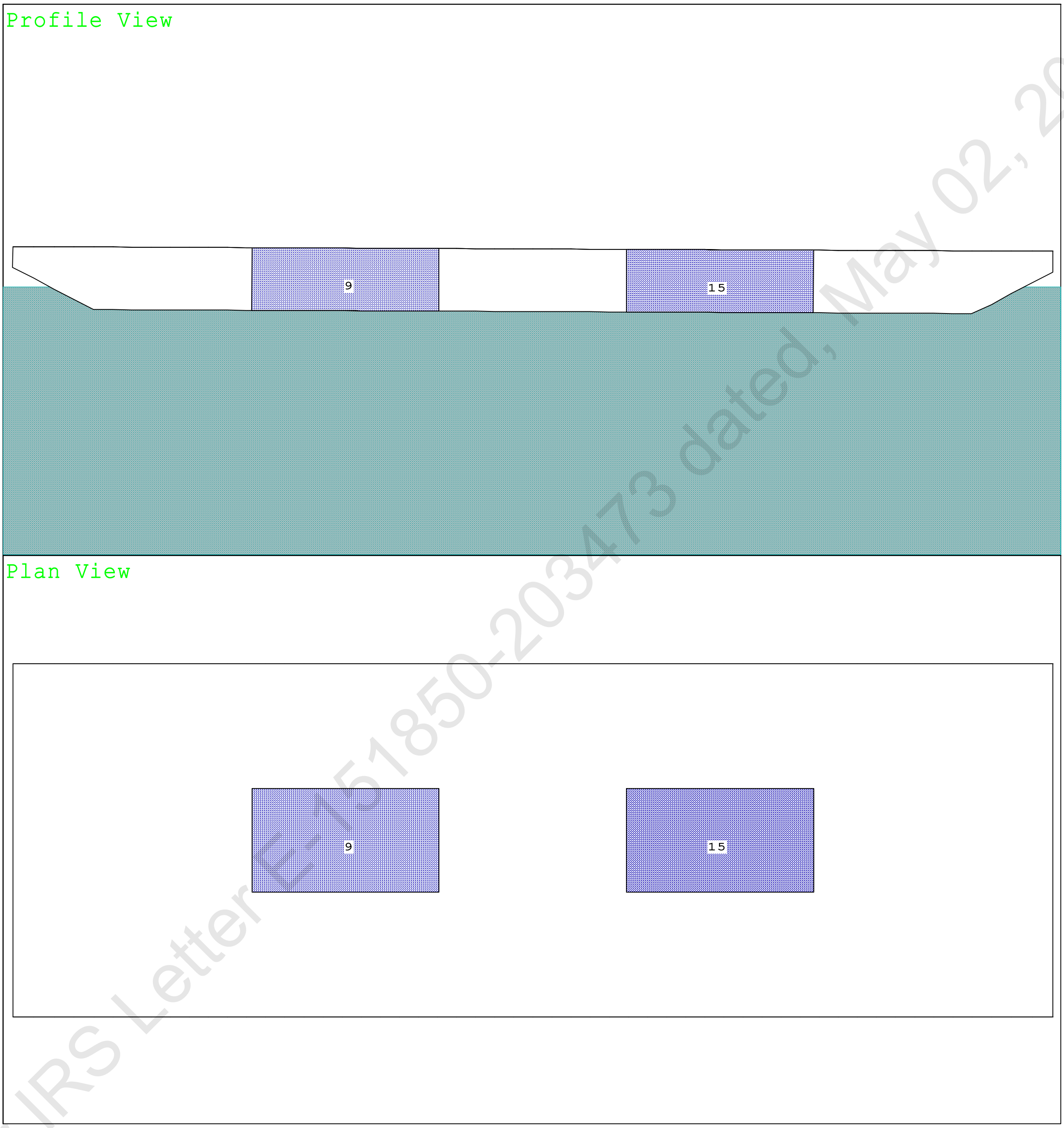


KB-24

Condition 5 : Lightship + Load at Fwd + Ballast

Crane Lifting 36.60T @ 18m Radius Fwd

Condition Graphic - Draft: 1.308 @ 50.000f, 1.084 @ 0.000 Heel: zero



Tanks 9 03_PFWBTK.C.100% FRESH WATER 15 05_PFWBTK.C.100% FRESH WATER

25-04-22 10:23:55 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 5 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

```
                  WEIGHT and DISPLACEMENT and WATERPLANE STATUS
             USK draft: 1.308 @ 50.00f, 1.196 @ 25.00f, 1.084 @ 0.00
                       Trim: Fwd 0.224/50.000,  Heel: zero
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                               356.33  25.000f  0.000   3.000
Crane                                    250.00  25.000f  0.000   5.200
Crane Load                                36.60  43.000f  0.000  76.977
  Total Fixed--------->                  642.93  26.025f  0.000   8.067
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000        132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000        132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->                  264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->                  907.53  25.726f  0.000   6.152
                                       Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025            907.49  25.752f  0.000   0.609
------------------------------------------------------------------------------
               Righting Arms:                    0.001f  0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025           794.9  25.206f  0.000    163.61  21.367*
                                        MT/cm--------m.-MT/cm-------GML-----GMT
                                         8.15           28.69    158.07  15.824
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            SUMMARY OF LOADING
      264.6 Cu.M.  (11%) FRESH WATER

     250.00 MT  Crane                       36.60 MT  Crane Load

DRAFT AT FP (M) =     1.308
DRAFT AT MS (M) =     1.196
DRAFT AT AP (M) =     1.084
DRAFT AT FWD MARK (M) =     1.290
DRAFT AT AFT MARK (M) =     1.102
                              HYDROSTATIC PROPERTIES
                 Trim: Fwd 0.224/50.000, No Heel, VCG = 6.152

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.197       907.49  25.752f  0.609     8.15  25.206f   28.69  158.07  15.824
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                              Trim is per 50.00m.
 Draft is from USK.                              Formal Free Surface included.

     Note:  GMT includes the formal free surface moment 183.7 m.-MT
```

25-04-22 10:23:55 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 5 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.308 @ 50.00f, 1.196 @ 25.00f, 1.084 @ 0.00
Trim: Fwd 0.224/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	89.4	0.920	1.502	0.05500	7.38
CRANE.C	32.0	3.816	4.398	0.05500	7.74
Total wind heeling moment to starboard------------->					15.12

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.726f TCG = 0.000 VCG = 6.152
Free Surface Adjustment: 0.202
Adjusted CG: LCG = 25.725f TCG = 0.000 VCG = 6.355

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.072	0.26f	0.00	907.48	0.000	-0.016	0.0000	5.963
1.072	0.26f	0.06s	907.53	0.000	0.000	-0.0000	5.908
1.059	0.25f	5.06s	907.52	0.000	1.392	0.0607	0.908
1.052	0.26f	5.96s	907.53	0.000	1.637	0.0847	0.000
0.982	0.28f	10.06s	907.50	0.000	2.582	0.2363	-4.092
0.750	0.37f	15.06s	907.53	0.000	2.984	0.4835	-9.092
0.729	0.38f	15.44s	907.53	0.000	2.986	0.5035	-9.476
0.467	0.50f	20.06s	907.53	0.000	2.820	0.7406	-14.092
0.178	0.63f	25.06s	907.53	0.000	2.431	0.9713	-19.092
-0.114	0.76f	30.06s	907.53	0.000	1.941	1.1628	-24.092
-0.409	0.90f	35.06s	907.53	0.000	1.397	1.3088	-29.092
-0.702	1.03f	40.06s	907.53	0.000	0.822	1.4058	-34.092
-0.991	1.15f	45.06s	907.53	0.000	0.228	1.4517	-39.092
-1.098	1.20f	46.95s	907.53	0.000	0.000	1.4555	-40.983
-1.273	1.27f	50.06s	907.53	0.000	-0.376	1.4453	-44.092
-1.544	1.37f	55.06s	907.53	0.000	-0.982	1.3861	-49.092
-1.803	1.47f	60.06s	907.53	0.000	-1.583	1.2742	-54.092

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

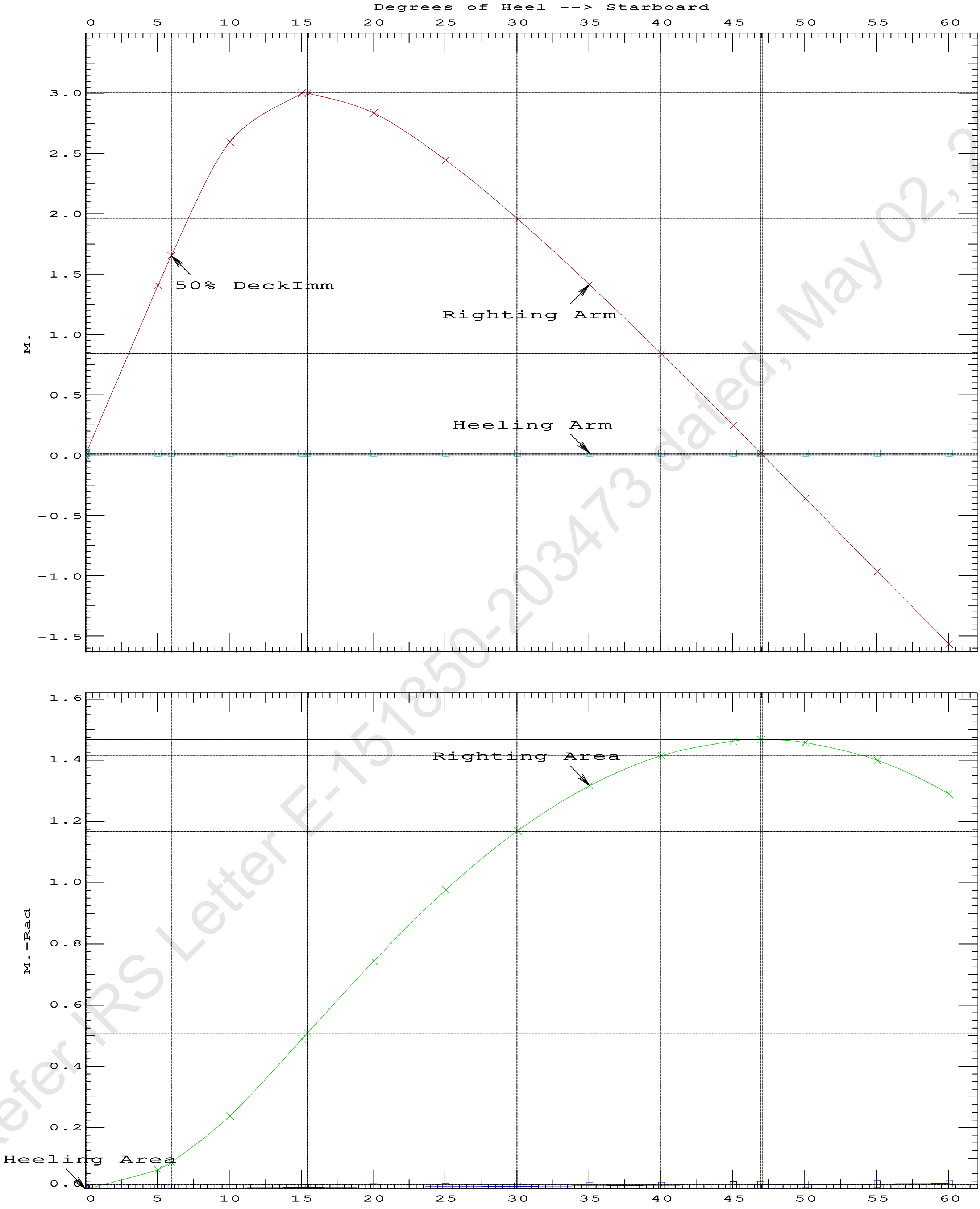
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 15.12 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.5035	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	46.89	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	5.91	P

25-04-22 10:23:55 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 5 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd



25-04-22 10:23:55 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 5 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.726f TCG = 0.000 VCG = 6.152
Free Surface Adjustment: 0.202
Adjusted CG: LCG = 25.725f TCG = 0.000 VCG = 6.355

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.072	0.26f	0.00	907.52	0.000	0.000	0.0000
1.059	0.25f	5.00s	907.53	0.000	1.392	0.0608
0.984	0.28f	10.00s	907.50	0.000	2.589	0.2359
0.753	0.37f	15.00s	907.53	0.000	2.999	0.4850
0.728	0.38f	15.45s	907.53	0.000	3.001	0.5085
0.471	0.50f	20.00s	907.53	0.000	2.839	0.7435
0.182	0.63f	25.00s	907.53	0.000	2.451	0.9769
-0.111	0.76f	30.00s	907.53	0.000	1.962	1.1701
-0.406	0.89f	35.00s	907.53	0.000	1.418	1.3180
-0.699	1.02f	40.00s	907.53	0.000	0.843	1.4169
-0.988	1.15f	45.00s	907.53	0.000	0.249	1.4648
-1.106	1.20f	47.07s	907.53	0.000	0.000	1.4693
-1.270	1.27f	50.00s	907.53	0.000	-0.354	1.4603
-1.541	1.37f	55.00s	907.53	0.000	-0.960	1.4030
-1.800	1.47f	60.00s	907.53	0.000	-1.561	1.2929

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

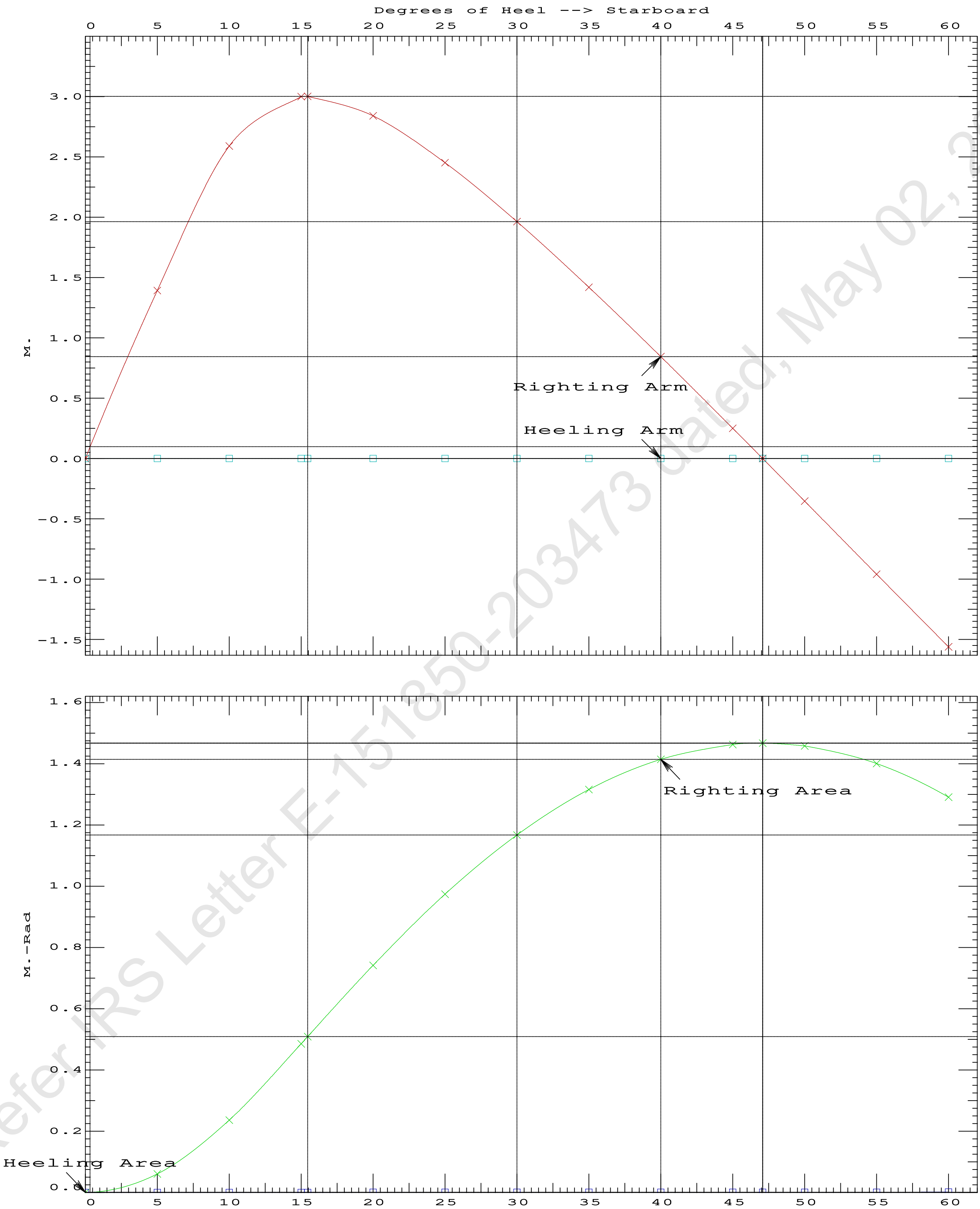
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	272382	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1623.97	P

25-04-22 10:23:55 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 5 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd



25-04-22 10:23:51 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

Condition 5 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.307 @ 50.00f, 1.196 @ 25.00f, 1.084 @ 0.00
Trim: Fwd 0.223/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			356.33	25.000f	0.000	3.000	
Crane			250.00	25.000f	0.000	5.200	
Crane Load			36.60	43.000f	0.000	76.977	
Total Fixed--------->			642.93	26.025f	0.000	8.067	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			907.53	25.726f	0.000	6.152	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	907.53	25.750f	0.001s	0.609

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001s
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	791.6	25.127f	0.025s	161.56	21.241*
		MT/cm		m.-MT/cm	GML	GMT
		8.11		28.32	156.02	15.698

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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25-04-22 10:23:51    Cybermarine Technologies PTE. Ltd.
GHS 18.00                        KB-24

                    After hook load drop
           WEIGHT and DISPLACEMENT and WATERPLANE STATUS
      USK draft: 1.148 @ 50.00f, 1.151 @ 25.00f, 1.154 @ 0.00
             Trim: Aft 0.006/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           356.33  25.000f  0.000   3.000
Crane                                250.00  25.000f  0.000   5.200
  Total Fixed--------->              606.33  25.000f  0.000   3.907
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000        132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000        132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->              264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->              870.93  25.000f  0.000   3.176
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        870.93  25.000f  0.001s  0.584
------------------------------------------------------------------------------
             Righting Arms:                   0.000   0.001s
             External Arms:                   0.000   0.001s
      Residual Righting Arms:                 0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         790.2  24.996f  0.014s   167.45  22.085*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.10           28.72    164.86  19.493
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

              Theta3 is 0.07 degrees, as demonstrated below.

                    RESIDUAL RIGHTING ARMS vs HEEL ANGLE
           Fixed CG:  LCG = 25.000f  TCG = 0.000   VCG =  3.907

      Origin    Degrees of   Displacement    Residual Arms       Res.
       Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
       1.142   0.01a   0.00       870.93     0.000    0.000    0.0000
       1.142   0.01a   0.07s      870.93     0.000    0.024    0.0000
       1.129   0.01a   5.00s      870.93     0.000    1.738    0.0759
       1.060   0.01a  10.00s      870.93     0.000    3.235    0.2947
       0.851   0.01a  15.00s      870.93     0.000    3.914    0.6126
       0.644   0.01a  19.22s      870.93     0.000    4.035    0.9077
       0.605   0.01a  20.00s      870.93     0.000    4.031    0.9627
```

0.353	0.01a	25.00s	870.70	0.000	3.903	1.3115
0.099	0.01a	30.00s	870.73	0.000	3.659	1.6423

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 606.33 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------------STABILITY CRITERION		Min/Max	Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	LARGE

25-04-22 10:23:51 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 25.000f TCG = 0.000 VCG = 3.907

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.142	0.01a	0.00	870.93	0.000	0.000	0.0000
1.142	0.01a	0.07s	870.93	0.000	0.024	0.0000
1.129	0.01a	5.00s	870.93	0.000	1.738	0.0759
1.060	0.01a	10.00s	870.93	0.000	3.235	0.2947
0.851	0.01a	15.00s	870.93	0.000	3.914	0.6126
0.644	0.01a	19.22s	870.93	0.000	4.035	0.9077
0.605	0.01a	20.00s	870.93	0.000	4.031	0.9627
0.353	0.01a	25.00s	870.70	0.000	3.903	1.3115
0.099	0.01a	30.00s	870.73	0.000	3.659	1.6423
-0.155	0.01a	35.00s	870.93	0.000	3.346	1.9485
-0.408	0.01a	40.00s	870.93	0.000	2.986	2.2251
-0.658	0.01a	45.00s	870.93	0.000	2.590	2.4686
-0.904	0.01a	50.00s	870.93	0.000	2.166	2.6763
-1.142	0.01a	55.00s	870.93	0.000	1.720	2.8460
-1.372	0.02a	60.00s	870.93	0.000	1.258	2.9761
-1.591	0.02a	65.00s	870.93	0.000	0.783	3.0652
-1.798	0.02a	70.00s	870.93	0.000	0.301	3.1126
-1.920	0.02a	73.10s	870.93	0.000	0.000	3.1207
-1.992	0.02a	75.00s	870.92	0.000	-0.185	3.1176

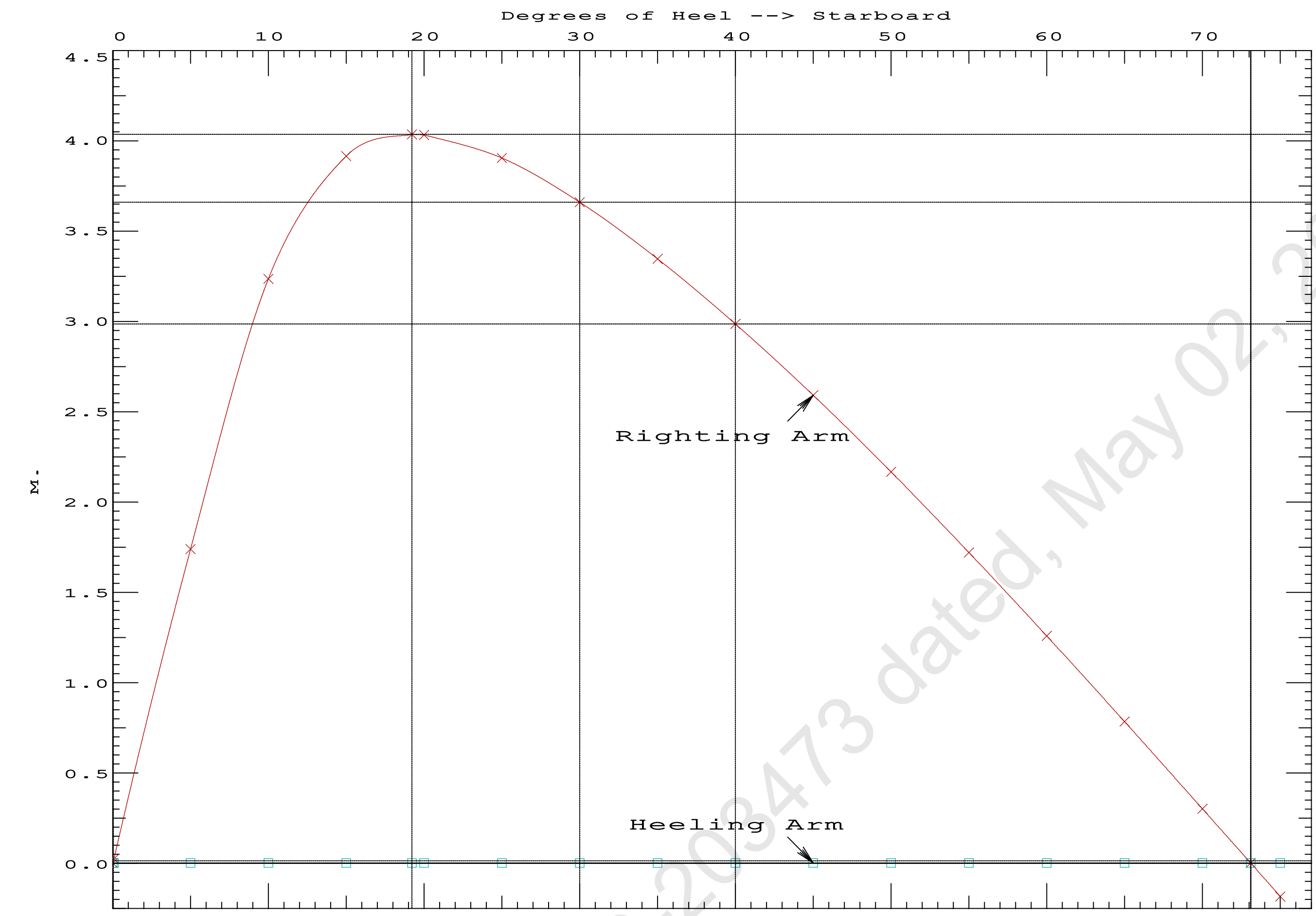
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

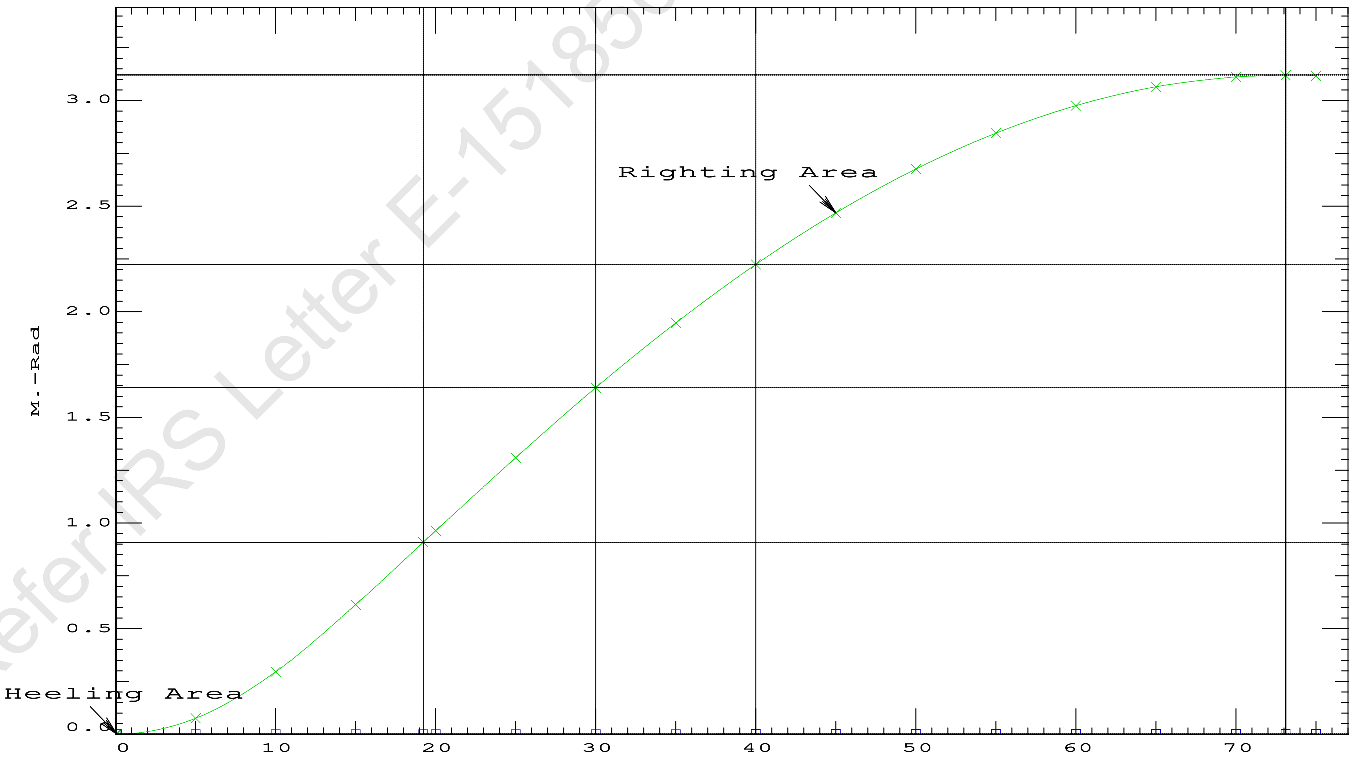
Note: The Center of Gravity shown above is for the Fixed Weight of 606.33 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1)	Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2)	Angle from abs 0.1 deg to RAzero	>	20.00	deg	73.03 P

25-04-22 10:23:51 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**





LIGHTSHIP PARTICULARS

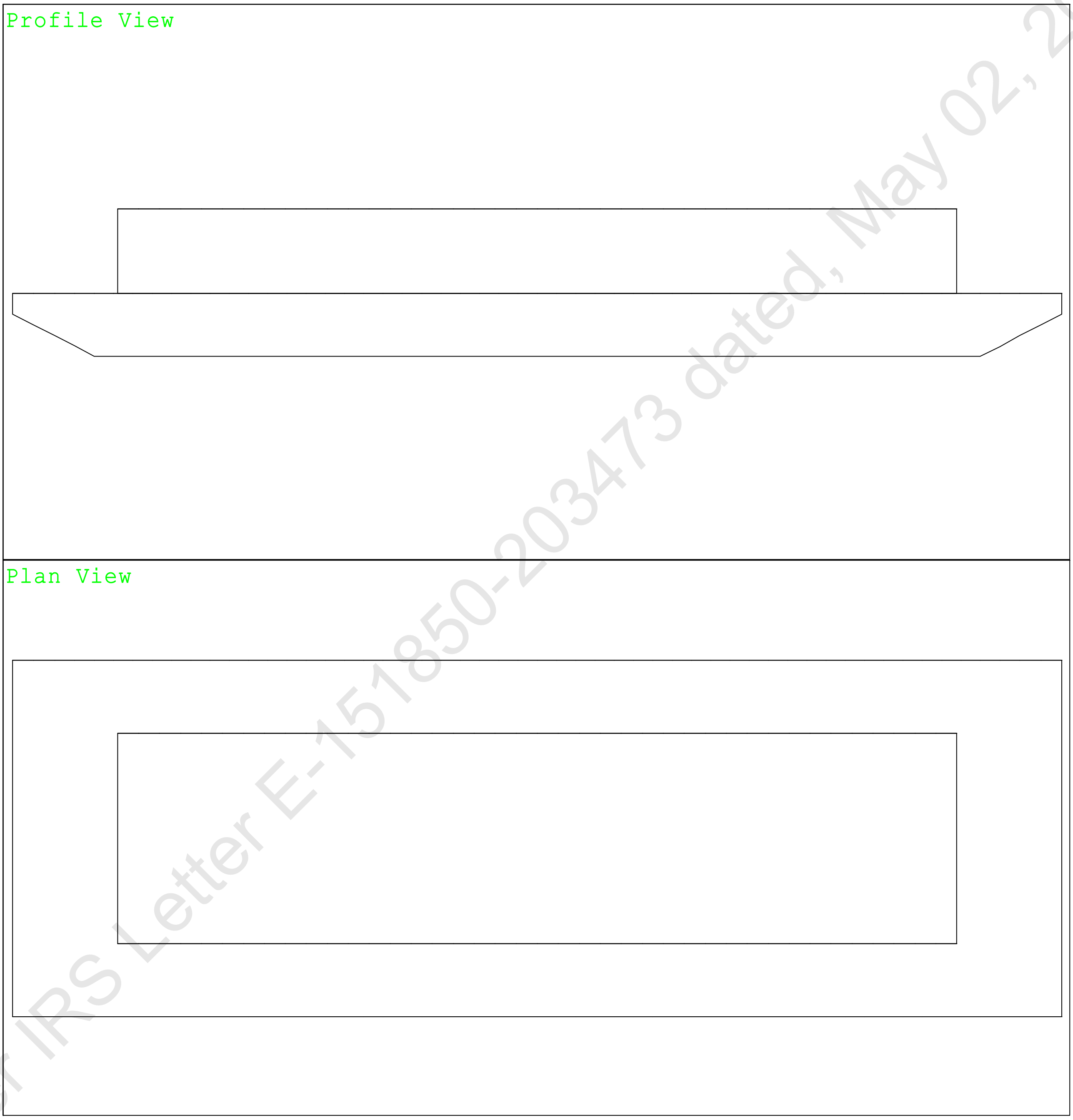
Lightship Weight	=	356.33	T
VCG	=	3.000	M above Baseline
LCG	=	25.000	M fwd of AP
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from IRS approved Draft Survey Report.)

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

WINDAGE AREA CALCULATION AT DIFFERENT DRAFTS

Condition Graphic



25-04-22 11:44:04 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

LATERAL PLANE STATUS
USK draft: 0.600 @ 50.00f, 0.600 @ 25.00f, 0.600 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	25.6	25.041f	-0.290	116.6	25.000f	1.232
DECKCARGO.C				160.0	25.000f	4.412
Total Lateral Plane->	25.6	25.041f	-0.290	276.6	25.000f	3.071

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.700 @ 50.00f, 0.700 @ 25.00f, 0.700 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	30.1	25.035f	-0.339	112.1	25.000f	1.179
DECKCARGO.C				160.0	25.000f	4.312
Total Lateral Plane->	30.1	25.035f	-0.339	272.1	25.000f	3.021

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.800 @ 50.00f, 0.800 @ 25.00f, 0.800 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	34.6	25.031f	-0.388	107.6	25.000f	1.126
DECKCARGO.C				160.0	25.000f	4.212
Total Lateral Plane->	34.6	25.031f	-0.388	267.6	25.000f	2.971

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.900 @ 50.00f, 0.900 @ 25.00f, 0.900 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	39.1	25.027f	-0.438	103.1	25.000f	1.073
DECKCARGO.C				160.0	25.000f	4.112
Total Lateral Plane->	39.1	25.027f	-0.438	263.1	25.000f	2.921

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.000 @ 50.00f, 1.000 @ 25.00f, 1.000 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	43.7	25.024f	-0.486	98.6	25.000f	1.020
DECKCARGO.C				160.0	25.000f	4.012
Total Lateral Plane->	43.7	25.024f	-0.486	258.6	25.000f	2.871

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.100 @ 50.00f, 1.100 @ 25.00f, 1.100 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	48.4	25.022f	-0.534	93.9	25.000f	0.968
DECKCARGO.C				160.0	25.000f	3.912
Total Lateral Plane->	48.4	25.022f	-0.534	253.9	25.000f	2.823

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.200 @ 50.00f, 1.200 @ 25.00f, 1.200 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	53.1	25.020f	-0.582	89.2	25.000f	0.917
DECKCARGO.C				160.0	25.000f	3.812
Total Lateral Plane->	53.1	25.020f	-0.582	249.2	25.000f	2.776

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.300 @ 50.00f, 1.300 @ 25.00f, 1.300 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	57.8	25.018f	-0.631	84.5	25.000f	0.865
DECKCARGO.C				160.0	25.000f	3.712
Total Lateral Plane->	57.8	25.018f	-0.631	244.5	25.000f	2.728

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.400 @ 50.00f, 1.400 @ 25.00f, 1.400 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	62.5	25.017f	-0.679	79.8	25.000f	0.813
DECKCARGO.C				160.0	25.000f	3.612
Total Lateral Plane->	62.5	25.017f	-0.679	239.8	25.000f	2.681

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.500 @ 50.00f, 1.500 @ 25.00f, 1.500 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	67.2	25.015f	-0.728	75.1	25.000f	0.760
DECKCARGO.C				160.0	25.000f	3.512
Total Lateral Plane->	67.2	25.015f	-0.728	235.1	25.000f	2.633

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.600 @ 50.00f, 1.600 @ 25.00f, 1.600 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	72.1	25.014f	-0.775	70.2	25.000f	0.710
DECKCARGO.C				160.0	25.000f	3.412
Total Lateral Plane->	72.1	25.014f	-0.775	230.2	25.000f	2.588

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.700 @ 50.00f, 1.700 @ 25.00f, 1.700 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	77.0	25.013f	-0.822	65.3	25.000f	0.659
DECKCARGO.C				160.0	25.000f	3.312
Total Lateral Plane->	77.0	25.013f	-0.822	225.3	25.000f	2.543

Distances in METERS.

25-04-22 11:44:04 Cybermarine Technologies PTE. Ltd.
GHS 18.00 **KB-24**

LATERAL PLANE STATUS

USK draft: 1.800 @ 50.00f, 1.800 @ 25.00f, 1.800 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	81.9	25.013f	-0.870	60.4	25.000f	0.608
DECKCARGO.C				160.0	25.000f	3.212
Total Lateral Plane->	81.9	25.013f	-0.870	220.4	25.000f	2.498

Distances in METERS.

LATERAL PLANE STATUS

USK draft: 1.900 @ 50.00f, 1.900 @ 25.00f, 1.900 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	86.8	25.012f	-0.918	55.5	25.000f	0.557
DECKCARGO.C				160.0	25.000f	3.112
Total Lateral Plane->	86.8	25.012f	-0.918	215.5	25.000f	2.454

Distances in METERS.

LATERAL PLANE STATUS

USK draft: 2.000 @ 50.00f, 2.000 @ 25.00f, 2.000 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	91.7	25.011f	-0.966	50.6	25.000f	0.506
DECKCARGO.C				160.0	25.000f	3.012
Total Lateral Plane->	91.7	25.011f	-0.966	210.6	25.000f	2.410

Distances in METERS.

LATERAL PLANE STATUS

USK draft: 2.100 @ 50.00f, 2.100 @ 25.00f, 2.100 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	96.6	25.011f	-1.013	45.6	25.000f	0.456
DECKCARGO.C				160.0	25.000f	2.912
Total Lateral Plane->	96.6	25.011f	-1.013	205.6	25.000f	2.367

Distances in METERS.

LATERAL PLANE STATUS

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	102.2	25.010f	-1.066	40.0	25.000f	0.400
DECKCARGO.C				160.0	25.000f	2.800
Total Lateral Plane->	102.2	25.010f	-1.066	200.0	25.000f	2.320

Distances in METERS.

IRCLASS
Indian Register of Shipping
CIN: U61100MH1975NPL018244
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Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

Our ref: E-105932-140827 06-July-2020

Company : Cybermarine Knowledge System Pvt. Ltd.

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project details	
Project number	NC002469
Project name	Chisti Marine Engineering Works -CHISTI/011
Project Manager	Dhruvesh Singh
Project In-Charge	Veerendra P

Document Details	
Plan No.	CM20-V-0205
Plan Title	Draft Survey Report
Rev.No.	0
Upload.No.	1
Submitted Date	27-June-2020
Review Status	Approved
Parent Plan Title	

Notes

1. The review status Approved indicates:

   The following Lightship particulars are approved:
   Lightship Displacement - 356.33 T
   Lightship L.C.G - 25.000 m from AP (Fr.0)
   Lightship V.C.G - 3.000 m above BL
   Lightship T.C.G - 0.000 m off-centerline

   Stability Documents prepared based on the approved Lightship particulars should be submitted for approval.

   Copy of the approved draft survey report should form part of the Trim & Stability booklet.

Thanking you,
Yours Sincerely,
For INDIAN REGISTER OF SHIPPING

Pradeepkumar K. V
Senior Surveyor

This is an electronically published document and does not require signature.